

# Module 17: Advanced Materials: Composites and Ceramics

## 17.1 Introduction to Advanced Materials

### Overview

Advanced materials represent a class of engineered substances offering superior performance characteristics compared to traditional metals and polymers. These materials—including composites, ceramics, and specialized alloys—enable applications requiring exceptional strength-to-weight ratios, high-temperature stability, corrosion resistance, or unique electrical and thermal properties.

CNC machining of advanced materials presents unique challenges requiring specialized knowledge of material science, tooling, process parameters, and safety considerations. This module explores the practical aspects of processing non-traditional materials on CNC equipment.

### Historical Context

**Early Development (1940s-1960s)** - Glass fiber reinforced plastics (GFRP) for marine and aerospace - Carbide and ceramic cutting tools enabled harder material machining - Initial applications in military and high-performance sectors

**Expansion (1970s-1980s)** - Carbon fiber composites for aerospace and motorsports - Advanced ceramics for electronics and wear applications - Development of diamond tooling for abrasive materials

**Modern Era (1990s-Present)** - Widespread adoption in automotive and consumer products - Nanocomposites and advanced ceramic matrix materials - 3D printing and hybrid manufacturing processes - Cost reduction enabling broader industrial use

### Material Categories

#### Composite Materials

**Definition:** Two or more constituent materials combined to achieve properties superior to individual components.

**Key Components:** - **Reinforcement:** Fibers providing strength (carbon, glass, aramid, ceramic) - **Matrix:** Polymer, metal, or ceramic binding reinforcement - **Interface:** Critical bond between fiber and matrix

**Common Types:** - Fiber-reinforced polymers (FRP): Carbon fiber, fiberglass, aramid - Metal matrix composites (MMC): Aluminum with ceramic reinforcement - Ceramic matrix composites (CMC): Silicon carbide fibers in ceramic - Sandwich structures: Composite skins with foam/honeycomb core

#### Ceramic Materials

**Definition:** Inorganic, non-metallic materials typically crystalline and formed through high-temperature processing.

**Categories:** - **Oxide ceramics:** Alumina ( $\text{Al}_2\text{O}_3$ ), zirconia ( $\text{ZrO}_2$ ) - **Non-oxide ceramics:** Silicon carbide (SiC), silicon nitride ( $\text{Si}_3\text{N}_4$ ) - **Glass ceramics:** Partially crystalline materials (Macor, Zerodur) - **Traditional ceramics:** Porcelain, clay-based materials

**Unique Properties:** - Extreme hardness (approaching diamond) - High temperature stability (>1000 degC) - Excellent wear and corrosion resistance - Electrical insulation or conductivity (depending on type) - Brittleness (low fracture toughness)

### Advanced Alloys

**Superalloys:** Nickel, cobalt, or iron-based alloys for extreme temperatures - Inconel, Waspaloy, Hastelloy - Aerospace turbines, chemical processing

**Titanium Alloys:** High strength-to-weight with corrosion resistance - Ti-6Al-4V (Grade 5) most common - Aerospace, medical implants, marine

**Tool Steels:** High hardness after heat treatment - M2, D2, H13 grades - Tooling, dies, cutting implements

## Why Advanced Materials?

### Performance Advantages

**Weight Reduction:** - Carbon fiber: 1.6 g/cm<sup>3</sup> vs. aluminum 2.7 g/cm<sup>3</sup> vs. steel 7.8 g/cm<sup>3</sup> - Aerospace fuel savings: 20-30% weight reduction possible - Automotive efficiency gains - Increased payload capacity

**Strength-to-Weight Ratio:** - Carbon fiber composites: 3-5x steel on strength-per-unit-weight basis - Enables lighter structures with equivalent or superior strength - Critical for aerospace, automotive, sports equipment

**Corrosion Resistance:** - Composites immune to electrochemical corrosion - Ceramics inert to most chemicals - Eliminates protective coatings and maintenance

**Thermal Stability:** - Ceramics maintain properties at extreme temperatures - Carbon-carbon composites for brake discs (>1500 degC) - Thermal barriers and insulation applications

**Tailored Properties:** - Fiber orientation controls directional strength - Composite layup optimized for load paths - Multi-functional structures (structural + thermal + electrical)

### Economic Considerations

**Material Costs:** - Carbon fiber prepreg: \$30-\$80/kg - Fiberglass: \$5-\$15/kg - Alumina ceramic: \$20-\$100/kg - Steel (comparison): \$1-\$3/kg

**Processing Costs:** - Specialized tooling (diamond, PCD): 5-20x carbide cost - Frequent tool changes increase labor - Dust collection systems: \$5,000-\$50,000 - Slower machining rates than metals

**Lifecycle Value:** - Reduced weight = fuel savings (aerospace, automotive) - Extended service life (wear resistance) - Eliminated corrosion maintenance - Performance advantages justify premium

**Applications Driving Adoption:** - Aerospace: Weight reduction, performance - Automotive: Fuel efficiency, emissions regulations - Wind energy: Large blade structures - Sports: Performance advantages - Medical: Biocompatibility, imaging transparency

## Machining Challenges

### Material-Specific Issues

**Composites:** - **Abrasive fiber wear:** Rapid tool dulling - **Delamination:** Layers separate under cutting forces - **Fiber pullout:** Incomplete cutting leaves protruding fibers - **Matrix melting:** Thermal damage from friction - **Anisotropic properties:** Strength varies with fiber direction

**Ceramics:** - **Extreme hardness:** Difficult to cut, rapid tool wear - **Brittleness:** Chipping and cracking - **Microcracking:** Subsurface damage reduces strength - **Thermal shock:** Sudden temperature changes cause fracture

**Advanced Alloys:** - **Work hardening:** Titanium and nickel alloys harden during cutting - **Heat generation:** Low thermal conductivity concentrates heat at tool - **Chemical reactivity:** Titanium reacts with oxygen and tool materials - **Built-up edge:** Material adheres to cutting edge

## Dust and Contamination

**Health Hazards:** - Carbon fiber dust: Respirable particles <10  $\mu\text{m}$ , potential carcinogen - Ceramic dust: Silicosis risk from crystalline silica - Resin decomposition: VOCs and toxic fumes

**Equipment Damage:** - Abrasive dust accelerates wear on guides, screws, seals - Electrical conductivity of carbon fiber causes shorts - Contamination of precision surfaces

**Environmental Concerns:** - Fine dust escapes standard filtration - HEPA and activated carbon filtration required - Disposal regulations for composite and ceramic waste

## Module Scope

This module focuses on CNC machining of advanced materials, covering:

**Material Science** (Sections 17.2-17.3): - Composite structures and behavior - Ceramic properties and classifications - Material selection for applications

**Processing Techniques** (Sections 17.4-17.5): - Cutting mechanics and tool selection - Process parameters and optimization - Fixtures and workholding

**Support Systems** (Section 17.6): - Dust collection and filtration - Environmental controls - Contamination prevention

**Quality and Finishing** (Sections 17.7-17.8): - Surface finishing techniques - Inspection and quality control - Defect identification and mitigation

**Safety and Maintenance** (Sections 17.9-17.10): - Health hazards and protection - Equipment maintenance - Regulatory compliance

**Practical Application** (Section 17.11-17.12): - Troubleshooting common issues - Industry case studies - Future trends

## Design for Manufacturability

**Composite Design Principles:** - Avoid through-holes perpendicular to fiber layers (delamination risk) - Design for net-shape manufacturing when possible - Include machining allowances for edge trimming - Consider fiber orientation relative to loads and machining directions

**Ceramic Design Rules:** - Generous radii (avoid stress concentrations) - Avoid thin walls (<2mm typical minimum) - Support during machining (brittle fracture risk) - Green or bisque machining before final sintering (when applicable)

**Tolerance Expectations:** - Composites:  $\pm 0.1\text{--}0.5$  mm typical (fiber orientation affects precision) - Ceramics:  $\pm 0.05\text{--}0.2$  mm achievable with diamond tooling - Tighter tolerances possible but expensive (slow feeds, frequent tool changes)

## Safety Preview

**Critical Hazards:** - Respiratory: Carcinogenic and toxic dusts - Skin/Eye: Irritation from fibers and particulates - Fire: Carbon fiber dust combustible, conductive - Chemical: Resin fumes, ceramic binders

**Required Controls** (Detailed in Section 17.9): - Engineering: Dust collection, ventilation, enclosures - Administrative: Training, procedures, exposure monitoring - PPE: Respirators, protective clothing, eye protection

## Economic Decision Framework

**When to Machine Advanced Materials In-House:** - High volume justifies equipment investment - Proprietary designs require confidentiality - Quick turnaround needed - Iterative design process benefits from direct control

**When to Outsource:** - Low volume, prototyping - Lack of specialized equipment or expertise - Regulatory compliance burden (dust, waste) - Risk mitigation for expensive materials

**Hybrid Approach:** - Near-net-shape manufacturing (molding, layup) + CNC finishing - Additive manufacturing + subtractive machining - Water jet or laser cutting + CNC edge finishing

## Module Objectives

By completing this module, you will be able to:

1. Understand composite and ceramic material structures and properties
2. Select appropriate tooling and cutting parameters for advanced materials
3. Design and implement effective dust collection systems
4. Apply proper safety protocols for hazardous material machining
5. Diagnose and troubleshoot common defects (delamination, chipping, etc.)
6. Perform quality inspection specific to composites and ceramics
7. Maintain CNC equipment operating in abrasive environments
8. Evaluate economic trade-offs for advanced material processing

## Prerequisites

This module builds on: - **Module 1-3:** Mechanical systems (rigidity important for brittle materials) - **Module 4:** Control systems (no advanced concepts, standard CNC) - **Module 6:** Spindle systems (high-speed operation for composites) - Basic knowledge of materials science helpful but not required

## Key Takeaways

- Advanced materials offer exceptional performance but require specialized processing
- Tooling costs and wear rates significantly exceed metal machining
- Dust control is mandatory for health, equipment protection, and quality
- Safety protocols must address respiratory, skin, fire, and chemical hazards
- Economic viability depends on application value and production volume
- Proper material handling and machining prevents costly defects

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**Next:** 17.2 Composite Materials Science

## 17.10 Maintenance and Tool Management for Advanced Materials

### Tool Wear Characteristics

#### Composite Machining Tool Wear

**Abrasive Wear** (dominant mechanism): - Carbon fibers: 10x harder than aluminum (Mohs 6-7) - Glass fibers: Very abrasive (Mohs 5-6) - Aramid fibers (Kevlar): Less abrasive but fibrous (difficult to cut cleanly)

**Wear Progression:** 1. **New tool:** Sharp edge, clean cuts 2. **Initial wear** (5-20% tool life): Edge radius increases slightly - Quality still acceptable - Cutting forces increase ~10% 3. **Accelerated wear** (20-80% tool life): Visible flank wear - Surface finish degrades - Forces increase 20-40% 4. **End of life** (80-100%): Rapid quality loss - Delamination, fuzzing - Burning (heat from friction) - Catastrophic failure possible (chip breakage)

#### Tool Life Examples:

| Material | Tool                       | Operation          | Parts Before Replacement |
|----------|----------------------------|--------------------|--------------------------|
| CFRP     | Carbide endmill            | Slotting           | 50-150                   |
| CFRP     | Diamond-coated compression | Trimming           | 500-2000                 |
| GFRP     | Carbide endmill            | Profiling          | 100-300                  |
| GFRP     | PCD endmill                | Production routing | 5000-15000               |

**Factors Affecting Tool Life:** - Fiber type: Carbon = moderate wear, Glass = high wear, Aramid = low wear but poor cut quality - Fiber content: 60% fiber volume approx 2x wear vs 40% - Layer thickness: Thick layers = less delamination stress - Cutting speed: Higher speed = more heat = shorter life - Tool coating: Diamond coating increases life 10-20x

#### Ceramic Machining Tool Wear

##### Grinding Wheel Wear:

**Mechanisms:** 1. **Grain fracture:** Diamond particles break (self-sharpening) 2. **Bond wear:** Bonding material erodes, particles fall out 3. **Glazing:** Particles dull but don't fracture (wheel loads)

**Wear Rate:** - G-ratio (grinding ratio): Volume material removed / volume wheel wear - Diamond on alumina: G-ratio 1000-10000 (excellent) - Diamond on SiC: G-ratio 100-500 (lower, SiC extremely hard)

**Example:** - Grinding alumina - Material removed:  $1.0 \text{ in}^3$  - Wheel wear:  $0.001 \text{ in}^3$  - G-ratio: 1000 - Wheel life: 1000  $\text{in}^3$  material per  $1 \text{ in}^3$  wheel (months of grinding)

**Dressing** (wheel sharpening): - Required when: Wheel glazes (dull), loading (pores filled with debris) - Frequency: Every 10-100 parts (depends on material, conditions) - Method: Single-point diamond dresser - Removes thin layer of wheel (exposes fresh diamonds)

**Wheel Life:** 500-5000 dressings before replacement (bond worn away)

### Tool Inspection and Monitoring

#### Visual Inspection

##### Composites:

**Inspection Intervals:** - Initial: Every 10 parts - Once pattern established: Every 50 parts or 2 hours

##### What to Look For:

**Flank Wear** (cutting edge): - Bright, worn flat behind edge - Measure width (wear land) - Acceptable:  $<0.005\text{-}0.010''$  (depends on application) - Replace when:  $>0.015''$

**Chipping:** - Small chips missing from cutting edge - Causes immediate quality loss - Replace immediately

**Built-up Edge (BUE):** - Resin accumulated on cutting edge - Dulls tool, causes poor finish - Clean with solvent (acetone) - If returns quickly -> tool too hot (reduce speed)

**Tool for Inspection:** - 10-20x magnifier or USB microscope - Good lighting - Compare to new tool (reference photo)

**Ceramics** (grinding wheels):

**Visual Check:** - **Glazed surface:** Shiny, smooth (should be slightly rough) - Solution: Dress wheel -

**Loading:** Pores filled with ceramic debris - Solution: Dress wheel, check coolant flow - **Cracks:** Radial cracks in wheel (dangerous!) - Solution: Replace immediately (burst risk)

**Ring Test** (wheel integrity): - Suspend wheel by hole (string) - Tap with plastic hammer - Sound: Clear ring = good, dull thud = cracked - Frequency: Before mounting, weekly during use

## Dimensional Monitoring

### Tool Wear Trending:

**Process:** 1. Measure critical dimension (first part after tool change) 2. Measure same dimension every N parts 3. Plot dimension vs part number 4. Trend line shows wear rate 5. Predict when tool reaches end of life

**Example** (CFRP hole drilling): - Specification: 0.250  $\pm$  0.002" diameter - Hole grows as drill wears (cutting edge erodes)

| Part Number   | Hole Diameter | Status                      |
|---------------|---------------|-----------------------------|
| 1 (new drill) | 0.2495"       | Good                        |
| 25            | 0.2500"       | Good                        |
| 50            | 0.2505"       | Good                        |
| 75            | 0.2512"       | Warning (approaching limit) |
| 100           | 0.2521"       | <b>Out of tolerance</b>     |

**Wear rate:** 0.0026" per 100 parts **Tool life:** ~70 parts (conservative replacement point)

**Benefit:** Prevents scrap (replace before out-of-tolerance)

## Force/Current Monitoring

**Spindle Load Monitoring:** - CNC controller measures spindle motor current - Dull tool -> higher cutting forces -> higher current - Set threshold (e.g., 120% of new tool current) - Alarm when exceeded

**Example:** - New tool current: 8.5 A (average during cut) - Threshold: 10.2 A (120%) - Part 60: Current 10.5 A -> alarm -> inspect tool

**Advantages:** - Automatic (no manual inspection) - Real-time (immediate notification) - Prevents quality issues

**Limitations:** - Requires CNC with load monitoring - Must establish baseline (new tool) - Feed rate variations affect reading

## Tool Inventory Management

### Tool Identification

**Labeling System:** - Tool number (unique ID) - Tool type (e.g., "0.250 CFRP drill") - Purchase date - First use date - Part count (cumulative)

### Example Label:

Tool #: C-042  
Type: 1/4" carbide drill, 118 deg point  
Material: CFRP  
Purchased: 2024-01-15  
First Use: 2024-01-20  
Parts Machined: 237  
Status: In service (limit 300 parts)

**Tracking Methods:** - Spreadsheet: Simple, manual - Tool management software: Automatic (scans bar-code, logs use) - ERP integration: Links tool use to jobs (cost accounting)

### Stock Levels

#### Determine Par Levels:

$$\text{Par Level} = \text{Weekly Usage} \times \text{Lead Time (weeks)} \times \text{Safety Factor}$$

**Example** (PCD router bit for CFRP): - Weekly usage: 2 bits (production rate / tool life) - Lead time: 3 weeks (order to delivery) - Safety factor: 2x (avoid stockouts) - **Par level:**  $2 \times 3 \times 2 = 12 \text{ bits}$

**When stock drops below par:** Reorder

**Cost Trade-off:** - Too much inventory: Cash tied up, tools expire (carbide doesn't, but resharpenable tools degrade) - Too little inventory: Production stops (lost revenue)

### Tool Cost Tracking

#### Cost Per Part:

$$\text{Tool Cost Per Part} = \frac{\text{Tool Price}}{\text{Parts Per Tool}}$$

**Example:** - Diamond-coated compression cutter: \$120 - Tool life: 1500 parts - **Cost per part:**  $\$120 / 1500 = \$0.08$

**Compare to Alternative:** - Carbide compression cutter: \$30 - Tool life: 150 parts - **Cost per part:**  $\$30 / 150 = \$0.20$

**Diamond cutter 60% cheaper per part** (despite 4x higher initial cost)

**Total Cost** includes: - Tool purchase price - Regrinding cost (if resharpenable) - Downtime for tool changes (labor) - Scrap from worn tools

### Tool Reconditioning

#### Resharpening Carbide Tools

**Candidates:** - Endmills, drills (simple geometry) - Significant material remaining (not chipped away) - Cost-effective if tool >\$50 (resharpening ~\$15-30)

**Process:** - Send to tool grinding service - Grind flutes, cutting edges back to new geometry - Inspect, measure (diameter reduced by 0.005-0.020") - Apply new coating (optional, extends life)

**Regrind Count:** - Typical: 2-5 regrinds before tool too small - 1/4" endmill: Can regrind to ~0.230" (0.015" under)

**Cost Savings:** - New 1/4" carbide endmill: \$50 - Regrind: \$20 - **Savings:** \$30 per regrind x 3 regrinds = \$90 savings over tool life

**Not Recommended:** - PCD tools (diamond cannot be easily reground) - Severely worn tools (more material to remove than cost-effective)

### **Diamond Tool Refurbishment**

**PCD Tools:** - Grinding PCD requires specialized equipment (diamond wheels, EDM) - Expensive (\$100-300 per regrind) - Only economical for large, expensive tools (>\$500)

**Diamond-Coated Tools:** - Cannot regrind (coating only a few microns thick) - Once worn -> discard - Some suppliers: Recoating service (strip old coating, reapply) - Cost: 40-60% of new tool price - Quality variable (coating adhesion issues possible)

## **Machine Maintenance for Advanced Materials**

### **Spindle Maintenance**

**Challenges:** - Fine dust ingests into bearings (abrasive wear) - Carbon fiber conductive (can short windings) - High-speed operation (thermal expansion, vibration)

#### **Preventive Maintenance:**

**Air Purge System:** - Positive pressure (clean, filtered air) blown into spindle housing - Pressure: 5-15 PSI - Flow rate: 5-10 CFM - Prevents dust entry (air flows out through seals)

**Seal Replacement:** - Frequency: Annually or per manufacturer (composites/ceramics accelerate wear) - Signs of wear: Dust inside spindle housing, increased spindle temperature - Cost: \$200-1000 (seals + labor)

**Bearing Inspection:** - Frequency: Per manufacturer schedule (1000-5000 hours typical) - Check: Play (radial/axial runout), noise, temperature - Replace if: Runout >0.0002", noisy, hot (>140 degF in operation)

**Rebuild Cost:** \$2,000-20,000 (depends on spindle size)

**Extend Spindle Life:** - Excellent dust collection (minimize ingestion) - Air purge system (mandatory for ceramics) - Avoid overheating (proper speeds, sharp tools, coolant) - Gentle handling (no impacts, avoid crash)

### **Way and Ballscrew Protection**

**Threats:** - Abrasive dust (ceramic, glass fiber) acts as grinding compound - Settles on ways -> ballscrew ingests -> rapid wear

#### **Protection:**

**Bellows/Covers:** - Accordion-style covers over ways (X, Y, Z axes) - Prevent dust settling on guideways - Inspect for tears (dust enters through holes) - Replace annually or as needed

**Wipers:** - Scraper at carriage/table (wipes dust off ways) - Made of plastic or felt - Inexpensive, replace quarterly

**Lubrication:** - Increase frequency (dust contaminates oil) - Composites/ceramics: Daily way oiling (vs weekly for metals) - Use centralized lube system (automatic, consistent)

**Inspection:** - Weekly: Check for dust accumulation on ways (vacuum if present) - Monthly: Check ballscrew (rotate by hand, feel for roughness)

**Ballscrew Replacement:** \$1,000-10,000 per axis (expensive!)

### **Coolant System Maintenance**

**Coolant Contamination:** - Ceramic dust settles in tank (sludge) - Composite dust floats (scum on surface) - Bacteria growth (water-based coolants)



## **Maintenance:**

**Daily:** - Skim surface (remove floating debris) - Check level (top off if low)

**Weekly:** - Measure concentration (refractometer) - Too dilute: Poor lubrication, bacteria growth - Too concentrated: Waste, possible residue - Target: Per manufacturer (typically 5-10%) - Measure pH - Target: 8.5-9.5 (alkaline inhibits bacteria) - Low pH: Add fresh concentrate

**Monthly:** - Clean tank (remove sludge) - Inspect pump, filters - Replace filters if clogged

**Quarterly** (or sooner if contaminated): - Drain tank completely - Clean tank, piping (remove biofilm) - Refill with fresh coolant

**Alternative:** Dry machining or MQL (minimal quantity lubrication) - Eliminates coolant maintenance - Good for composites (dust suppression less critical than metals) - Ceramics: Water spray for cooling, dust suppression

## **Dust Collection System Maintenance**

**Daily:** - Empty dust hopper (don't let overfill) - Check pressure drop gauge (clean filter if high)

**Weekly:** - Pulse-clean filters (manual if no automatic system) - Inspect ductwork (loose connections, clogs)

**Monthly:** - Inspect filter cartridges (tears, excessive loading) - Check blower (unusual noise, vibration)

**Quarterly:** - Replace intake filters (control cabinet) - Inspect blower impeller (dust buildup)

**Annually:** - Replace filter cartridges (even if pressure OK, efficiency degrades) - Blower maintenance (bearings, belts)

**Cost:** \$150-900/year (filters), \$200-500 (blower maintenance)

## **Preventive Maintenance Schedule**

### **Daily Tasks (Operator)**

- ☐ Vacuum machine surfaces (HEPA vac)
- ☐ Empty dust collector hopper
- ☐ Wipe control panel (damp cloth)
- ☐ Check coolant level
- ☐ Inspect tools (visual, magnifier)

**Time:** 10-15 minutes

### **Weekly Tasks (Operator/Technician)**

- ☐ Clean machine thoroughly (remove guards, vacuum inside)
- ☐ Check coolant concentration, pH
- ☐ Inspect way covers (tears)
- ☐ Pulse-clean dust collector filters
- ☐ Check dust collection pressure drop
- ☐ Review tool wear logs (trending)

**Time:** 30-60 minutes

### **Monthly Tasks (Technician)**

- ☐ Deep clean coolant tank (remove sludge)
- ☐ Inspect ballscrews (rotate, check for rough spots)
- ☐ Grease ballscrew bearings (if required)
- ☐ Inspect spindle seals (look inside housing for dust)

- ☐ Check machine level (spirit level on table, ways)
- ☐ Test interlocks (door switches, E-stop)

**Time:** 2-4 hours

### **Quarterly Tasks (Technician)**

- ☐ Replace control cabinet intake filters
- ☐ Inspect/clean dust collection blower impeller
- ☐ Replace coolant entirely (drain, clean tank, refill)
- ☐ Inspect/replace way wipers
- ☐ Lubricate machine (detailed per manual: hinges, latches, etc.)

**Time:** 4-8 hours

### **Annual Tasks (Specialist)**

- ☐ Replace dust collector filter cartridges
- ☐ Spindle seal replacement
- ☐ Spindle bearing inspection (runout, play, noise)
- ☐ Ballscrew backlash measurement
- ☐ Laser alignment check (if available)
- ☐ Electrical connections inspection (tighten, check for corrosion)

**Time:** 8-16 hours

**Cost** (parts + labor): \$2,000-5,000 per machine

## **Tool Storage and Organization**

### **Storage Conditions**

**Temperature:** Room temperature (60-80 degF) - Extreme temperatures: Epoxy bonding (PCD) can fail

**Humidity:** Low (<50% RH) - High humidity: Carbide corrosion (cobalt binder rusts) - Store in desiccant cabinet (silica gel packs)

**Protection:** - Individual tool boxes (plastic tubes) - Foam inserts (prevent tools contacting each other) - Never: Loose in drawer (edges chip)

### **Organization System**

**Shadow Board:** - Outline of each tool drawn on board - Tool hangs in specific spot - Missing tool immediately visible - Good for frequently used tools

**Tool Crib:** - Locked cabinet - Tools checked out (logged) - Enforces inventory tracking - Reduces loss/theft

**Color Coding** (material-specific tools): - Blue labels: Aluminum tools (don't use on composites) - Red labels: Composite tools - Yellow labels: Ceramic tools (diamond) - Prevents cross-contamination (aluminum chips on composite = corrosion)

## **Cost-Benefit Analysis**

### **Preventive Maintenance ROI**

**Cost of Prevention:** - Labor: 50 hours/year @ \$40/hour = \$2,000 - Parts: Filters, seals, coolant = \$1,000 - **Total:** \$3,000/year

**Cost of Reactive Maintenance** (failure-based): - Spindle failure: \$10,000 (rebuild) + \$5,000 (downtime) - Ballscrew replacement: \$5,000 + \$2,000 (downtime) - Probability without PM: 10-20% per year - **Expected cost:** \$15,000 x 15% = \$2,250/year

**Breakeven:** But doesn't include: - Lost production (downtime for emergency repairs) - Scrap (worn tools produce bad parts before caught) - Reputation damage (late deliveries)

**Realistic ROI:** 3:1 to 5:1 (every \$1 spent on PM saves \$3-5 in reactive costs)

### Tool Management ROI

**Without Tool Management:** - Replace tools when they break or quality issues occur - 5-10% scrap from worn tools (not caught in time) - Downtime: 30 min per tool failure (find replacement, load, restart)

**With Tool Management:** - Replace tools at 70-80% of life (before failure) - Scrap: 1-2% (proactive replacement) - Downtime: Planned (during shift breaks)

**Example** (small shop, \$200k/year revenue): - Scrap reduction: 5% -> 1.5% (saves 3.5%) - \$200k x 3.5% = **\$7,000/year saved** - Tool management cost: 2 hours/week x \$30/hour x 50 weeks = \$3,000/year - **Net benefit:** \$4,000/year

**Payback:** Immediate (first year)

### Summary

Effective tool and machine maintenance is critical for advanced materials machining:

**Tool Wear:** - Composites: Abrasive fibers wear tools rapidly (50-2000 parts typical) - Ceramics: Grinding wheels last long but require dressing (G-ratio 100-10000) - Monitor: Visual inspection (wear land), dimensional trending, force monitoring

**Tool Management:** - Inventory: Track usage, maintain par levels (avoid stockouts) - Cost tracking: Tool cost per part (diamond often cheaper than carbide long-term) - Resharpening: Carbide tools 2-5 regrinds (\$20-30 per regrind)

**Machine Maintenance:** - Spindle: Air purge system (prevents dust ingestion), annual seal replacement - Ways/ballscrews: Covers, wipers, frequent lubrication - Coolant: Clean weekly, replace quarterly (bacteria growth) - Dust collection: Daily hopper emptying, annual filter replacement

**Preventive Maintenance:** - Daily (10-15 min): Vacuum, empty dust collector, inspect tools - Monthly (2-4 hours): Deep clean, inspect seals, test interlocks - Annual (8-16 hours): Replace filters, seals, bearings - Cost: \$3,000/year (saves \$10,000+ in reactive repairs)

**ROI:** - Preventive maintenance: 3:1 to 5:1 return - Tool management: \$4,000/year savings (small shop example) - Avoiding spindle failure alone justifies program

**Next:** Troubleshooting common problems in advanced materials machining

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**Next:** 17.11 Troubleshooting Common Issues

## 17.11 Troubleshooting Common Issues in Advanced Materials Machining

### Composite Machining Problems

#### Delamination

**Symptom:** Plies separate at edges, visible gaps between layers

**Entry Delamination** (top surface):

```
      down Tool entry
===== <- Top ply lifts
-----
----- Intact plies
-----
=====
```

**Exit Delamination** (bottom surface):

```
=====
-----
----- Intact plies
-----
===== <- Bottom ply peels away
      down Tool exit (unsupported)
```

**Causes:**

| Cause                     | Why It Happens                                  | Evidence                               |
|---------------------------|-------------------------------------------------|----------------------------------------|
| Dull tool                 | High forces peel plies apart                    | Delamination worsens over run          |
| Feed too high             | Excessive force per revolution                  | Consistent delamination, even new tool |
| No backing support        | Exit ply unsupported (bends, tears)             | Exit side only, entry side clean       |
| Wrong tool (conventional) | Upcut pushes bottom ply down, downcut lifts top | Entry or exit specific                 |
| Insufficient clamping     | Part vibrates (peeling action)                  | Random locations                       |

**Solutions:**

**Immediate:** 1. **Reduce feed rate:** Cut feed by 30-50% - Example: 100 IPM -> 50 IPM 2. **Replace tool:** If dull (check wear land) 3. **Add backing:** Sacrificial MDF, phenolic, or same material underneath - Clamp sandwich (backing + part) together 4. **Use compression cutter:** Upcut flutes at tip (push bottom ply up), downcut at top (push top ply down) - Both surfaces compressed into part

**Long-term:** - Establish tool life (replace proactively) - Use compression or diamond-coated tools - Vacuum table (distributed clamping, no point loads)

**Repairing Delamination:** - Minor (<0.030"): Inject thin epoxy (syringe), clamp, cure, sand flush - Major (>0.030"): Part likely scrap (structural weakness)

#### Fuzzing

**Symptom:** Loose fibers standing up from surface (hairy appearance)

**Causes:** - Dull tool (fibers torn, not cut cleanly) - Cutting against fiber direction (fibers pulled) - Wrong tool geometry (too aggressive) - Aramid fibers (Kevlar): Naturally fuzzy (fiber is tough, hard to cut)

### Solutions:

**Prevention:** 1. **Sharp tools:** Replace when fuzzing starts 2. **Cut with fiber direction:** If unidirectional, feed along fibers 3. **Higher speed, lower feed:** Less force per fiber - Example: 18,000 RPM, 50 IPM (vs 12,000 RPM, 100 IPM) 4. **Compression cutter:** Shears fibers cleanly

**Cleanup** (if fuzzing occurs): 1. Light sanding (220-320 grit) removes loose fibers 2. Flame polish (very brief propane torch pass): Melts fuzz (dangerous, practice required) 3. Sharp blade scraping (carefully)

**Aramid-Specific:** - Extremely fuzzy by nature (tough fiber resists cutting) - Scissors-type cutters work better than mills - Ultrasonic cutting (high-frequency vibration) reduces fuzzing

### Burning/Heat Damage

**Symptom:** Darkened, charred surface; melted resin; smoking during cut

**Causes:** - Dull tool (friction generates heat) - Speed too high (heat from rubbing) - Feed too low (tool dwells in cut, heat accumulates) - Insufficient chip evacuation (chips recutting = heat)

### Solutions:

**Immediate:** 1. **Reduce spindle speed:** Cut RPM by 20-30% - Example: 24,000 RPM -> 18,000 RPM 2. **Increase feed rate:** Reduce dwell time - Example: 50 IPM -> 80 IPM 3. **Replace tool:** If dull 4. **Air blast:** Compressed air (coolant) at cut zone

**Long-term:** - Optimize feeds/speeds (high feed, moderate speed) - Sharp tools (proactive replacement) - Chip evacuation (vacuum near cutter, air blast)

**Temperature Limits:** - Epoxy: 250-350 degF (starts softening) - Phenolic: 300-400 degF - Visual smoke = >400 degF (resin burning)

**Damage Assessment:** - Light discoloration: Cosmetic (usually acceptable) - Heavy charring: Resin degraded (structural damage, reject) - Matrix Analysis: DSC (differential scanning calorimetry) detects thermal damage

### Fiber Pullout

**Symptom:** Holes where fibers torn out (not cut cleanly)

**Causes:** - Dull tool - Wrong fiber orientation (cutting perpendicular to fibers) - Tool geometry (negative rake pulls instead of shears)

**Solutions:** 1. **Sharp tool:** Positive rake geometry preferred 2. **Reduce feed:** Less force per fiber 3. **Higher speed:** Shearing action (not pulling) 4. **Climb milling:** Cutting fibers at entry (not exit)

**Repair** (if minor): - Fill with epoxy (color-matched), sand flush - Major pullout: Structural concern, possibly reject

### Uneven Edge Quality

**Symptom:** Some edges clean, others ragged (same part)

**Causes:** - Fiber orientation variation (woven fabrics have 0 deg and 90 deg fibers) - Uneven clamping (part vibrates in some areas) - Tool runout (cutting edge wobbles)

**Solutions:** 1. **Check fiber orientation:** Adjust feed direction if possible 2. **Improve clamping:** Add clamps, use vacuum table 3. **Check tool runout:** Collet, holder, spindle taper - Acceptable: <0.0005" TIR (total indicator reading) - Poor: >0.001" TIR -> replace collet/holder

**Runout Measurement:** - Dial indicator on tool shank (near cutting edge) - Rotate spindle by hand, observe reading - Causes: Dirty taper, worn collet, bent tool

## Ceramic Machining Problems

### Edge Chipping

**Symptom:** Small chips (0.001-0.020") missing from edges, corners

**Causes:**

| Cause                 | Mechanism                                    | Evidence                          |
|-----------------------|----------------------------------------------|-----------------------------------|
| Depth of cut too high | Excessive force (brittle fracture)           | Chipping size correlates with DOC |
| Wheel too coarse      | Large grains create large chips              | Chipping size matches grit size   |
| Exit edge unsupported | Edge breaks off (no backing)                 | Exit side only                    |
| Grinding burn         | Localized heating -> thermal stress -> crack | Discoloration near chip           |

**Solutions:**

**Immediate:** 1. **Reduce depth of cut:** Cut DOC by 50% - Example: 0.002" -> 0.001" per pass 2. **Finer grit wheel:** - If using 120 grit, switch to 220-320 grit 3. **Add support:** Magnetic chuck, adhesive backing 4. **Spark-out passes:** 2-5 passes at zero depth (wheel/part spring back)

**Long-term:** - Dress wheel regularly (sharp particles) - Optimize parameters (lower DOC, finer grit for finishing) - Post-grinding chamfer (0.002-0.005" x 45 deg): Removes fragile edge

**Repair** (if necessary): - Light chipping (<0.005"): May be acceptable (check drawing) - Diamond honing: Remove sharp edges of chip (cosmetic improvement) - Large chips: Part likely scrap (stress concentrator)

### Cracking

**Symptom:** Visible cracks (usually radial from holes, corners)

**Causes:** - Thermal shock (rapid heating/cooling) - Coolant application: Intermittent = cycling = cracks - Grinding burn: Localized heating - Mechanical stress (excessive force) - Pre-existing flaws (propagate under stress)

**Thermal Shock Mechanism:** 1. Grinding heats surface locally 2. Surface expands (thermal expansion) 3. Cooler interior restrains expansion (stress) 4. Rapid cooling (coolant splash): Surface contracts 5. Tensile stress exceeds fracture strength -> crack

**Solutions:**

**Thermal Management:** 1. **Flood coolant:** Continuous, not intermittent - Flow rate: 5-10 GPM minimum 2. **Reduce heat generation:** - Lower wheel speed (reduce friction) - Lighter DOC (less energy input) - Sharp wheel (less rubbing) 3. **Slow down:** Lower table speed (longer coolant contact, better cooling)

**Mechanical Stress Reduction:** 1. **Support part:** Avoid point clamping (stress concentrations) 2. **Reduce forces:** Lighter cuts, finer wheel

**Prevention:** - Stress relief anneal (after grinding): Heat to 50-70% sintering temp, slow cool - Removes residual stresses - Example (alumina): 1000-1200 degC, 2 hours, slow cool - Quality raw material (low defect content)

**Inspection:** - Dye penetrant (surface cracks) - Ultrasonic (internal cracks)

### Grinding Burn

**Symptom:** Discoloration (yellow, brown, blue on white ceramics); surface damage

**Causes:** - Excessive heat from grinding (friction) - Dull wheel (rubbing, not cutting) - Insufficient coolant - Too high wheel speed or DOC

**Damage:** - Localized phase transformation (material structure changes) - Residual tensile stress (weakens part) - Microcracks (subsurface)

**Detection:** - Visual: Discoloration - Nital etch (metallographic): Burned layer appears different - Hardness test: Burned zone often softer (over-tempered) or harder (re-hardened)

**Solutions:**

**Immediate:** 1. **Dress wheel:** Expose sharp particles 2. **Increase coolant flow:** 10+ GPM 3. **Reduce DOC:** Cut by 50% (0.001" -> 0.0005") 4. **Lower wheel speed:** Reduce by 10-20%

**Prevention:** - Sharp wheel (frequent dressing) - Flood coolant (proper application, aimed at cutting zone) - Conservative parameters (slower = cooler)

**Repair:** - Grind off burned layer (0.002-0.010" deep typically) - Re-grind part to spec (if enough stock remains) - Otherwise: Scrap

### Wheel Loading

**Symptom:** Grinding wheel clogs (pores filled with debris), grinding slows, poor finish

**Causes:** - Soft material (relatively): Some ceramics smear into wheel - Inadequate coolant (chips not flushed) - Wheel too hard (bond doesn't release dulled particles) - Wheel too fine (pores too small, fill quickly)

**Evidence:** - Increasing grinding forces - Surface finish degrades - Wheel face appears clogged (gray, glazed)

**Solutions:**

**Immediate:** 1. **Dress wheel:** Removes loaded material 2. **Increase coolant flow:** Better chip flushing 3. **Dressing stick:** Aluminum oxide stick (cleans wheel during grinding) - Hold stick against rotating wheel (breaks up loading)

**Long-term:** 1. **Softer wheel grade:** Self-dressing (particles release sooner) - Example: N grade -> M grade 2. **Coarser grit:** Larger pores (resist loading) - Example: 320 grit -> 220 grit 3. **Lower concentration:** Fewer diamonds = more pore space - Example: 150 -> 100 concentration

### Poor Surface Finish

**Symptom:** Scratches, roughness above specification

**Causes:**

| Symptom                | Cause                           | Solution                         |
|------------------------|---------------------------------|----------------------------------|
| Coarse, even scratches | Grit too coarse                 | Finer grit wheel                 |
| Random deep scratches  | Contamination (chip re-cutting) | Better coolant flow, dressing    |
| Chatter marks (wavy)   | Vibration                       | Reduce DOC, check setup rigidity |
| Dull, smeared surface  | Wheel loaded                    | Dress wheel                      |

**Solutions:**

**Grit Selection:** - Current: 120 grit (Ra 100 muin) - Target: Ra 32 muin - Solution: 320 grit wheel (Ra 20-40 muin capable)

**Contamination Control:** - Flood coolant (flush chips) - Clean coolant (filter, change regularly) - Dress wheel (remove embedded chips)

**Vibration Reduction:** - Check workholding (loose part?) - Machine rigidity (level machine, tighten gibs) - Wheel balance (out-of-balance = vibration) - Reduce DOC (less force = less vibration)

**Final Finish Improvement:** - Spark-out passes (2-5x at zero depth) - Lapping (if ultra-smooth required)  
- Polishing (colloidal silica)

## Dust Collection Problems

### Insufficient Capture

**Symptom:** Visible dust in air, accumulation on surfaces

**Causes:** - Insufficient airflow (CFM too low) - Poor hood design (dust escapes capture zone) - Leaks in ductwork (lose velocity) - Blocked filter (high resistance)

**Diagnosis:**

**Airflow Test:** - Anemometer: Measure velocity at hood face - Calculate CFM:  $\text{Area (ft}^2\text{)} \times \text{Velocity (FPM)}$  - Compare to design CFM

**Example:** - Hood: 6" x 6" ( $0.25 \text{ ft}^2$ ) - Measured velocity: 120 FPM - Actual CFM:  $0.25 \times 120 = 30 \text{ CFM}$   
- Design CFM: 200 CFM - **Conclusion:** 85% loss (blockage or leak)

**Solutions:**

1. **Check filter:** Replace if loaded (pressure drop  $>6'' \text{ WC}$ )
2. **Inspect ductwork:** Look for disconnections, holes
3. **Seal leaks:** Aluminum tape, gaskets
4. **Increase blower:** Larger motor or faster speed (if possible)
5. **Reduce duct length:** Shorter run = less resistance

**Hood Improvement:** - Closer to source (capture zone =  $\sim 1\text{x}$  hood diameter) - Larger hood (more area, lower velocity needed) - Flanges around hood (improve capture efficiency 25%)

### Filter Blinding

**Symptom:** Pressure drop very high ( $>8'' \text{ WC}$ ), airflow drops rapidly

**Causes:** - Fine dust (plugs pores) - Moist dust (sticks to filter) - Pulse cleaning inadequate (dust cake not removed)

**Solutions:**

**Immediate:** 1. **Manual cleaning:** Shake filters (if safe to access) 2. **Compressed air:** Blow filters clean (outside, wear respirator)

**Long-term:** 1. **More filter area:** Add cartridges (lower velocity, longer life) 2. **Pre-filter:** Cyclone separator (removes coarse dust before filter) - Captures 80-90% of dust by weight (centrifugal force) - Filter sees only fine dust (longer life) 3. **Better pulse system:** Higher pressure (60-90 PSI), more frequent pulses 4. **Nanofiber filters:** Surface-loading (dust cake on surface, not in pores) - Easier to pulse-clean - Cost: 2-3x standard filters

### Dust Escaping System

**Symptom:** Clean air side has visible dust (filter failure)

**Causes:** - Filter torn (hole) - Poor seal (gasket leaking) - Filter saturated (dust passes through)

**Diagnosis:** - Visual: White cloth on exhaust, run system 1 min, check cloth (dust?) - Light test: Shine flashlight inside collector (look for light leaks = dust leaks)

**Solutions:** 1. **Inspect filters:** Replace torn filters immediately 2. **Check seals:** Gaskets between filter and housing 3. **Replace filters:** Even if not torn (efficiency degrades over time)

**Safety:** - Dust escaping = exposure - Investigate immediately (respiratory hazard)



## Equipment Malfunctions

### Spindle Issues

**Symptom:** Vibration, noise, overheating, loss of power

**Diagnosis:**

| Symptom                 | Probable Cause                       | Test                         | Solution                     |
|-------------------------|--------------------------------------|------------------------------|------------------------------|
| Vibration (new)         | Tool imbalance, runout               | Check runout (indicator)     | Balance tool, replace collet |
| Vibration (gradual)     | Bearing wear                         | Temperature (IR gun), noise  | Replace bearings             |
| Noise (grinding)        | Dust in bearings                     | Spin by hand (rough?)        | Rebuild spindle              |
| Overheating (>140 degF) | Seal failure, lubrication loss       | Thermal camera               | Replace seals, re-lubricate  |
| Loss of power           | Winding failure (carbon fiber short) | Resistance test (multimeter) | Rewind or replace            |

**Runout Measurement:** - Indicator on tool (near tip) - Rotate spindle by hand - TIR >0.001" = excessive (check collet, holder, taper)

**Bearing Temperature:** - IR gun (non-contact thermometer) - Normal: 100-130 degF - Warning: 130-150 degF - Shutdown: >150 degF (risk of seizure)

**Prevention:** - Dust control (air purge) - Regular maintenance (seal replacement) - Avoid crashes (bearing damage)

### Coolant System Problems

**Symptom:** Coolant not flowing, low pressure, foaming, smell

**Diagnosis:**

**No Flow:** - Check pump (running?) - Check filter (clogged?) - Check lines (kinked, blocked?)

**Low Pressure:** - Measure at nozzle (pressure gauge) - Normal: 50-200 PSI (depending on system) - Low: Clogged nozzle, weak pump, leak

**Foaming:** - Cause: Concentration too high, contamination (soap, oil) - Test: Measure concentration (refractometer) - Normal: 5-10% - High: >12% (foaming) - Solution: Dilute (add water), skim foam

**Smell** (rotten, sulfur): - Cause: Bacteria growth (anaerobic) - Solution: Add biocide, change coolant, clean system - Prevention: Maintain pH 8.5-9.5 (inhibits bacteria)

### Machine Accuracy Loss

**Symptom:** Parts out of tolerance, drift over time

**Causes:** - Thermal drift (machine warming/cooling) - Ballscrew wear (backlash) - Scale contamination (dust on encoders) - Foundation issues (machine not level)

**Diagnosis:**

**Thermal Drift:** - Measure first part (cold machine) vs 10th part (warm) - If dimensions drift consistently -> thermal - Solution: Warm-up cycle (run machine 15-30 min before production)

**Backlash Test:** - Dial indicator on table - Command +0.100" move, then -0.100" - Measure actual movement in each direction - Backlash = difference (should be <0.0005") - High backlash -> ballscrew/nut wear

**Scale Contamination:** - Clean linear encoders (compressed air, lint-free cloth) - Fine dust causes position errors

**Leveling:** - Precision level (0.0005"/ft resolution) - Check machine bed (should be within 0.001"/ft) - Shim feet if needed

## Process Optimization

### Long Cycle Times

**Symptom:** Machining too slow, low productivity

#### Optimization Strategies:

**Increase Feed Rate:** - Current: Conservative (safe but slow) - Test: Increase feed by 20% (monitor quality) - If quality OK: Continue increasing until quality degrades or forces excessive - Example: CFRP routing 60 IPM -> test 72 IPM -> if OK, test 86 IPM

**Increase Depth of Cut:** - Composites: Can often double DOC if feed reduced proportionally - Example: 0.050" DOC @ 100 IPM -> 0.100" DOC @ 70 IPM - Material removal rate increased 40%

**Optimize Tool Path:** - Reduce rapid moves (non-cutting time) - Use trochoidal milling (arcs instead of linear plunge) - Minimize tool changes (group similar operations)

**Better Tools:** - Diamond-coated: 10x tool life (fewer changes) - Compression cutters: Clean entry/exit (less cleanup)

**Measurement:** - Cycle time (current): 15 minutes - Improvement: 20% faster feed = 12 minutes - Savings: 3 min per part x 100 parts/day = 300 min/day (5 hours)

### Excessive Tool Wear

**Symptom:** Tools wearing faster than expected, high tool cost per part

#### Diagnosis:

**Compare to Baseline:** - Expected tool life: 500 parts - Actual: 150 parts - **3x faster wear** -> find cause

#### Possible Causes:

1. **Material harder than expected:**
  - Test hardness (different batch?)
  - Check material cert (correct material?)
2. **Feed/speed incorrect:**
  - Too high speed: Excessive heat (oxidation wear)
  - Too low feed: Rubbing (abrasive wear)
  - Optimize: Manufacturer recommendations
3. **Coolant issues:**
  - Concentration low: Poor lubrication
  - Contamination: Chips re-cutting
4. **Runout:**
  - High runout: One flute does all work (overloads)
  - Uneven wear on flutes
  - Check: TIR <0.0005"

**Solutions:** - Optimize parameters (feeds/speeds database) - Maintain coolant (concentration, cleanliness) - Check setup (runout, clamping, vibration) - Better tools (coating, geometry, material)

### Poor Part Quality

#### Systematic Approach:

1. **Define Problem:**
  - What feature(s) out of spec?
  - How much (measurement)?
  - When did it start (sudden or gradual)?
2. **Check Setup:**
  - Tool: Correct tool, sharp, properly seated?
  - Work offset: Confirmed (probe or edge finder)?
  - Clamping: Secure, not deforming part?
3. **Check Machine:**
  - Accuracy: Run test part or calibration routine
  - Maintenance: Lubrication, covers, seals OK?
4. **Check Process:**
  - Parameters: Feed/speed appropriate?
  - Tool path: Optimized (no chatter, good lead-in/out)?
  - Coolant: Flowing, correct concentration?
5. **Check Material:**
  - Correct material (verify cert)?
  - Consistent (same batch as previous good parts)?
6. **Check Environment:**
  - Temperature stable (thermal growth)?
  - Vibration (nearby equipment)?

**Document Findings:** - Keep log (problem -> investigation -> solution) - Build tribal knowledge (next time faster)

## Emergency Response

### Fire

**Types:** - **Class A:** Ordinary combustibles (wood, paper) - Extinguisher: Water or ABC dry chemical - **Class B:** Flammable liquids (resin, solvent) - Extinguisher: ABC dry chemical, CO<sub>2</sub>, foam - **Class C:** Electrical - Extinguisher: ABC dry chemical, CO<sub>2</sub> (non-conductive)

**Small Fire** (<3 ft): 1. Evaluate: Safe to fight? 2. Get extinguisher 3. Pull pin, aim at base, squeeze, sweep 4. If grows: Evacuate, call 911

**Large Fire:** 1. Evacuate immediately (activate fire alarm) 2. Close doors (contain) 3. Call 911 (from safe location) 4. Do not re-enter

### Dust Explosion

**Indicators** (rarely warning): - Flash, fireball, pressure wave - Usually sudden (no time to react)

**After Explosion:** 1. Evacuate (secondary explosion possible) 2. Account for personnel 3. Call 911 (injuries, fire) 4. Do not re-enter (structural damage, fire risk)

**Prevention** (critical): - Housekeeping (no dust accumulation) - Dust collection (remove fuel) - No ignition sources (spark-proof tools, grounding)

### Major Coolant Spill

**Hazards:** Slip hazard, contamination

**Response:** 1. Stop source (shut off pump) 2. Contain (absorbent booms) 3. Absorb (pads, absorbent, kitty litter) 4. Dispose (per regulations, likely hazardous waste) 5. Clean/decontaminate area

**Prevent:** - Berm around machines (contain spills) - Regular inspection (hoses, connections)

## Summary

Troubleshooting advanced materials machining requires systematic diagnosis:

**Composites:** - Delamination: Dull tool, high feed, no backing -> compression cutter, backing support - Fuzzing: Dull tool, wrong feed direction -> sharp tools, cut with fibers - Burning: Excessive heat -> reduce speed, increase feed, sharp tools

**Ceramics:** - Edge chipping: Excessive force, coarse wheel -> lighter DOC, finer grit, spark-out - Cracking: Thermal shock -> flood coolant, reduce heat generation - Grinding burn: Dull wheel, inadequate coolant -> dress wheel, increase coolant flow

**Dust Collection:** - Insufficient capture: Low CFM, poor hood -> check filters, seal leaks, improve hood - Filter blinding: Fine dust -> pre-separator (cyclone), more filter area

**Equipment:** - Spindle issues: Vibration, noise -> check runout, bearings, dust ingress - Coolant problems: No flow, foaming -> check pump, filter, concentration

**Optimization:** - Long cycle times: Conservative parameters -> increase feed/DOC incrementally - Excessive tool wear: Incorrect parameters -> optimize feeds/speeds, check setup

**Emergency Response:** - Fire: ABC extinguisher for small fires, evacuate for large - Dust explosion: Evacuate immediately, prevention critical (housekeeping)

**Systematic Approach:** 1. Define problem (measurement, observation) 2. Check setup (tool, offsets, clamping) 3. Check machine (accuracy, maintenance) 4. Check process (parameters, tool path) 5. Document (build knowledge base)

**Next:** Conclusion and future trends in advanced materials machining

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**Next:** 17.12 Conclusion and Future Trends

## 17.12 Conclusion and Future Trends in Advanced Materials Machining

### Key Takeaways from Module 17

#### Material Characteristics Drive Process Selection

**Composites:** - Fiber-reinforced structures combine high strength with low weight - Anisotropic properties: Strength depends on fiber orientation - Machining challenges: Delamination, fiber pullout, abrasive wear - Tool selection critical: Diamond-coated or PCD for production - Dust control mandatory: Carbon fiber possibly carcinogenic (IARC 2B)

**Ceramics:** - Extreme hardness with brittleness: High compressive strength, low tensile - Subsurface damage from grinding reduces strength 20-50% - Green machining (before firing) economical: 10-100x faster than fired grinding - Diamond grinding dominant: Slow material removal (0.0001-0.001" per pass) - Silica-containing ceramics: OSHA regulated (PEL 0.05 mg/m<sup>3</sup>, medical surveillance)

#### Safety Is Non-Negotiable

**Health Hazards:** - Respiratory: Carbon fiber (carcinogen concern), silica (silicosis), glass fiber (irritant) - Exposure limits extremely low: Engineering controls (dust collection) mandatory - Personal protective equipment: P100 respirator or PAPR required for production work - Long-term consequences: Silicosis incurable, carbon fiber effects unknown (treat conservatively)

**Equipment Protection:** - Fine dust destroys spindles: Air purge systems essential - Abrasive contamination: Ceramic dust accelerates wear on ways, ballscrews - Cost of poor dust control: \$10,000-50,000 equipment damage + health claims

**Regulatory Compliance:** - OSHA Silica Rule (29 CFR 1910.1053): Medical surveillance, 30-year record-keeping - Respiratory Protection Program: Fit testing, medical evaluation, training - Combustible dust: Housekeeping prevents explosions

**Investment in Safety:** - Dust collection system: \$3,000-9,000 (small shop) - Operating cost: \$2,800-4,000/year - ROI: Cheap compared to \$500,000+ health claims, \$15,000+ OSHA fines

#### Process Optimization Balances Quality, Speed, Cost

**Tool Life Management:** - Composites: 50-2000 parts per tool (abrasive fiber wear) - Ceramics: G-ratio 100-10,000 (diamond grinding wheels last months) - Proactive replacement: 70-80% of life (prevents quality issues, scrap) - Cost analysis: Diamond tools often cheaper per part despite higher initial cost

**Surface Finish Affects Performance:** - Ceramics: Polished surface 3x stronger than as-ground (removes subsurface cracks) - Composites: Sealed edges prevent moisture ingress (fatigue life) - Finishing cost: \$100-150 per part (labor + materials) justified by functional requirements

**Maintenance Prevents Failures:** - Preventive maintenance: \$3,000/year (small shop) - Reactive maintenance: \$10,000+ per failure (spindle rebuild + downtime) - ROI: 3:1 to 5:1 return on preventive maintenance investment

### Emerging Technologies

#### Additive Manufacturing Integration

**Hybrid Manufacturing** (additive + subtractive):

**Concept:** - 3D print near-net shape - CNC machine critical features (precision) - Combines speed of printing with accuracy of machining

**Example Applications:**

**Ceramic Parts:** - Binder jet 3D printing: Print ceramic powder + binder in complex shape - Sinter: Fire part (binder burns out, ceramic densifies) - CNC grind: Precision surfaces (bearing bores, seal faces) to tight tolerance - **Advantage:** Complex internal geometries (impossible to machine) + precision external features

**Composite Parts:** - Automated fiber placement (AFP): Robot lays composite tape in programmed pattern - Cure: Autoclave or oven - CNC trim: Cut to final dimensions, drill holes - **Advantage:** Complex shapes (double curvature), optimized fiber orientation, precision edges

**Market Growth:** - Hybrid machines: \$200,000-2,000,000 (DMG MORI, Mazak, Okuma) - Adoption: Aerospace (Boeing, Airbus), medical (implants), tooling - Forecast: 15-20% annual growth (2024-2030)

## Advanced Tooling Materials

**PCD Evolution:** - Current: Polycrystalline diamond (PCD) brazed to carbide substrate - Emerging: Thick-film PCD (entire tool body diamond) - Advantages: No braze joint (failure point), longer life (more regrind cycles) - Cost: 2-3x conventional PCD - Applications: High-volume production (automotive carbon fiber parts)

**CVD Diamond Coatings:** - Chemical Vapor Deposition: Diamond grown on carbide substrate (atom by atom) - Thickness: 10-30  $\mu\text{m}$  (thicker than PVD coatings) - Adhesion: Excellent (chemical bond, not mechanical) - Tool life: 10-50x uncoated carbide - Cost: Decreasing (broader adoption as CVD equipment improves)

**Nanocrystalline Diamond:** - Grain size: <100 nm (vs 2-30  $\mu\text{m}$  for PCD) - Advantages: Sharper edge (finer grains), better surface finish - Applications: Ultra-precision machining (optics, medical devices)

## Automation and Industry 4.0

### In-Process Monitoring:

**Force Monitoring:** - Real-time cutting force measurement (dynamometer, spindle current) - Detects tool wear (forces increase), breakage (forces drop suddenly) - Automatic tool replacement triggered - Reduces scrap (worn tool caught before quality degrades)

**Acoustic Emission:** - Sensors detect ultrasonic sound from cutting (material fracture, tool wear) - Machine learning: Pattern recognition (normal vs abnormal) - Early warning: Tool wear, chatter, delamination onset

**Vision Systems:** - Cameras inspect tool before/after machining - Edge detection: Measure wear land, chipping - Automatic decision: Continue, dress (grinding wheel), or replace

**Implementation:** - High-value production (aerospace): ROI justified - Small shops: Simple load monitoring (already on many CNCs)

### Digital Twin:

**Concept:** Virtual model of physical process - Physics-based simulation: Predict forces, temperatures, tool wear - Updated with real data: Sensors feed actual conditions - Optimization: Software suggests parameter improvements - Predictive maintenance: Forecast failures before they occur

**Example:** - Input: Part geometry, material (CFRP properties), tool (PCD endmill), parameters - Simulation: Predicts cutting forces, temperatures, tool wear rate - Output: Optimal feed/speed, estimated tool life (1500 parts +/-100) - Monitor: Real machine data compared to prediction (if deviation -> investigate)

**Status:** Research -> early commercial adoption (Siemens, GE, Autodesk)

## Sustainable Manufacturing

### Waste Reduction:

**Composite Recycling:** - Traditional: Thermoset composites (epoxy) not recyclable (landfill or incinerate) - Emerging: Thermoplastic composites (PEEK, PEKK) melt, reform (recyclable) - Challenge: Higher processing temps (>650 degF), more expensive resin - Adoption: Automotive (BMW, Toyota experimenting)

**Ceramic Powder Reclamation:** - Grinding swarf (dust + coolant): Traditionally waste - Reclamation: Filter, dry, re-sinter into blanks (not critical parts) - Savings: \$5-20 per pound powder (depends on material)

#### **Energy Efficiency:**

**Cryogenic Machining:** - Liquid nitrogen coolant (vs water-based) - Benefits: Superior cooling (tool life 2-5x), no bacteria, no disposal (evaporates) - Energy: Nitrogen production energy-intensive (net energy higher) - Economics: Tool savings offset nitrogen cost (high-volume production)

**Dry Machining:** - Composites: Often machined dry (minimal heat compared to metals) - Dust collection: More critical (no coolant to suppress dust) - Energy savings: No coolant pump, chiller, disposal

**Renewable Diamond Tools:** - Lab-grown diamond (vs mined): Same properties, lower environmental impact - Cost parity achieved (2020s): Market shifting to synthetic

#### **Material Innovations**

##### **Next-Generation Composites:**

**Natural Fiber Composites:** - Fibers: Flax, hemp, bamboo (renewable, biodegradable) - Matrix: Bio-resins (plant-based epoxies) - Properties: Lower strength than carbon fiber, but sufficient for many applications - Machining: Less abrasive (easier on tools), less health concern - Markets: Automotive interiors, consumer goods, packaging

**Self-Healing Composites:** - Microcapsules of resin embedded in matrix - Damage (microcrack) -> capsules break -> resin flows, cures (fills crack) - Status: Laboratory -> early testing (aerospace) - Machining impact: Unknown (microcapsules may complicate cutting)

**Nanocomposites:** - Carbon nanotubes, graphene dispersed in polymer - Tiny amounts (0.1-1%) improve properties: Strength, conductivity, thermal - Machining: Similar to conventional composites (but very expensive, limited production)

##### **Advanced Ceramics:**

**Ultra-High Temperature Ceramics (UHTCs):** - Materials: Hafnium carbide, tantalum carbide (melting point >6000 degF) - Applications: Hypersonic vehicles (Mach 5+), rocket nozzles - Machining: Extremely difficult (hardness near diamond) - Diamond grinding only practical method - Laser machining (thermal damage concerns)

**Transparent Ceramics:** - Aluminum oxynitride (ALON): Transparent, scratch-resistant, strong - Applications: Armor (bulletproof windows), optics, semiconductors - Machining: Similar to sapphire (very hard, brittle) - Requires ultra-precision grinding, polishing (optical quality)

**MAX Phase Ceramics:** - Ternary carbides/nitrides: Machinable like metals, heat-resistant like ceramics - Example: Ti<sub>3</sub>SiC<sub>2</sub> (titanium silicon carbide) - Machining: Carbide tools work! (unlike most ceramics) - Status: Research -> niche applications (coatings, electrical contacts)

#### **Artificial Intelligence and Machine Learning**

##### **Process Optimization:**

**Adaptive Control:** - AI monitors sensor data (forces, vibration, temperature) - Detects patterns: Tool wear signature, chatter onset - Adjusts parameters in real-time: Reduce feed if forces spike - Result: Optimal process without operator intervention

**Parameter Databases:** - Machine learning trains on thousands of cuts - Input: Material, tool, desired features - Output: Optimal feed, speed, DOC (based on similar historical jobs) - Continuous improvement: System learns from every part

**Predictive Maintenance:** - AI analyzes machine data: Vibration, temperature, power consumption - Detects anomalies: Bearing wear signature, spindle imbalance - Predicts failure: “Spindle bearings will fail in 200 hours” - Schedule maintenance proactively (avoid unplanned downtime)

**Quality Prediction:** - AI correlates process data with inspection results - Learns: What sensor signatures produce defects - Predicts: “This part likely has delamination” (before inspection) - Action: Re-make part or inspect more closely

**Status:** - Research: Universities, national labs (NIST, Fraunhofer) - Commercialization: Starting (Autodesk Fusion 360, Siemens NX use AI) - Adoption: Early (aerospace, medical), expanding to general manufacturing

## Skills for the Future

### Technical Competencies

**Multi-Material Expertise:** - Future machinist: Must understand metals, composites, ceramics - Different materials in single assembly (hybrid structures) - Example: Ceramic bearing in aluminum housing with carbon fiber cover - Training: Broader material science education, cross-training

**Programming Complexity:** - Simple 2.5D -> complex 5-axis simultaneous - Composite draping: Simulate layup, account for thickness variation - Adaptability: Modify programs based on in-process measurements

**Data Analysis:** - Interpret sensor data (force plots, vibration spectra) - Statistical process control (X-bar charts, Cpk calculations) - Root cause analysis (troubleshoot with data, not just intuition)

### Soft Skills

**Collaboration:** - Advanced materials = interdisciplinary teams - Machinist + materials engineer + quality inspector + programmer - Communication: Explain machining constraints to designers

**Continuous Learning:** - Technology evolving rapidly (new materials, tools, techniques) - Professional development: Webinars, trade shows (IMTS, CAMX), certifications - Mindset: Lifelong learner (comfortable with change)

**Problem-Solving:** - Complex systems: Many variables interact - Systematic approach: Diagnose, test hypotheses, document - Creativity: Non-traditional solutions (process innovations)

## Career Opportunities

### Growing Industries

**Aerospace:** - Composites dominate: 50-80% of airframe (Boeing 787, Airbus A350) - Demand: Skilled composite machinists (trim, drilling, routing) - Certifications: AS9100 (aerospace quality), NADCAP (special processes)

**Automotive:** - Lightweighting: Regulations (fuel economy) drive composite adoption - Carbon fiber: High-end vehicles (Lamborghini, BMW i-series) - Glass fiber: Mass market (under-hood, structural) - Ceramics: Brakes (carbon-ceramic), bearings (silicon nitride)

**Medical Devices:** - Bioceramics: Implants (zirconia hip, alumina knee), dental - Composites: Prosthetics (carbon fiber limbs), surgical instruments - Precision: Tolerances  $\pm 0.0001$ ”, surface finish  $< 5 \mu\text{in}$  - Regulation: FDA, ISO 13485 (quality management)

**Energy:** - Wind turbines: Blades (50-100m long) glass fiber/carbon hybrid - Machining: Trim, drill holes (lightning protection), surface finish - Oil & gas: Ceramic wear parts (valves, seals)

**Electronics:** - Substrates: Ceramic (alumina, AlN) for chips, LED packages - Machining: Laser cutting, precision grinding - Miniaturization: Features  $< 1 \text{ mm}$  (micro-machining)



## Salary and Demand

**Entry-Level** (0-2 years): - CNC operator (composites/ceramics): \$35,000-50,000/year - Training: Community college, apprenticeship (2-year programs)

**Mid-Level** (3-7 years): - CNC machinist/programmer: \$50,000-75,000/year - Responsibilities: Setup, programming, troubleshooting - Skills: Multi-axis programming, tooling selection, process optimization

**Senior-Level** (8+ years): - Manufacturing engineer, process specialist: \$75,000-120,000/year - Responsibilities: Process development, automation, team leadership - Education: Often Bachelor's degree (engineering, manufacturing technology)

**Demand Forecast:** - BLS (Bureau of Labor Statistics): CNC machinist -3% growth (2023-2033) overall - Advanced materials: +10-15% growth (subset, high demand) - Shortage: Skilled workers (aging workforce, fewer entering trades) - Opportunity: Secure employment, competitive wages for specialized skills

## Final Thoughts

### Challenges and Rewards

**Challenges:** - Steep learning curve: Advanced materials less forgiving than aluminum - Health hazards: Dust control vigilance required - Equipment cost: Dust collection, specialized tooling adds investment - Process development: Less established knowledge (trial-and-error)

**Rewards:** - Cutting-edge technology: Work with materials in high-performance applications - Problem-solving: Intellectual challenge (optimize complex processes) - Career security: Specialized skills in demand - Pride: See your work in aircraft, medical devices, sports equipment

### The Path Forward

**For Machinists:** 1. **Learn fundamentals:** Start with metals (easier to learn principles) 2. **Cross-train:** Add composites or ceramics (increases value) 3. **Specialize:** Become expert in one material class (depth valuable) 4. **Stay current:** Follow industry trends (trade publications, forums) 5. **Document:** Keep notebooks (processes, troubleshooting, solutions)

**For Shop Owners:** 1. **Assess market:** Is there demand for advanced materials in your region? 2. **Invest incrementally:** Start with one material (dust collection, tooling) 3. **Train employees:** Send to courses, hands-on learning projects 4. **Quality systems:** Implement SPC, NDT (customers expect it) 5. **Network:** Join industry groups (SAMPE, American Ceramic Society)

**For Engineers:** 1. **Design for manufacturing:** Understand machining constraints (delamination, chipping) 2. **Collaborate early:** Involve machinists in design reviews 3. **Specify realistically:** Tolerances, surface finish appropriate for material 4. **Test early:** Prototype -> iterate (find issues before production)

## Conclusion

Advanced materials machining is a specialized, growing field combining material science, precision machining, and safety engineering. Success requires:

- **Technical knowledge:** Material properties, tool selection, process parameters
- **Safety commitment:** Dust control, PPE, regulatory compliance
- **Continuous improvement:** Process optimization, quality control
- **Adaptability:** New materials, technologies emerging constantly

The skills developed in machining composites and ceramics are highly transferable and valuable. As industries demand lighter, stronger, more durable components, machinists with advanced materials expertise will be essential to manufacturing's future.

**Thank you for engaging with Module 17.** Apply these principles carefully, prioritize safety, and never stop learning. The field of advanced materials is evolving rapidly—those who master it will be at the forefront of manufacturing innovation.

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## Additional Resources

### Professional Organizations

- **SAMPE** (Society for the Advancement of Material and Process Engineering): Composites focus
- **American Ceramic Society**: Ceramics technical resources, conferences
- **SME** (Society of Manufacturing Engineers): General manufacturing, certification programs
- **ASM International**: Materials science, heat treating, testing

### Certifications

- **NIMS** (National Institute for Metalworking Skills): Machining certifications
- **ASNT** (American Society for Nondestructive Testing): NDT Level I/II/III
- **AS9100/NADCAP**: Aerospace quality systems (company-level)

### Publications

- **Composites World**: Industry news, technical articles
- **American Machinist**: General machining, new technologies
- **Ceramic Industry**: Ceramics manufacturing processes
- **Modern Machine Shop**: CNC techniques, tooling

### Training

- **Community colleges**: 2-year CNC/manufacturing technology programs
  - **Manufacturer training**: Harvey Tool, Kennametal, Sandvik offer courses
  - **Webinars**: SPE (Society of Plastics Engineers), SAMPE host online sessions
  - **Trade shows**: IMTS (Chicago), CAMX (Composites), Ceramics Expo
- 

## End of Module 17

**Next Module:** Module 18 - Industry 4.0 and Smart Manufacturing

## 17.2 Composite Materials Science

### Definition and Structure

**Composite Material:** A macroscopic combination of two or more distinct materials with a recognizable interface between them, engineered to achieve superior properties compared to individual constituents.

**Three Essential Components:** 1. **Reinforcement:** Load-bearing phase (fibers, particles, whiskers) 2. **Matrix:** Continuous phase binding reinforcement 3. **Interface:** Critical bond zone transferring stress between phases

### Fiber-Reinforced Composites

#### Carbon Fiber

**Production Methods:** - **PAN-based** (polyacrylonitrile): 90% of production, general purpose - Precursor stretched and oxidized at 200-300 degC - Carbonized at 1000-1500 degC in inert atmosphere - Graphitization at 2500-3000 degC for high modulus

- **Pitch-based:** Lower cost, specialty applications
  - Petroleum or coal tar pitch precursor
  - Higher thermal conductivity

#### Fiber Types and Properties:

| Type                       | Tensile Strength | Modulus | Elongation | Density                | Cost         |
|----------------------------|------------------|---------|------------|------------------------|--------------|
| Standard Modulus (T300)    | 3,530 MPa        | 230 GPa | 1.5%       | 1.76 g/cm <sup>3</sup> | \$20-30/kg   |
| Intermediate Modulus (IM7) | 4,310 MPa        | 276 GPa | 1.9%       | 1.78 g/cm <sup>3</sup> | \$40-60/kg   |
| High Modulus (M55J)        | 4,020 MPa        | 540 GPa | 0.7%       | 1.91 g/cm <sup>3</sup> | \$200-400/kg |
| Ultra High Modulus (K13D)  | 3,430 MPa        | 930 GPa | 0.4%       | 2.20 g/cm <sup>3</sup> | \$1000+/kg   |

**Key Properties:** - Negative coefficient of thermal expansion (CTE) - Excellent fatigue resistance (maintains 90% strength at 10<sup>7</sup> cycles) - Electrically conductive (10<sup>-3</sup> to 10<sup>-5</sup> Ohms\*cm) - Inert to most chemicals - Degrades above 400 degC in air (oxidation)

**Typical Applications:** - Aerospace primary structures (wing skins, fuselage) - Automotive (monocoque chassis, drive shafts) - Sporting goods (bicycle frames, fishing rods) - Medical (prosthetics, X-ray tables - radiolucent)

#### Glass Fiber

##### Composition Types:

**E-Glass (Electrical):** - Composition: 54% SiO<sub>2</sub>, 15% Al<sub>2</sub>O<sub>3</sub>, 21% CaO+MgO, 10% B<sub>2</sub>O<sub>3</sub> - Most common, good electrical insulation - Tensile strength: 3,400 MPa - Modulus: 72 GPa - Cost: \$2-4/kg

**S-Glass (Strength):** - Higher SiO<sub>2</sub> and Al<sub>2</sub>O<sub>3</sub>, no boron - 40% stronger than E-glass (4,580 MPa tensile) - Modulus: 86 GPa - Aerospace and ballistic applications - Cost: \$15-25/kg

**C-Glass (Chemical):** - Enhanced acid resistance - Chemical processing equipment - Cost: \$8-12/kg

**Properties Comparison to Carbon:** - Density: 2.54 g/cm<sup>3</sup> (44% heavier than carbon) - Lower strength and stiffness (1/5 modulus of carbon) - Non-conductive (insulator) - Transparent to radio waves (radomes) - Less abrasive to tooling - Cost: 5-10x cheaper than carbon fiber

**Common Applications:** - Marine (boat hulls, tanks) - Wind turbine blades - Automotive body panels - Electrical enclosures

### Aramid Fiber (Kevlar)

**Structure:** - Aromatic polyamide polymer chains - Highly oriented molecular structure - Produced by DuPont (Kevlar), Teijin (Twaron)

**Properties:**

| Property         | Kevlar 29              | Kevlar 49              | Kevlar 149             |
|------------------|------------------------|------------------------|------------------------|
| Tensile Strength | 3,620 MPa              | 3,620 MPa              | 3,450 MPa              |
| Modulus          | 83 GPa                 | 131 GPa                | 175 GPa                |
| Elongation       | 3.6%                   | 2.4%                   | 1.9%                   |
| Density          | 1.44 g/cm <sup>3</sup> | 1.44 g/cm <sup>3</sup> | 1.47 g/cm <sup>3</sup> |
| Application      | Rope, cables           | Aerospace              | Electronics            |

**Unique Characteristics:** - Exceptional impact resistance and toughness - High tensile strength, poor compression (30-40% of tensile) - Difficult to cut (fibers fray and pull out) - Absorbs moisture (affects properties) - Degrades under UV exposure - Non-conductive

**Machining Challenges:** - Cannot be cut cleanly with conventional tools - Requires shearing action or specialized scissors - Abrasive waterjet or laser cutting preferred - Dulls tools rapidly

**Applications:** - Body armor and ballistic protection - Cut-resistant gloves and clothing - High-strength ropes and cables - Aerospace pressure vessels

## Matrix Materials

### Polymer Matrix Composites (PMC)

**Thermoset Resins:**

**Epoxy:** - Most common aerospace matrix - Excellent mechanical properties - Good adhesion to fibers - Low shrinkage during cure (<2%) - Cure temperature: 120-180 degC typical - Service temperature: Up to 150 degC continuous - Cost: \$8-20/kg

**Polyester:** - Lower cost than epoxy (\$3-8/kg) - Faster cure, room temperature possible - Higher shrinkage (4-8%) - Marine and automotive applications - Moderate mechanical properties

**Vinyl Ester:** - Intermediate properties between epoxy and polyester - Excellent corrosion resistance - Chemical processing applications - Cost: \$6-12/kg

**Thermoplastic Resins:**

**PEEK (Polyether Ether Ketone):** - High-performance engineering plastic - Service temperature: 250 degC continuous - Excellent chemical resistance - Weldable and recyclable - Cost: \$80-150/kg - Medical implants, aerospace

**PPS (Polyphenylene Sulfide):** - Lower cost than PEEK (\$40-80/kg) - Service temperature: 200 degC - Chemical and flame resistance - Automotive under-hood components

**Matrix Functions:** 1. Transfer loads to fibers 2. Protect fibers from environment 3. Maintain fiber orientation 4. Provide surface finish 5. Distribute stress around damaged fibers

## Metal Matrix Composites (MMC)

**Aluminum MMC:** - Matrix: 6061, 2024, or 7075 aluminum alloys - Reinforcement: SiC particles (10-25% volume fraction) - Properties: - Increased stiffness: 50-100% over base alloy - Lower CTE: Better thermal stability - Improved wear resistance - Applications: Automotive pistons, brake rotors, bicycle components - Cost: \$15-40/kg

**Titanium MMC:** - Matrix: Ti-6Al-4V typical - Reinforcement: SiC or TiB whiskers - Ultra-high temperature applications (>500 degC) - Aerospace turbine blades - Cost: \$200-500/kg

**Machining Considerations:** - Abrasive reinforcement particles - Rapid tool wear (PCD or CBN tooling required) - Difficult chip formation - High cutting forces

## Ceramic Matrix Composites (CMC)

**Silicon Carbide/Silicon Carbide (SiC/SiC):** - SiC fibers in SiC matrix - Ultra-high temperature (>1300 degC) - Jet engine components, thermal protection - Brittle but fracture-tolerant (fiber bridging) - Extremely difficult to machine

**Carbon/Carbon (C/C):** - Carbon fibers in carbon matrix - Highest temperature capability (>2000 degC in inert atmosphere) - Oxidizes above 400 degC in air (requires coating) - Aircraft brakes, rocket nozzles - Machined green, then carbonized

## Composite Architectures

### Fiber Orientations

**Unidirectional (UD):** - All fibers aligned in single direction - Maximum strength and stiffness along fiber axis - Weak in transverse direction (matrix-dominated) - Typical properties (carbon/epoxy): - Longitudinal tensile: 2,000 MPa - Transverse tensile: 50 MPa (40x weaker!) - Longitudinal modulus: 140 GPa - Transverse modulus: 10 GPa

**Woven Fabric:** - Fibers interlaced in warp and fill directions - Balanced properties in two directions - Easier to handle than UD tape - Slight crimping reduces properties (10-20% vs UD) - Plain weave, twill, satin weaves

**Braided:** - Fibers interlaced at  $\pm\theta$  deg to axis - Complex 3D shapes (tubes, I-beams) - Good impact resistance - Automated production

**Random Mat:** - Short chopped fibers randomly oriented - Isotropic in-plane properties - Lower strength (30% of UD) - Low cost, ease of manufacturing - Non-structural applications

### Layup Design

**Quasi-Isotropic Layup:** -  $[0/+/45/90]_s$  - balanced properties in all directions - Subscript 's' indicates symmetric layup - Prevents warping during cure - General-purpose structural applications

**Angle-Ply Layup:** -  $[+/-\theta]_n$  - optimized for specific load direction -  $[+/-45]$  for shear loading -  $[0/90]$  for orthogonal loading - 'n' indicates number of repeating units

### Laminate Theory:

Classical Laminated Plate Theory (CLPT) predicts composite behavior:

**Stiffness Matrix [ABD]:** - [A]: Extensional stiffness - [B]: Bending-extension coupling - [D]: Bending stiffness

For symmetric laminates,  $[B] = 0$  (no coupling - desirable).

### Example Calculation:

For a  $[0/90/0]$  carbon/epoxy laminate with ply thickness  $t = 0.125$  mm:

Properties of single UD ply: -  $E_1 = 140$  GPa (longitudinal) -  $E_2 = 10$  GPa (transverse) -  $G_{12} = 5$  GPa (shear) -  $\nu_{12} = 0.3$  (Poisson's ratio)

Transformed reduced stiffness matrix  $[Q]$  for each ply, then:

$$A_{ij} = \sum_{k=1}^n \bar{Q}_{ij}^{(k)} (z_k - z_{k-1})$$

Results in laminate modulus approx 75 GPa (halfway between  $E_1$  and  $E_2$ ).

## Sandwich Structures

**Configuration:** - Two thin, stiff face sheets (composite) - Lightweight core separating faces - Core resists shear, faces resist bending

### Bending Stiffness:

For sandwich beam with face thickness  $t_f$ , core thickness  $c$ , face modulus  $E_f$ , width  $b$ :

$$D = \frac{E_f b t_f (c + t_f)^2}{2}$$

Example: - 1 mm carbon faces, 10 mm foam core, 100 mm wide -  $E_f = 70$  GPa (woven carbon) -  $D = \frac{70 \times 10^9 \times 0.1 \times 0.001 \times 0.011^2}{2} = 421 \text{ N}\cdot\text{m}^2$

Solid plate of same weight (2.2 mm carbon):  $D = 26 \text{ N}\cdot\text{m}^2$  (16x less stiff!)

### Core Materials:

| Core Type                            | Density (kg/m <sup>3</sup> ) | Shear Strength (MPa) | Shear Modulus (MPa) | Cost (\$/m <sup>2</sup> ) |
|--------------------------------------|------------------------------|----------------------|---------------------|---------------------------|
| PVC Foam<br>(Divinycell H80)         | 80                           | 1.4                  | 30                  | \$30-50                   |
| PMI Foam<br>(Rohacell 110)           | 110                          | 2.5                  | 54                  | \$100-200                 |
| Aluminum<br>Honeycomb<br>(1/8" cell) | 80                           | 1.7                  | 280                 | \$50-100                  |
| Nomex<br>Honeycomb<br>(1/8" cell)    | 48                           | 1.2                  | 140                 | \$80-150                  |
| Balsa Wood<br>(end grain)            | 150                          | 2.0                  | 200                 | \$20-40                   |

**Machining Considerations:** - Core easily damaged (crushing, heat) - Edge routing requires sharp tools, high speed - Dust collection critical (foam dust hazardous) - Potential for face/core debonding

## Manufacturing Processes

### Hand Layup

**Process:** 1. Mold preparation and release agent 2. Gel coat application (if needed) 3. Resin application to mold 4. Fabric placement 5. Consolidation (roller, brush) 6. Cure at room temperature or elevated temperature

**Advantages:** - Low tooling cost - Flexible for prototypes - Large structures possible

**Disadvantages:** - Labor intensive - Variable quality - Health hazards (styrene exposure) - Fiber volume fraction: 30-50%

### **Vacuum Bagging**

**Process Enhancement to Hand Layup:** - Sealed plastic bag over layup - Vacuum applied (-0.8 to -1.0 bar) - Atmospheric pressure consolidates laminate - Results in higher fiber volume (50-60%)

**Materials:** - Release film (perforated for air escape) - Bleeder cloth (absorbs excess resin) - Breather cloth (air distribution) - Vacuum bag (nylon or polyester film) - Sealant tape

### **Prepreg Layup**

**Material Form:** - Fabric pre-impregnated with partially cured resin - Stored frozen (-18 degC) to prevent full cure - Shelf life: 6-12 months frozen, days at room temperature

**Process:** 1. Thaw prepreg to room temperature 2. Cut plies (by hand or automated cutter) 3. Layup on tool 4. Vacuum bag 5. Autoclave cure (120-180 degC, 5-7 bar pressure)

**Advantages:** - Consistent resin content and fiber volume (55-65%) - Cleaner process - Superior mechanical properties - Reduced voids

**Disadvantages:** - Higher material cost (\$40-100/kg vs \$20-40 for dry fabric+resin) - Requires autoclave or oven - Limited out-time before tack is lost

### **Resin Transfer Molding (RTM)**

**Process:** 1. Dry fabric preform placed in closed mold 2. Resin injected under pressure (2-5 bar) 3. Cure in heated mold 4. Part removed

**Advantages:** - Both sides of part have smooth finish - Lower VOC emissions (closed mold) - Faster cycle time than hand layup - Suitable for moderate production (10-1000 parts/year)

**Tooling Requirements:** - Matched male/female molds - Injection ports and vents - Resin distribution channels - Tooling cost: \$10,000-\$100,000 depending on size/complexity

### **Pultrusion**

**Process:** - Continuous fiber pulled through resin bath - Then through heated die (120-150 degC) - Constant cross-section profiles (rods, tubes, beams) - Production rate: 0.1-5 m/min

**Applications:** - Structural shapes (I-beams, channels, gratings) - Utility poles - Handrails

**Properties:** - High fiber volume (60-70%) - Unidirectional or mat reinforcement - Excellent longitudinal properties - Limited to constant cross-section

### **Rule of Mixtures**

Simple model for estimating composite properties from constituent properties.

#### **Longitudinal Properties (Parallel to Fibers):**

Modulus:

$$E_c = E_f V_f + E_m V_m$$

Strength (fiber-dominated):

$$\sigma_c = \sigma_f V_f + \sigma'_m V_m$$

where  $\sigma'_m$  is matrix stress at fiber failure strain.

### Transverse Properties (Perpendicular to Fibers):

Modulus (series model):

$$\frac{1}{E_c} = \frac{V_f}{E_f} + \frac{V_m}{E_m}$$

### Example Calculation:

Carbon/epoxy with: -  $E_f = 230$  GPa,  $E_m = 3$  GPa -  $V_f = 0.60$  (60% fiber volume)

Longitudinal modulus:

$$E_c = 230(0.6) + 3(0.4) = 138 + 1.2 = 139.2 \text{ GPa}$$

Transverse modulus:

$$\frac{1}{E_c} = \frac{0.6}{230} + \frac{0.4}{3} = 0.00261 + 0.133 = 0.136$$
$$E_c = 7.4 \text{ GPa}$$

Note: Rule of mixtures provides upper bound. Actual properties often 10-20% lower due to imperfect fiber alignment, voids, and interface quality.

## Failure Modes

### Fiber-Dominated Failures

**Fiber Breakage:** - Tensile overload parallel to fibers - Brittle fracture, sudden failure - Occurs at ~1.5% strain for carbon, 2% for glass

**Fiber Buckling:** - Compression parallel to fibers - Wavy fibers collapse - Compression strength 40-60% of tensile for carbon

### Matrix-Dominated Failures

**Matrix Cracking:** - Transverse tension or in-plane shear - First ply failure (FPF) but not ultimate failure - Multiple cracks distribute load

**Delamination:** - Separation between plies - Caused by interlaminar shear or normal stress - Often from impact damage or free edge stresses - Grows under fatigue loading

### Interface Failures

**Fiber/Matrix Debonding:** - Loss of adhesion at interface - Reduces load transfer efficiency - Fiber pullout during fracture

## Testing Standards

**Tensile Testing:** - ASTM D3039: Tension test of polymer matrix composites - Specimen: 250 mm long x 25 mm wide - End tabs prevent grip damage - Strain measurement via extensometer or strain gauge

**Compression Testing:** - ASTM D6641: Combined loading compression - ASTM D3410: Compression with antibuckling fixture - Difficult test (premature buckling failure)

**Interlaminar Shear:** - ASTM D2344: Short beam shear test - Span/thickness ratio of 4:1 - Induces shear failure between plies

**Impact Testing:** - ASTM D7136: Drop-weight impact - Evaluates damage resistance - Followed by compression-after-impact (CAI) test



## Material Selection Guide

**High Stiffness Applications:** - Choose high-modulus carbon fiber (M40J, M55J) - UD or quasi-isotropic depending on load paths - Prepreg for maximum fiber volume - Examples: Satellite structures, precision instruments

**High Strength Applications:** - Intermediate or standard modulus carbon (T700, IM7) - Consider impact resistance (hybrid with aramid or glass) - Woven fabric for complex loads - Examples: Automotive components, sporting goods

**Cost-Sensitive Applications:** - E-glass fiber - Vinyl ester or polyester resin - Hand layup or RTM - Examples: Marine, wind energy, chemical tanks

**Harsh Environment:** - Carbon or glass (inert to most chemicals) - Vinyl ester resin for superior chemical resistance - Surface protection (gel coat, UV-resistant topcoat) - Examples: Chemical processing, offshore platforms

## Summary

Composite materials offer exceptional performance through intelligent combination of reinforcing fibers and matrix materials. Understanding fiber types (carbon, glass, aramid), matrix systems (thermoset, thermoplastic), and architecture (UD, woven, sandwich) enables informed material selection for CNC machining applications.

Key considerations for CNC processing: - Fiber orientation affects machining forces and tool wear - Matrix type determines thermal sensitivity - Sandwich structures require specialized fixturing - Material cost varies 100x from glass/polyester to carbon/PEEK

Proper material selection balances performance requirements against manufacturing cost and machinability.

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**Next:** 17.3 Ceramic Materials Science

## 17.3 Ceramic Materials Science

### Definition and Classification

**Ceramic:** Inorganic, non-metallic material typically consisting of metallic and non-metallic elements bonded primarily through ionic and covalent bonds.

**Defining Characteristics:** - High melting points (typically >2000 degC) - Extreme hardness (Mohs 7-10) - Chemical stability and corrosion resistance - Low electrical and thermal conductivity (most types) - Brittleness - low fracture toughness - High compressive strength, low tensile strength

### Atomic Bonding and Structure

**Ionic Bonding** (e.g., Al<sub>2</sub>O<sub>3</sub>, MgO): - Electron transfer from metal to non-metal - Strong electrostatic attraction - Non-directional bonding - High melting points (2050 degC for alumina)

**Covalent Bonding** (e.g., SiC, Si<sub>3</sub>N<sub>4</sub>): - Electron sharing between atoms - Directional bonding - Very high strength and hardness - Difficult to process and sinter

**Crystal Structures:** - Face-centered cubic (MgO) - Hexagonal close-packed (Al<sub>2</sub>O<sub>3</sub>) - Complex structures (many silicates)

### Oxide Ceramics

#### Alumina (Al<sub>2</sub>O<sub>3</sub>)

**Properties:** - Melting point: 2050 degC - Density: 3.95 g/cm<sup>3</sup> - Hardness: 9 Mohs (1800-2100 Vickers) - Flexural strength: 300-550 MPa (varies with purity and grain size) - Modulus: 350-380 GPa - Fracture toughness: 3-5 MPa\*sqrtm - Excellent electrical insulation - Chemical inertness

#### Purity Grades:

| Grade | Al <sub>2</sub> O <sub>3</sub> Content | Properties                       | Applications                        | Cost (\$/kg) |
|-------|----------------------------------------|----------------------------------|-------------------------------------|--------------|
| 85%   | 85-90%                                 | Lower cost, higher machinability | Electrical insulators, general wear | \$5-15       |
| 95%   | 95-96%                                 | Good balance properties          | Seals, valves, wear parts           | \$15-30      |
| 99%   | 99.0-99.5%                             | High strength, purity            | High-temp furnace parts             | \$30-60      |
| 99.9% | >99.9%                                 | Ultra-pure, semiconductor        | Crucibles, substrates               | \$80-200     |

**Grain Size Effects:** - Fine grain (<1 μm): Higher strength (550 MPa flexural) - Coarse grain (>10 μm): Easier to machine, lower strength (350 MPa)

**Machining Notes:** - Requires diamond tooling (PCD, diamond wheels) - Green machining before sintering preferred when possible - Prone to edge chipping - requires support - Coolant essential for heat removal

**Applications:** - Cutting tool inserts - Wear-resistant components (seals, valves, bearings) - High-temperature insulators - Biomedical implants (hip/knee replacements) - Ballistic armor plates

#### Zirconia (ZrO<sub>2</sub>)

**Crystal Phases:** - Monoclinic (room temperature) - Tetragonal (>1170 degC) - Cubic (>2370 degC)

**Phase Transformation:** Volume change on cooling from tetragonal to monoclinic causes cracking - solved by stabilization.

### Stabilized Grades:

**PSZ (Partially Stabilized Zirconia):** - MgO addition (3 mol%) - Tetragonal precipitates in cubic matrix - Fracture toughness: 8-12 MPa $\sqrt{\text{m}}$  (highest of structural ceramics) - Transformation toughening mechanism

**TZP (Tetragonal Zirconia Polycrystal):** - Y<sub>2</sub>O<sub>3</sub> addition (3 mol% = 3Y-TZP) - Fine grain (<1  $\mu\text{m}$ ) all tetragonal - Flexural strength: 900-1200 MPa - Superior wear resistance

**FSZ (Fully Stabilized Zirconia):** - CaO or Y<sub>2</sub>O<sub>3</sub> (8-10 mol%) - Cubic structure stable at all temperatures - Lower strength (600 MPa) but excellent thermal shock resistance - Thermal barrier coatings

**Properties (3Y-TZP):** - Density: 6.05 g/cm<sup>3</sup> - Hardness: 1200 Vickers - Flexural strength: 900-1200 MPa - Fracture toughness: 9-10 MPa $\sqrt{\text{m}}$  - Modulus: 210 GPa

**Transformation Toughening:** Crack tip stress induces tetragonal  $\rightarrow$  monoclinic transformation, absorbing energy and creating compressive stress that resists crack propagation.

**Applications:** - Dental crowns and implants (biocompatible, tooth-colored) - Oxygen sensors (ionic conductivity) - Thermal barrier coatings (jet engines) - High-wear components (cutting tools, dies) - Hip joint balls (ultra-smooth, wear-resistant)

### Other Oxide Ceramics

**Magnesia (MgO):** - Refractory material (melting point 2800 degC) - Basic oxide (reacts with acidic slags) - Furnace linings

**Silicon Dioxide (SiO<sub>2</sub>) - Fused Silica:** - Amorphous structure - Very low thermal expansion ( $0.5 \times 10^{-6}/\text{degC}$ ) - Excellent thermal shock resistance - Optical applications, crucibles

**Titania (TiO<sub>2</sub>):** - High refractive index - Pigments, coatings - Photocatalytic properties

### Non-Oxide Ceramics

#### Silicon Carbide (SiC)

**Production:** - Acheson process: SiO<sub>2</sub> + C  $\rightarrow$  SiC + CO at 2500 degC - Single crystals grown for semiconductors - Sintered for structural applications

**Properties:** - Melting point: 2730 degC (decomposes) - Density: 3.21 g/cm<sup>3</sup> - Hardness: 9.5 Mohs (2500 Vickers) - Flexural strength: 400-550 MPa - Modulus: 410 GPa - Fracture toughness: 4-5 MPa $\sqrt{\text{m}}$  - *Thermal conductivity: 120 W/(mK)* - very high for ceramic - Low thermal expansion:  $4.3 \times 10^{-6}/\text{degC}$

**Forms:** - Reaction bonded SiC (RBSC): Si infiltrated into SiC+C preform - Sintered SiC: Hot pressed or sintered with additives - CVD SiC: Chemical vapor deposition, ultra-pure

**Machining Characteristics:** - Extremely hard and abrasive - Requires diamond grinding - Low fracture toughness  $\rightarrow$  chipping risk - Excellent thermal conductivity aids heat removal

**Applications:** - Mechanical seals (chemical pumps) - Semiconductor processing equipment (wafer carriers) - High-temperature bearings - Bulletproof armor plates - Abrasives and grinding media

#### Silicon Nitride (Si<sub>3</sub>N<sub>4</sub>)

**Production:** - Reaction bonding: Si + N<sub>2</sub>  $\rightarrow$  Si<sub>3</sub>N<sub>4</sub> - Sintering with additives (Y<sub>2</sub>O<sub>3</sub>, Al<sub>2</sub>O<sub>3</sub>) - Hot isostatic pressing (HIP) for full density

**Properties:** - Density: 3.20 g/cm<sup>3</sup> - Hardness: 1700-2000 Vickers - Flexural strength: 600-1000 MPa (highest of engineering ceramics) - Fracture toughness: 6-8 MPa $\sqrt{\text{m}}$  - *Modulus: 310 GPa* - *Thermal conductivity: 30 W/(mK)* - Excellent thermal shock resistance

**Unique Characteristics:** - Elongated grain structure -> enhanced toughness - Superior high-temperature strength retention (up to 1200 degC) - Oxidation resistance (forms protective SiO<sub>2</sub> layer) - Low density for ceramic

**Applications:** - Cutting tool inserts (high-speed machining cast iron) - Turbocharger rotors (low inertia) - Bearings (high-speed, high-temp) - Engine components (valves, cam followers) - Metal casting crucibles

### Boron Carbide (B<sub>4</sub>C)

**Properties:** - Third hardest material (after diamond, cubic BN) - Hardness: 2900 Vickers - Density: 2.52 g/cm<sup>3</sup> (lightest of technical ceramics) - Melting point: 2450 degC - Modulus: 450 GPa - Neutron absorber

**Challenges:** - Difficult to sinter (covalent bonding) - Requires hot pressing or HIP - Very expensive (\$100-500/kg) - Brittle, low toughness (3.5 MPa\*sqrtm)

**Applications:** - Ballistic armor (lightweight body armor, vehicle protection) - Abrasive powders (lapping, grinding) - Nuclear reactor control rods - Sandblasting nozzles

### Aluminum Nitride (AlN)

**Unique Property:** Exceptional thermal conductivity for electrical insulator: - Thermal conductivity: 150-180 W/(mK) (*comparable to aluminum metal!*) - *Electrical resistivity: >10<sup>13</sup> Ohmcm*

**Applications:** - Power electronics substrates (heat spreading) - LED packages - Semiconductor processing

### Glass-Ceramics

**Definition:** Hybrid materials initially formed as glass, then heat-treated to develop controlled crystallization.

#### Macor(R)

**Composition:** Fluorophlogopite mica crystals in borosilicate glass matrix

**Key Properties:** - Machinable with carbide tools (no diamond required!) - Density: 2.52 g/cm<sup>3</sup> - Flexural strength: 94 MPa (much lower than engineering ceramics) - Modulus: 67 GPa - Max service temp: 800 degC continuous, 1000 degC short term - Excellent electrical insulation - Zero porosity (vacuum compatible)

**Machinability Mechanism:** Mica crystals act as “chip breakers,” preventing crack propagation. Material machines similar to brass.

**Machining Parameters:** - Carbide tooling (no diamond needed) - Speeds: 300-600 SFM - Feeds: 0.001-0.005 IPR - Coolant or dry machining - Fine finish achievable (Ra 0.8 μm)

**Applications:** - Vacuum feedthroughs - Precision fixtures and jigs - Electrical insulators - Prototype ceramic parts (machine first, then replace with engineering ceramic)

**Cost:** \$50-100/kg (expensive but eliminates diamond grinding for prototypes)

#### Other Glass-Ceramics

**Zerodur(R)** (Schott): - Near-zero thermal expansion (+/-0.05 x 10<sup>-6</sup>/ degC) - Telescope mirrors, optical benches - Not as easily machinable as Macor

**Pyroceram(R)** (Corning): - High strength glass-ceramic - Cookware, radomes

### Traditional Ceramics

#### Porcelain

**Composition:** Kaolin (Al<sub>2</sub>Si<sub>2</sub>O<sub>5</sub>(OH)<sub>4</sub>) + feldspar + quartz

**Properties:** - Density:  $2.4 \text{ g/cm}^3$  - Flexural strength: 60-100 MPa - Glassy phase reduces strength vs engineering ceramics - Porosity in low-fire grades

**Machining:** - Green machining before firing (easy) - Post-fire machining (difficult, diamond required) - Prone to chipping

**Applications:** - Electrical insulators (power lines) - Sanitary ware - Dental applications - Decorative

### Steatite (Talc-Based)

**Composition:** Magnesium silicate

**Properties:** - Good electrical insulation - Moderate strength (100 MPa flexural) - Easier to machine than high-performance ceramics - Low cost (\$5-20/kg)

**Applications:** - Electrical insulators - Kiln furniture

## Ceramic Processing and Sintering

### Powder Preparation

**Particle Size Effects:** - Finer powder ( $<1 \text{ }\mu\text{m}$ ): Better sintering, higher strength, more expensive - Coarser powder ( $>10 \text{ }\mu\text{m}$ ): Lower cost, easier to handle, lower strength

**Powder Production Methods:** - Chemical precipitation (fine, pure) - Solid-state reaction (variable purity) - Sol-gel (ultra-fine, expensive)

### Green Forming

**Pressing:** - Uniaxial die pressing (simple shapes) - Isostatic pressing (complex shapes, uniform density) - Green density: 50-60% theoretical

**Slip Casting:** - Ceramic slurry poured into porous mold - Water absorbed, part solidifies - Complex shapes, hollow parts

**Extrusion:** - Plastic mass forced through die - Tubes, rods, bricks

**Injection Molding:** - Ceramic powder + polymer binder - High-volume, complex shapes - Requires binder burnout before sintering

### Sintering

**Solid-State Sintering:** Temperature: 1200-1800 degC (50-80% of melting point)

**Driving Force:** Surface energy reduction

**Mechanisms:** 1. **Surface diffusion:** Neck growth, no densification 2. **Grain boundary diffusion:** Densification 3. **Volume diffusion:** Densification

**Densification:** Green density 50-60%  $\rightarrow$  Sintered density 95-99.5%

**Grain Growth:** - Occurs simultaneously with densification - Larger grains reduce strength - Control via temperature, time, dopants

**Liquid-Phase Sintering:** - Additive forms liquid at sintering temperature - Enhances densification (capillary forces) - Silicon nitride with  $\text{Y}_2\text{O}_3/\text{Al}_2\text{O}_3$  additives - Lowers sintering temperature

**Hot Pressing / HIP:** - Pressure applied during sintering - Achieves full density (99.9%+) - Finer grain size  $\rightarrow$  higher strength - More expensive

## Green Machining vs Fired Machining

**Green Machining:** - Before sintering, material is soft (chalk-like) - Conventional tooling (carbide, HSS) - Fast, easy, low cost - Must account for shrinkage (15-20% linear typical) - Risk: Handling damage to fragile green body

**Fired Machining:** - After sintering, material is fully hard - Requires diamond tooling - Slow, expensive - Achieves exact final dimensions - Necessary for features added after sintering

**Hybrid Approach:** 1. Near-net-shape green machining (oversized) 2. Sintering 3. Minimal fired grinding to final dimensions

## Mechanical Properties in Detail

### Strength

**Flexural (Bend) Strength:** Most common test for ceramics (tensile testing difficult due to grip damage).

**3-Point Bend Test** (ASTM C1161):

$$\sigma_f = \frac{3FL}{2bh^2}$$

where: -  $F$  = load at fracture -  $L$  = support span -  $b$  = specimen width -  $h$  = specimen height

**Weibull Statistics:** Ceramic strength is statistical (depends on flaw population).

**Weibull Distribution:**

$$P_f = 1 - \exp \left[ - \left( \frac{\sigma}{\sigma_0} \right)^m \right]$$

where: -  $P_f$  = probability of failure -  $\sigma$  = applied stress -  $\sigma_0$  = characteristic strength (63.2% failure probability) -  $m$  = Weibull modulus (shape parameter)

**Weibull Modulus Interpretation:** -  $m = 5-10$ : High scatter (porous ceramics, large flaws) -  $m = 10-15$ : Typical structural ceramics -  $m = 15-25$ : High-quality ceramics (fine grain, minimal flaws) -  $m \rightarrow \infty$ : No scatter (ideal deterministic material)

**Example:** Alumina with  $\sigma_0 = 400$  MPa,  $m = 12$

At 300 MPa:  $P_f = 1 - \exp[-(300/400)^{12}] = 0.09$  (9% failure probability) At 400 MPa:  $P_f = 0.632$  (63.2% by definition) At 500 MPa:  $P_f = 0.96$  (96% failure probability)

Design stress must account for this variability (safety factor 3-5 typical).

### Fracture Toughness

**Definition:** Resistance to crack propagation.

**Measurement:**  $K_{IC}$  (mode I critical stress intensity factor)

**Typical Values:** - Glass: 0.7-0.8 MPa $\sqrt{\text{m}}$  - Porcelain: 1.0 MPa $\sqrt{\text{m}}$  - Alumina: 3-5 MPa $\sqrt{\text{m}}$  - Silicon carbide: 4-5 MPa $\sqrt{\text{m}}$  - Silicon nitride: 6-8 MPa $\sqrt{\text{m}}$  - Zirconia (PSZ): 8-12 MPa $\sqrt{\text{m}}$  - Metals (comparison): 50-100+ MPa $\sqrt{\text{m}}$

**Griffith Fracture Criterion:**

$$\sigma_f = K_{IC} \sqrt{\frac{1}{\pi a}}$$

where  $a$  is flaw size.

**Example:** Alumina ( $K_{IC} = 4 \text{ MPa}\sqrt{\text{m}}$ ) with 100  $\mu\text{m}$  flaw:

$$\sigma_f = 4\sqrt{\frac{1}{\pi \times 0.0001}} = 225 \text{ MPa}$$

Same alumina with 10  $\mu\text{m}$  flaw (better processing):

$$\sigma_f = 712 \text{ MPa}$$

**Key Insight:** Strength inversely proportional to  $\sqrt{\text{flaw size}}$ . High-quality processing (minimize flaws) critical for high strength.

## Hardness

**Vickers Hardness Test:** Diamond pyramid indenter, square impression.

$$HV = \frac{1.854F}{d^2}$$

where: -  $F$  = applied load (kg) -  $d$  = diagonal length (mm)

**Typical Values:** - Macor: 250 HV - Alumina (99%): 1800-2100 HV - Silicon carbide: 2500 HV - Boron carbide: 2900 HV - Cubic boron nitride: 4500 HV - Diamond: 10,000 HV

**Relationship to Wear:** Harder ceramics resist abrasive wear. For cutting tools and wear parts, hardness critical.

## Compressive vs Tensile Strength

**Compressive Strength:** Typically 10-15x tensile strength

Example (99% alumina): - Flexural strength: 400 MPa - Compressive strength: 4000 MPa

**Design Implication:** Design ceramic components to load in compression, not tension.

## Thermal Properties

### Thermal Expansion

**Linear Coefficient (CTE):**

$$\Delta L = L_0 \alpha \Delta T$$

**Values** ( $\times 10^{-6} / ^\circ\text{C}$ ): - Fused silica: 0.5 (excellent thermal shock resistance) - Borosilicate glass: 3.3 - Alumina: 8.0 - Zirconia: 10.5 - Silicon carbide: 4.3 - Silicon nitride: 3.2 - Steel (comparison): 11-13

**Thermal Shock Resistance:** Low CTE  $\rightarrow$  better thermal shock resistance.

**Thermal Shock Parameter:**

$$R = \frac{\sigma_f(1 - LFu)}{\alpha E}$$

Higher  $R$  = better thermal shock resistance.

Materials with low  $\alpha$  and high  $\sigma_f$  resist thermal shock.

## Thermal Conductivity

**Values** (W/(m\*K)): - Zirconia: 2 (insulator - thermal barriers) - Alumina: 30 - Silicon nitride: 30 - Silicon carbide: 120 (semi-metallic) - Aluminum nitride: 170 (exceptional for electrical insulator) - Diamond: 2000 (highest known) - Copper (comparison): 400

**Machining Significance:** High thermal conductivity ceramics (SiC, AlN) dissipate cutting heat better -> easier to machine than alumina or zirconia.

## Electrical Properties

### Electrical Resistivity

**Insulators** (Ohms\*cm): - Alumina:  $10^{14}$  - Silicon nitride:  $10^{14}$  - Macor:  $10^{16}$

**Semiconductors:** - Silicon carbide:  $10$ - $10^5$  (varies with doping)

**Ionic Conductors:** - Stabilized zirconia:  $10^{-2}$  to  $10^{-3}$  (temperature dependent, oxygen ion conductor)

### Dielectric Properties

**Relative Permittivity** (epsilon\_r): - Fused silica: 3.8 - Alumina: 9.8 - Silicon nitride: 7.5

**Dielectric Strength** (kV/mm): - Alumina: 15-30

High dielectric strength -> high-voltage insulators.

## Material Selection for CNC Machining

**Easiest to Machine:** 1. Macor (carbide tools, machines like metal) 2. Soft ceramics (low-fire porcelain, steatite green state) 3. Green state engineering ceramics (before sintering)

**Moderate Difficulty:** 1. Alumina (85-95%) 2. Glass-ceramics 3. Fired porcelain

**Most Difficult:** 1. Silicon carbide 2. Boron carbide 3. Silicon nitride 4. High-purity alumina (99.9%)

**Key Factors:** - Hardness: Harder = more tool wear - Toughness: Lower toughness = more chipping risk - Thermal conductivity: Higher = easier heat removal from cutting zone

## Summary

Ceramic materials offer extreme hardness, high-temperature capability, and corrosion resistance unmatched by metals or polymers. However, brittleness and processing difficulty require specialized CNC machining approaches.

Material selection depends on: - Required properties (strength, hardness, temperature, electrical) - Machinability requirements (green vs fired) - Cost constraints - Production volume

Understanding ceramic science enables informed decisions about tooling, parameters, and quality expectations for CNC processing.

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**Next:** 17.4 Machining Composites - Cutting Mechanics



## 17.4 Machining Composites - Cutting Mechanics and Tooling

### Composite Machining Challenges

#### Unique Characteristics

**Anisotropic Material:** - Properties vary with fiber direction - Cutting forces different parallel vs perpendicular to fibers - Different failure modes in each direction

**Heterogeneous Structure:** - Two distinct phases (fiber + matrix) - Different hardness and ductility - Interface between phases creates challenges

**Abrasive Fibers:** - Carbon fiber: Hardness 2000-3000 HV (comparable to hardened steel) - Glass fiber: Hardness 500-600 HV (harder than aluminum) - Silicon carbide fiber: Hardness 2500 HV - Rapid tool wear on conventional tooling

**Low Thermal Conductivity:** - Polymer matrix:  $0.2\text{--}0.5 \text{ W/(mK)}$  - Cf. *aluminum*:  $200 \text{ W/(mK)}$  (400x better) - Heat concentrates at cutting zone -> matrix degradation

#### Cutting Mechanisms

**Fiber-Dominated Cutting** (parallel to fibers): - Fibers bend ahead of cutting edge - Shear failure of fibers - Matrix supports fibers - Good surface finish possible

**Matrix-Dominated Cutting** (perpendicular to fibers): - Matrix cut first - Fibers exposed and cut individually - Risk of fiber pullout and delamination - Rougher surface finish

**Multi-Directional Laminates:** - Each ply cuts differently - 0 deg plies: Longitudinal cutting (smooth) - 90 deg plies: Transverse cutting (rough) - +/-45 deg plies: Intermediate behavior

**Interlaminar Cutting:** - Cutting between plies - Weak interface -> delamination risk - Critical during drilling exit and edge trimming

### Defects in Composite Machining

#### Delamination

**Definition:** Separation of plies, creating void between layers.

#### Types:

**Entry Delamination** (drilling): - Drill pushes first ply down - Bending stress exceeds interlaminar strength - First ply separates from second ply

**Exit Delamination** (drilling): - Most common and severe - Drill breakthrough pushes last ply outward - Unsupported material peels away

**Edge Delamination** (milling): - Cutting forces peel plies apart at free edge - Worse with dull tools - Climb milling reduces risk

**Mechanisms:** - Peel stress at ply interface - Exceeds Mode I fracture toughness - Propagates if not arrested

**Critical Thrust Force** (drilling):

For zero delamination:

$$F_{crit} = \pi \sqrt{\frac{8G_{IC}Et^3}{3(1-LFu^2)}}$$

where: -  $G_{IC}$  = Mode I fracture toughness (N/mm) -  $E$  = flexural modulus (MPa) -  $t$  = uncut laminate thickness (mm) -  $LFu$  = Poisson's ratio

**Example** (carbon/epoxy): -  $G_{IC} = 0.2 \text{ N/mm}$  -  $E = 70,000 \text{ MPa}$  -  $t = 3 \text{ mm}$  -  $LFu = 0.3$

$$F_{crit} = \pi \sqrt{\frac{8 \times 0.2 \times 70000 \times 27}{3 \times 0.91}} = 486 \text{ N (109 lbf)}$$

Exceed this thrust -> delamination initiates.

**Prevention:** - Sharp tools (low cutting forces) - Backup support (exit side drilling) - Reduced feed rate - Compression tooling (milling)

### Fiber Pullout

**Mechanism:** - Fiber-matrix interface fails - Fiber pulled from matrix instead of cut - Leaves protruding fiber and void

**Causes:** - Dull cutting edge - Insufficient fiber support (wrong fiber angle) - Poor fiber-matrix adhesion - Excessive cutting speed (matrix softens)

**Effect on Quality:** - Rough surface (protruding fibers) - Reduced mechanical properties - Poor cosmetic appearance - Assembly problems (mating surfaces)

**Prevention:** - Sharp tools (minimize cutting forces) - Proper cutting geometry - Support fibers during cutting (backing)

### Matrix Burning/Melting

**Problem:** Polymer matrix degrades at cutting temperature.

**Typical Matrix Limits:** - Epoxy: 150-180 degC continuous, begins degrading at 200 degC - Polyester: 120 degC continuous - PEEK: 250 degC continuous (high-temp thermoplastic)

**Causes:** - Excessive cutting speed - Dull tool (friction) - Insufficient coolant - Tool rubbing (inadequate clearance)

**Signs:** - Discoloration (brown/black) - Burned smell - Gummy residue on tool - Surface porosity (matrix vaporized)

**Prevention:** - Moderate cutting speeds (500-800 SFM typical) - Sharp tools - Adequate coolant/air blast - Proper chip evacuation

### Fuzzing

**Definition:** Loose fiber ends standing up from cut surface.

**Causes:** - Fiber bending instead of cutting (dull tool) - Improper fiber orientation relative to cutting direction - Insufficient tool support angle

**Effect:** - Poor surface finish - Reduced mechanical properties - Difficult to paint/coat - Handling hazard (fiber splinters)

**Prevention:** - Sharp tools changed frequently - Proper tool geometry - Climb milling preferred - Abrading/sanding post-machining

### Microcracking

**Definition:** Fine cracks in matrix, not visible to naked eye.

**Detection:** - Ultrasonic C-scan - Microscopic examination - Dye penetrant

**Causes:** - Thermal stress (heating/cooling cycles) - Mechanical stress (excessive cutting forces) - Impact damage from chip strikes

**Concern:** Reduces mechanical properties, moisture ingress path.

## Tool Materials for Composites

### Diamond Tooling

#### Polycrystalline Diamond (PCD):

**Structure:** Diamond particles (2-30  $\mu\text{m}$ ) sintered onto carbide substrate - Diamond layer: 0.5-1.0 mm thick - Carbide backing: Provides toughness

**Properties:** - Hardness: 8000-10,000 HV (hardest cutting tool material) - Thermal conductivity: 500-2000 W/(m\*K) (excellent heat removal) - Low friction coefficient: 0.05-0.1 - Abrasion resistance: 100x carbide in composite machining

**Grades:** - **Coarse grain** (20-30  $\mu\text{m}$ ): Toughest, rougher edge, roughing - **Medium grain** (10  $\mu\text{m}$ ): General purpose - **Fine grain** (2-5  $\mu\text{m}$ ): Sharpest edge, best finish, finishing - **Extra fine** (<2  $\mu\text{m}$ ): Finest finish, most brittle

**Tool Life:** - In carbon fiber: 50-200x longer than carbide - Example: Carbide drill = 50 holes, PCD drill = 5000+ holes

**Cost:** - 10-30x carbide cost - PCD router bit: \$200-800 vs \$20-50 carbide - Justified by tool life in production

**Limitations:** - Cannot cut ferrous metals (carbon diffuses into steel at cutting temp) - Brittle (sensitive to impact) - Cannot be resharpened in most cases ( brazed tip)

**Applications:** - Carbon fiber machining (primary choice) - Fiberglass machining - Abrasive plastics - Wood composites (MDF, particleboard)

#### Chemical Vapor Deposition (CVD) Diamond

**Structure:** Pure diamond film deposited on carbide substrate - Diamond layer: 10-30  $\mu\text{m}$  thick - Smoother than PCD (no grain boundaries)

**Advantages over PCD:** - Sharper edge (smoother structure) - Better surface finish - Lower friction

**Disadvantages:** - Thinner coating (less total diamond) - More expensive - Edge failure if coating breached

**Applications:** Precision finishing of composites

### Solid Carbide

**For Composites:** Uncoated micrograin carbide preferred - Grain size: 0.5-0.8  $\mu\text{m}$  (fine grain) - Binder: 6-10% cobalt (tough)

**Advantages:** - Lower cost than diamond (\$20-100 per tool) - Can be resharpened - Good for low-volume work

**Disadvantages:** - Wears rapidly (10-100 holes vs 5000+ for PCD) - Requires frequent changes

**Coating Not Recommended:** - TiN, TiAlN coatings wear off quickly in composites - Coating can delaminate, creating rough edge - Uncoated carbide performs better

**Application:** - Prototyping, low-volume production - When diamond not justified economically

### High-Speed Steel (HSS)

**Generally Not Recommended** for composites: - Wears extremely rapidly (5-10 holes typical) - Softens at temperatures generated in composite cutting - Poor surface finish

**Exception:** Very low-speed operations (hand drilling, countersinking)

## Tool Geometry for Composites

### Rake Angle

**Positive Rake** (5-15 deg): - Slicing action - Lower cutting forces - Cleaner cut - **Preferred for most composite machining**

**Zero to Negative Rake** (0 to -5 deg): - Stronger cutting edge (less prone to chipping) - Used for highly abrasive materials - Higher cutting forces

**Example:** PCD router bit for carbon fiber - Rake angle: +10 deg - Reduced cutting force vs negative rake - Cleaner fiber cutting

### Helix Angle

**Standard Helix** (30-35 deg): - General purpose - Good chip evacuation

**High Helix** (45-55 deg): - Better chip evacuation - Shearing cut (reduces force) - **Recommended for carbon fiber**

**Low Helix** (15-25 deg): - Stronger cutting edge - Used for highly abrasive ceramics - Not ideal for composites

### Compression Tooling

**Purpose:** Prevent delamination by compressing laminate during cutting.

**Up-Cut/Down-Cut Geometry:** - Lower section: Up-cut flute (pushes material down) - Upper section: Down-cut flute (pushes material down) - Transition: At mid-thickness of material

**Effect:** - Top surface pushed down - Bottom surface pushed down - Laminate compressed in middle -> no delamination - **Critical for through-cutting composites**

**Application:** - Edge trimming of laminates - Slotting operations - Any through-cut where both surfaces critical

**Example:** 1/4" compression end mill for 0.125" carbon fiber - Lower 0.100": Up-cut geometry - Upper 0.100": Down-cut geometry - Transition at 0.062" (mid-thickness)

### Point Angle (Drilling)

**Standard Twist Drill** (118 deg): - General purpose metal drilling - NOT ideal for composites (high thrust force)

**Low Point Angle** (90 deg or less): - Reduces thrust force - Less delamination risk - **Preferred for composite drilling**

**Brad Point / Spur Point:** - Outer spurs cut cleanly - Center point provides guidance - Excellent for composites - **Recommended for fiberglass, carbon fiber**

**Step Drill:** - Multiple diameters on one tool - Pilots small hole, then enlarges - Reduces exit delamination - Excellent for countersinking/counterboring

### Clearance Angle

**Larger Clearance** (10-15 deg): - Reduces rubbing - Important in composites (low thermal conductivity) - Prevents heat buildup

**Smaller Clearance** (5-7 deg): - Stronger edge (for ceramics) - Not necessary for composites

## Cutting Parameters for Composites

### Cutting Speed (Surface Speed)

**Carbon Fiber:** - Roughing: 500-800 SFM (150-240 m/min) - Finishing: 800-1200 SFM (240-370 m/min) - Diamond tooling: Can run up to 2000 SFM

**Fiberglass:** - Roughing: 400-700 SFM - Finishing: 700-1000 SFM

**Aramid (Kevlar):** - Very difficult to machine (fibers don't cut, they fray) - Low speed: 200-400 SFM - Sharp tools mandatory

**Formula for RPM:**

$$N = \frac{12 \times SFM}{\pi \times D} = \frac{3.82 \times SFM}{D}$$

**Example:** 1/2" PCD endmill in carbon fiber at 800 SFM

$$N = \frac{3.82 \times 800}{0.5} = 6112 \text{ RPM}$$

### Feed Rate

**Feed Per Tooth:** Lower than metals (reduce cutting forces)

| Material               | Feed per Tooth | Notes                        |
|------------------------|----------------|------------------------------|
| Carbon fiber (PCD)     | 0.002-0.006"   | Higher for roughing          |
| Carbon fiber (carbide) | 0.001-0.004"   | Lower to extend life         |
| Fiberglass             | 0.003-0.008"   | Less abrasive than carbon    |
| Aramid                 | 0.001-0.003"   | Very low to minimize fraying |

**Feed Rate Calculation:**

$$F = f_z \times Z \times N$$

**Example:** 4-flute PCD tool, 6000 RPM,  $f_z = 0.004$ "

$$F = 0.004 \times 4 \times 6000 = 96 \text{ IPM}$$

**Drilling Feed Rates:** - Carbon fiber: 2-8 IPM - Fiberglass: 3-10 IPM - Lower feed -> lower thrust force -> less delamination

### Depth of Cut

**Roughing:** - ADOC: 0.050-0.200" (depends on material thickness) - Full slot width: Avoid when possible (use trochoidal/adaptive)

**Finishing:** - ADOC: 0.010-0.030" - RDOC: 0.010-0.040" (light radial)

**Key Principle:** Multiple light passes better than single heavy pass - Reduces cutting forces - Minimizes delamination - Better surface finish - Longer tool life

**Example Strategy** (0.125" carbon fiber panel): - Pass 1: 0.100" depth (roughing) - Pass 2: 0.030" depth (finishing) Total: Two passes with light finish cut

## Coolant / Lubrication

**Compressed Air:** - Most common for composites - Blows chips away - Some cooling effect - No mess, no cleanup

**Mist Coolant:** - Better cooling than air alone - Vegetable-based oil preferred (5-10% solution) - Reduces dust - Cleaner edges

**Flood Coolant:** - Maximum cooling - Effective dust suppression - Water-based (no oil on carbon fiber - contamination risk) - Matrix absorption concern (especially for honeycomb core)

**Vacuum / Dust Collection:** - **Mandatory for health and safety** - Carbon fiber dust carcinogenic (IARC Group 2B) - HEPA filtration required - Downdraft table or tool-mounted shroud

**Dry Cutting:** - Acceptable only with excellent dust collection - Higher tool wear - Risk of matrix burning

**Recommendation:** Compressed air + dust collection for most applications

## Drilling Composites

### Drill Selection

**Best Choices:** 1. **PCD brad-point drill:** Sharpest, longest life, best quality 2. **Carbide brad-point:** Good quality, moderate life, lower cost 3. **Carbide twist drill** (90-100 deg point): Acceptable, frequent changes

**Avoid:** - Standard HSS twist drills (wear out immediately) - Step drills in carbon fiber (too much heat)

### Drilling Strategy

**Peck Drilling:** - Retract frequently (every 0.5-1.0x diameter) - Clears chips - Reduces heat buildup - **Mandatory for deep holes** (depth > 3x diameter)

**Pilot Hole:** - Drill small pilot (1/4 final diameter) - Enlarge with final drill - Reduces thrust force on final pass - **Significantly reduces exit delamination**

**Example:** 1/2" hole - Pilot: 1/8" drill - Final: 1/2" drill - Thrust force reduced ~60% on final pass

### Backup Support

#### Critical for Exit Delamination Prevention:

**Solid Backup Plate:** - Place rigid plate under workpiece (wood, aluminum) - Drill through workpiece into backup - Backup supports last ply during breakthrough - **Most effective method**

**Sacrificial Layer:** - Tape thin aluminum foil or plastic film to exit side - Provides minimal support - Better than nothing, not as good as solid backup

**Tape Method:** - Apply packing tape to exit side - Helps hold fibers down - Minimal support - Quick/easy for field repairs

**Two-Sided Drilling** (best quality): - Drill from one side until point just breaks through - Flip workpiece - Drill from other side, using breakthrough as center - Both sides are "entry" (no exit delamination) - Time-consuming but highest quality

## Routing / Milling Composites

### Edge Trimming

**Flush Trim Bit** (bearing-guided): - Bearing rides on template or finished edge - Carbide or PCD cutting edge - Up-cut, down-cut, or compression geometry

**Compression Router:** - Ideal for composite laminates - Prevents top and bottom surface delamination  
- Requires tool length > material thickness

**Tabbed Parts:** - Leave small tabs connecting part to sheet - Route all edges leaving tabs - Remove tabs by hand (sanding, filing) - Prevents part from shifting during final cut

### Contouring

**Climb Milling** (down-milling): - Cutting force pushes part onto table - Cleaner edge (fiber cutting vs pulling) - **Preferred for composites** - Requires rigid fixturing

**Conventional Milling** (up-milling): - Cutting force lifts part from table - More fiber pullout risk - Use only if climb milling causes vibration

**Corner Radius:** - Composites hate sharp internal corners (stress concentration) - Minimum radius: 2-3x material thickness - Larger radius reduces stress concentration

### Slotting

**Full Slot Milling** (slot width = tool diameter): - High radial engagement - High cutting forces - Short tool life - Risk of delamination

**Better: Trochoidal Milling:** - Small radial engagement (10-20% diameter) - Circular arc paths overlapping to create slot - Lower forces, longer tool life - Smoother edges

**Example:** 0.500" slot with 0.500" endmill - Traditional: Full slot, RDOC = 0.500" - Trochoidal: Multiple passes, RDOC = 0.050-0.100" each - 5-10x longer tool life with trochoidal

## Waterjet and Laser Cutting

### Abrasive Waterjet

**Process:** High-pressure water (40,000-90,000 PSI) + abrasive (garnet) cuts material.

**Advantages for Composites:** - No heat (cold cutting) - No delamination - No tool wear - Any shape possible - Cuts thick laminates easily (up to 6"+)

**Disadvantages:** - Slow (2-10 IPM typical) - Rough edge (taper, striation marks) - Expensive equipment (\$50,000-300,000) - Abrasive disposal/mess

**Edge Quality:** - Kerf taper: 0.002-0.010" per inch thickness - Surface roughness: Ra 100-300  $\mu\text{in}$  (rougher than machining) - May require post-machining for precision edges

**Applications:** - Rough cutting blanks - Prototyping - One-off parts - Thick laminates (>1")

### Laser Cutting

**CO<sub>2</sub> Laser** (10.6  $\mu\text{m}$  wavelength): - Absorbed well by polymer matrix - Cuts carbon fiber, fiberglass, aramid - Heat-affected zone (HAZ): 0.010-0.050" - Matrix charring/burning

**Fiber Laser** (1.06  $\mu\text{m}$  wavelength): - Absorbed poorly by most polymers - Better for metals - Not ideal for composites

**Advantages:** - Fast (100-500 IPM) - Narrow kerf (0.005-0.015") - No tool wear - Complex shapes

**Disadvantages:** - Heat-affected zone (matrix damage) - Charred edges (cosmetic issue) - Fiber ends exposed (fuzzing) - Toxic fumes (polymer pyrolysis products)

**Applications:** - Thin laminates (<0.125") - Non-structural parts (cosmetic OK) - Rapid prototyping - When edge quality not critical

## Cutting Method Comparison

| Method                | Speed     | Edge Quality | Delamination Risk   | Cost     | Best For                |
|-----------------------|-----------|--------------|---------------------|----------|-------------------------|
| CNC Routing (PCD)     | Fast      | Excellent    | Low (if done right) | Medium   | Production, precision   |
| CNC Routing (carbide) | Fast      | Good         | Moderate            | Low      | Low-volume              |
| Abrasive Waterjet     | Slow      | Fair         | None                | High     | Thick, prototypes       |
| Laser                 | Very Fast | Poor (HAZ)   | None                | High     | Thin, non-structural    |
| Hand Tools (diamond)  | Very Slow | Fair         | Moderate            | Very Low | Field repairs, one-offs |

## Special Considerations

### Sandwich Structures

**Core Types:** - Foam (PVC, PU, PMI): Soft, crushes easily - Honeycomb (aluminum, Nomex): Hollow cells, delicate - Balsa: Natural wood, grain direction matters

**Challenges:** - Core crushes under clamping pressure - Core pulls away from skin (debonding) - Chip packing in honeycomb cells

**Drilling:** - Use brad-point drill (prevents walking) - Low feed rate (prevent crushing) - Clear chips frequently (peck cycle) - Solid backup mandatory

**Edge Routing:** - Compression router essential - Core exposed at edge -> structural weakness - Edge sealing required after machining

**Facing (Surfacing Core):** - Very light depth of cut (0.010-0.020") - Sharp tools only - Balsa: Cut with grain when possible

### Honeycomb Core

**Specific Issues:** - Cells collapse under point loads - Chips pack into cells (difficult to remove) - Thin cell walls tear easily

**Potting Compound:** - Fill cells with epoxy/microsphere mixture - Cure before drilling - Provides support during drilling - Required for fastener holes (prevents pull-through)

**Honeycomb Router Bits:** - Specialized geometry - Multiple close-spaced flutes - Shears cell walls cleanly - PCD tipped

### Cured vs Uncured

**Green (Uncured) Machining:** - Composite laid up but not cured - Soft, easy to cut with conventional tools - Allows complex features - Must cure after machining (shrinkage, distortion risk)

**Post-Cure Machining:** - Composite fully cured (hard) - Requires diamond tooling - Accurate final dimensions - Standard practice for production parts

## Summary

Machining composites requires specialized knowledge:



**Key Principles:** 1. Sharp tools mandatory (dull tools cause delamination) 2. Diamond tooling for production (PCD primary choice) 3. Compression geometry prevents delamination 4. Climb milling preferred 5. Dust collection non-negotiable (health hazard) 6. Backup support for drilling 7. Lower cutting forces than metals (sharp tools, light cuts)

**Tool Life:** - PCD: 50-200x carbide - Carbide: 10-50x HSS - Justifies diamond tooling cost in production

**Quality Defects:** - Delamination (most critical) - Fiber pullout (surface finish) - Matrix burning (heat damage) - Fuzzing (loose fibers)

**Next:** Advanced techniques and ceramic machining strategies

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**Next:** 17.5 Machining Ceramics - Diamond Grinding and Special Processes

## 17.5 Machining Ceramics - Diamond Grinding and Special Processes

### Ceramic Machining Overview

#### Why Ceramics Are Difficult to Machine

**Extreme Hardness:** - Alumina: 1800-2100 HV (approaching diamond) - Silicon carbide: 2500 HV - Boron carbide: 2900 HV - Compare: Hardened steel 800-900 HV

**Brittleness:** - Low fracture toughness (3-12 MPa $\sqrt{\text{m}}$  vs 50-100 for steel) - Cannot deform plastically - Chip or crack instead of bending - Edge chipping primary concern

**Low Thermal Shock Resistance:** - Most ceramics sensitive to rapid temperature changes - Intermittent cutting with coolant causes thermal cycling - Can lead to crack propagation

**Grinding Dominant Process:** - Conventional machining (turning, milling) very limited - Diamond grinding primary method - Material removed as tiny chips (microchipping)

### Green Machining vs Fired Machining

#### Green State Machining

**Material State:** Pressed powder, not yet sintered - Consistency: Like hard chalk or graphite - Strength: 5-20 MPa (very weak) - Machinability: Excellent (conventional tools work)

**Advantages:** - Fast material removal (100-500x faster than fired) - Conventional carbide tooling - Complex features easily created - Low cost

**Challenges:** - Fragile (handle carefully) - Sintering shrinkage: 15-20% linear typical - Must oversize features to compensate - Dimensional changes during firing (warping possible)

#### Shrinkage Calculation:

Final dimension after firing:

$$D_{\text{fired}} = D_{\text{green}} \times (1 - S)$$

where  $S$  = linear shrinkage fraction

**Example:** - Target fired dimension: 2.000" - Shrinkage: 18% (0.18) - Required green dimension:  $2.000 / (1 - 0.18) = 2.439$ "

**Shrinkage Variation:** - Typical tolerance:  $\pm 0.5\%$  of dimension - 2.000" part:  $\pm 0.010$ " variation possible - Requires fired machining for tight tolerances

**Process Flow:** 1. Green machine (oversize) 2. Sintering (shrinks, hardens) 3. Light fired grinding (final dimensions)

**Tool Materials for Green Machining:** - HSS: Acceptable for very soft greens - Carbide: Standard choice - Diamond: Unnecessary expense

#### Fired State Machining

**Material State:** Fully sintered, densified ceramic - Hardness: Near-maximum for material - Density: 95-99.9% theoretical - Machinability: Very difficult

**Only Option When:** - Features must be added post-sintering - Tolerances tighter than firing can achieve - Green machining not possible (pre-sintered blanks)

**Process:** Diamond grinding almost exclusively

**Challenges:** - Very slow (0.0001-0.001" per pass typical) - Expensive tooling (diamond wheels) - Edge chipping risk - Requires precision machine (grinding center)

**Cost Impact:** - Fired grinding: \$50-300/hour (equipment + labor + tooling) - Green machining: \$30-80/hour - 10-100x time difference for same material removal

**Economic Strategy:** Maximize green machining, minimize fired grinding

## Diamond Grinding

### Grinding Wheel Specifications

**Notation:**

D 120 N 100 B

| | | | +- Bond type (B = resin, V = vitrified, M = metal)

| | | +----- Concentration (25-200, higher = more diamond)

| | +----- Grade (hardness: A-Z, soft to hard)

| +----- Grit size (mesh number, higher = finer)

+----- Abrasive type (D = diamond, B = CBN)

**Example:** D 180 M 150 B - Diamond abrasive - 180 grit (fine, ~80  $\mu\text{m}$  particles) - Medium grade - 150 concentration (medium) - Resin bond

### Grit Size Selection

| Grit      | Particle Size ( $\mu\text{m}$ ) | Application     | Surface Finish ( $R_a$ )                       |
|-----------|---------------------------------|-----------------|------------------------------------------------|
| 80-120    | 125-180                         | Rough grinding  | 100-200 $\mu\text{in}$ (2.5-5 $\mu\text{m}$ )  |
| 150-220   | 63-100                          | General purpose | 50-100 $\mu\text{in}$ (1.2-2.5 $\mu\text{m}$ ) |
| 320-400   | 38-45                           | Fine grinding   | 20-40 $\mu\text{in}$ (0.5-1 $\mu\text{m}$ )    |
| 600-1200  | 15-25                           | Finishing       | 5-15 $\mu\text{in}$ (0.12-0.4 $\mu\text{m}$ )  |
| 2000-8000 | 3-8                             | Polishing       | <5 $\mu\text{in}$ (<0.12 $\mu\text{m}$ )       |

**Selection Guide:** - **Roughing:** 80-150 grit (fast stock removal, rougher finish) - **Finishing:** 220-600 grit (balance removal and finish) - **Polishing:** 1200+ grit (mirror finish)

**Example Sequence** (alumina part): 1. Rough grind: 120 grit, remove 0.020" ->  $R_a$  100  $\mu\text{in}$  2. Finish grind: 320 grit, remove 0.005" ->  $R_a$  25  $\mu\text{in}$  3. Polish: 1200 grit, remove 0.001" ->  $R_a$  5  $\mu\text{in}$

### Bond Type

**Resin Bond** (phenolic): - Resilient, slight cushioning - Good surface finish - Wears faster (free-cutting) - **Best for general ceramic grinding** - Cost: Moderate

**Vitrified Bond** (glass): - Rigid, precise - Excellent form retention - Requires dressing - High-precision applications - Cost: Higher

**Metal Bond** (bronze, sintered): - Most durable - Highest concentration possible - Requires electrolytic dressing (complex) - Production grinding - Cost: Highest

**For DIY/Small Shop:** Resin bond, 150-320 grit most versatile

### Concentration

**Definition:** Volume fraction of diamond in bond (carats per  $\text{cm}^3$ )

| Concentration | Diamond Content | Application                    |
|---------------|-----------------|--------------------------------|
| 25-50         | Low             | Soft materials, aggressive cut |
| 75-125        | Medium          | General purpose, most ceramics |
| 150-200       | High            | Hard materials, finishing      |

**Higher Concentration:** - More diamond particles in cutting - Longer wheel life - Finer finish - More expensive

**Typical:** 100-150 concentration for alumina, SiC

### Grade (Wheel Hardness)

**Soft Grade (E-J):** - Bond releases dull diamond particles easily - Self-sharpening - Soft, gummy materials (aluminum)

**Medium Grade (K-P):** - General purpose - **Most ceramics ground with M-N grade**

**Hard Grade (Q-Z):** - Retains particles longest - Hard, brittle materials - Requires manual dressing more frequently

**Rule of Thumb:** Harder workpiece -> softer wheel (self-dresses)

## Grinding Parameters

### Surface Grinding

**Wheel Speed:** 4000-6000 SFPM (20-30 m/s) - Diamond wheels run slower than aluminum oxide - Higher speeds increase heat, risk thermal damage

**Table Speed:** 30-100 FPM - Slower for roughing (more contact time) - Faster for finishing (lighter cuts)

**Depth of Cut:** - Roughing: 0.0005-0.002" per pass - Finishing: 0.0001-0.0005" per pass - **Much lighter than metal grinding**

**Crossfeed (stepover):** - 50-80% of wheel width - Overlap ensures uniform finish

**Coolant:** - Water-based, 5-10% concentration - Flood application (5-10 GPM minimum) - Prevents thermal damage - Flushes chips (sludge)

**Example Roughing Pass:** - Material: 99% alumina - Wheel: D 120 M 100 B, 6" diameter x 1/2" wide - Wheel speed: 5000 SFPM -> 3185 RPM - Table speed: 50 FPM - Depth of cut: 0.001" - Crossfeed: 0.350" (70% wheel width) - Coolant: 7 GPM flood

**Material Removal Rate:**

$$MRR = \text{table speed} \times \text{crossfeed} \times \text{DOC}$$

$$MRR = 50 \times 0.350 \times 0.001 = 0.0175 \text{ in}^3/\text{min}$$

Compare to metal grinding: 0.5-5 in<sup>3</sup>/min (30-300x faster!)

### Cylindrical Grinding

**Work Speed:** 50-150 SFPM (workpiece rotation) - Slower than metal grinding

**Wheel Speed:** 4000-6000 SFPM

**Feed Rate:** 0.1-0.5x wheel width per revolution - Example: 1/2" wheel -> 0.050-0.250" per rev

**Plunge Rate:** 0.0001-0.0005" per pass

**Spark-Out Passes:** 2-5 passes at zero depth - Allows wheel/work deflection to recover - Improves dimensional accuracy - Reduces residual stress

## Centerless Grinding

**Advantages for Ceramics:** - No centerpoint (no fragile ends to support) - Continuous processing - High production volume

**Setup:** - Grinding wheel (driven) - Regulating wheel (controls work rotation, feed) - Workrest blade (supports work)

**Applications:** - Ceramic rods, shafts - Bearings (alumina, zirconia) - Seal faces

**Challenges:** - Complex setup (wheel angles, position) - Lobing (out-of-roundness) can occur

## Internal Grinding

**Small Hole Challenge:** - Hole diameter: 0.125-1.0" typical - Wheel diameter: 80-90% of hole ID - Very small wheel -> less rigid, more deflection

**Spindle Speed:** Very high (10,000-40,000 RPM) - Small diameter requires high RPM for proper surface speed - Example: 0.500" wheel, 5000 SFPM -> 38,200 RPM

**Reciprocating Motion:** - Wheel moves in/out of hole - Prevents wheel loading - Distributes wear

**Challenges:** - Wheel deflection (poor roundness) - Wheel loading (diamond particles covered with debris) - Chatter (small wheel less rigid)

**Solution:** Frequent wheel dressing, rigid machine

## Ultrasonic Machining (USM)

### Process

**Mechanism:** - Tool vibrates at 20-40 kHz (ultrasonic frequency) - Amplitude: 0.001-0.003" (25-75  $\mu\text{m}$ ) - Abrasive slurry (water + boron carbide, silicon carbide, diamond powder) - Abrasive particles hammered into workpiece by vibrating tool - Material removed by microchipping

**Tool Material:** Soft metal (brass, mild steel) - Tool doesn't cut; it transmits vibration - Wears slowly (abrasive particles do cutting)

### Setup:

Transducer (piezoelectric) -> Horn (amplitude amplifier) -> Tool  
down  
Abrasive slurry  
down  
Ceramic workpiece

### Advantages

- Complex shapes possible (tool shape copied into workpiece)
- No cutting forces (gentle process)
- No heat generated
- Can machine any hard, brittle material

### Disadvantages

- Very slow (0.001-0.01 in<sup>3</sup>/min)
- Tool wear (must periodically redress)
- Expensive equipment (\$20,000-100,000)
- Limited to small parts/features

## Applications

- Complex holes (square, hexagonal, shaped)
- Very hard ceramics (boron carbide, alumina)
- Fragile parts (thin walls)
- Prototype parts

**Example:** - Hole: 0.250" square, 0.500" deep in alumina - Machining time: 2-4 hours - Alternative (drilling): Not possible (square hole)

## Laser Machining

### CO<sub>2</sub> Laser

**Wavelength:** 10.6  $\mu\text{m}$  (far infrared)

**Absorption:** Poor for most ceramics (transparent or reflective) - Alumina: Poorly absorbed - Zirconia: Poorly absorbed - Silicon carbide: Moderate absorption

**Result:** Not effective for most technical ceramics

### Nd:YAG / Fiber Laser

**Wavelength:** 1.06  $\mu\text{m}$  (near infrared)

**Better Absorption:** Many ceramics absorb better at this wavelength

**Mechanism:** - Localized melting/vaporization - Thermal stress  $\rightarrow$  microcracking - Material spallation (chunks flake off)

**Advantages:** - No tool wear - Complex 2D shapes - Fast for thin materials

**Disadvantages:** - Heat-affected zone (HAZ)  $\rightarrow$  microcracks - Reduced mechanical properties near cut - Taper on through-cuts - Limited to thin sections ( $<0.125"$ )

**Applications:** - Trim cuts on thin ceramic substrates - Scribing (partial depth cuts for breaking) - Prototyping

**Not Recommended For:** Structural parts (cracks reduce strength)

## Electrical Discharge Machining (EDM)

### Conductive Ceramics Only

**Requirement:** Material must be electrically conductive - Silicon carbide: Conductive (can EDM) - Titanium carbide: Conductive - Graphite: Conductive - **Alumina, zirconia, silicon nitride:** Insulators (cannot EDM)

### Process (for Conductive Ceramics)

**Mechanism:** - Electrical discharge (spark) between electrode and workpiece - Localized melting/vaporization - Material removed in tiny craters

**Electrode:** Copper or graphite - Wears slowly - Negative shape of desired feature

**Dielectric Fluid:** Deionized water or oil - Flushes debris - Cools workpiece

**Advantages:** - Complex 3D shapes - No cutting forces (gentle) - Excellent for hard ceramics (hardness irrelevant)

**Disadvantages:** - Very slow (0.001-0.1 in<sup>3</sup>/hr depending on material) - Only conductive ceramics - Expensive equipment - Altered surface layer (recast layer with microcracks)

**Applications:** - Dies and punches (silicon carbide) - Cutting tool inserts (titanium carbide) - Complex shapes in conductive ceramics

## Abrasive Waterjet

### Process

High-pressure water (40,000-90,000 PSI) + abrasive (garnet, aluminum oxide) cuts by erosion.

### Advantages for Ceramics

- No heat (cold cutting)
- No mechanical forces (gentle)
- Any material (no hardness limit)
- Complex 2D shapes
- Thick sections possible (up to 6"+)

### Disadvantages

- Slow (1-5 IPM in ceramics)
- Rough edge (Ra 100-300  $\mu$ in)
- Kerf taper (0.005-0.020" per inch thickness)
- Surface microcracking possible
- Expensive equipment and operating costs

### Edge Quality

**Top Surface:** Clean (entry)

**Bottom Surface:** Ragged (exit) - Jet loses energy through thickness - Last material abraded less aggressively

**Taper:** - Kerf wider at top than bottom - 2-5 deg taper typical - Reduces with slower speeds (more expensive)

### Applications

- Rough cutting blanks
- Prototyping
- Artistic/decorative (edge quality not critical)
- Thick plates (where grinding impractical)

**Post-Machining:** Often requires diamond grinding on critical edges

## Lapping and Polishing

### Lapping

**Process:** Loose abrasive (slurry) on flat lap plate - Lap plate: Cast iron, copper, or glass - Abrasive: Diamond paste, aluminum oxide, silicon carbide - Part pressed onto rotating lap with light force

**Removes:** 0.0001-0.001" material

**Achieves:** - Very flat surfaces (flatness 0.0001" possible) - Good surface finish (Ra 5-20  $\mu$ in) - Parallel surfaces

**Grit Progression:** 1. Coarse lap: 30-60  $\mu$ m grit, remove 0.001-0.005" 2. Fine lap: 9-15  $\mu$ m grit, remove 0.0005" 3. Polishing: 1-3  $\mu$ m grit, remove 0.0001"

**Lap Maintenance:** - Re-flatten periodically (lap wears) - Use three-plate method (plates lap each other flat)

**Applications:** - Gage blocks - Optical flats - Mechanical seal faces - Precision spacers

## Polishing

**Objective:** Mirror finish, minimal subsurface damage

**Process:** Similar to lapping but finer abrasives - Polishing cloth (neoprene, felt, polyurethane) - Diamond paste: 0.25-3  $\mu\text{m}$  - Colloidal silica: 0.05  $\mu\text{m}$  (final polish)

**Results:** - Surface finish:  $R_a < 5 \mu\text{in}$  ( $< 0.12 \mu\text{m}$ ) - Mirror-like appearance - Minimal subsurface damage ( $< 1 \mu\text{m}$  depth)

**Applications:** - Optical components - Metallographic samples (microscopy) - Biomedical implants (ultra-smooth for tissue contact)

**Example Polishing Sequence** (alumina): 1. 9  $\mu\text{m}$  diamond on hard cloth, 5 min  $\rightarrow R_a 15 \mu\text{in}$  2. 3  $\mu\text{m}$  diamond on medium cloth, 5 min  $\rightarrow R_a 8 \mu\text{in}$  3. 1  $\mu\text{m}$  diamond on soft cloth, 10 min  $\rightarrow R_a 4 \mu\text{in}$  4. 0.05  $\mu\text{m}$  colloidal silica, 15 min  $\rightarrow R_a 2 \mu\text{in}$  (mirror)

## Machining-Induced Damage

### Subsurface Cracks

**Mechanism:** Grinding induces stress  $\rightarrow$  microcracks beneath surface

**Depth:** 1-50  $\mu\text{m}$  depending on grinding conditions

**Effect on Strength:** Can reduce strength 20-50% - Cracks act as stress concentrators - Propagate under load  $\rightarrow$  fracture

**Mitigation:** - Finer grit (distributes stress) - Lighter cuts (less force) - Sharp wheel (dull wheel crushes more) - Final stress relief (low-temp anneal)

### Residual Stress

**Tensile Stress** (bad): - Surface in tension - Cracks open more easily - Reduces strength

**Compressive Stress** (good): - Surface in compression - Resists crack opening - Increases strength

**Grinding Usually Creates Tension:** - Heat generation  $\rightarrow$  thermal expansion  $\rightarrow$  quenching  $\rightarrow$  tension

**Solutions:** 1. **Annealing:** Heat to 50-70% of sintering temp, slow cool - Relieves residual stress - 99% alumina: Anneal at 1000-1200  $^{\circ}\text{C}$

2. **Shot Peening:** Bombard surface with small balls

- Induces compressive stress
- Can double strength

3. **Optimize Grinding:** Cooler = less stress

- Sharp wheels
- Light cuts
- Flood coolant

### Surface Roughness Effects

#### Strength vs Roughness:

Rough surface ( $R_a 100 \mu\text{in}$ ) vs smooth ( $R_a 10 \mu\text{in}$ ): - Rough surface: Deeper scratches act as crack initiation sites - Strength reduction: 30-50%

**Polished Ceramic:** - Highest strength (fewer/smaller flaws) - Required for high-stress applications



## Macor Machining (Special Case)

### Why Macor is Different

**Machinable Glass-Ceramic:** Can be machined with carbide tools (no diamond required!)

**Mechanism:** Mica crystals act as chip breakers - Prevent crack propagation - Allow conventional cutting

**Hardness:** 67 GPa (much softer than engineering ceramics)

### Machining Parameters

**Turning:** - Speed: 300-600 SFM - Feed: 0.002-0.010 IPR - Depth: 0.020-0.100" - Tool: Carbide insert (TCMT, CCMT)

**Milling:** - Speed: 200-500 SFM - Feed per tooth: 0.001-0.005" - Depth: 0.050-0.200" - Tool: Carbide endmill (2-4 flute)

**Drilling:** - Speed: 200-500 SFM - Feed: 2-8 IPM - Peck drilling for deep holes - Tool: Carbide twist drill

**Coolant:** Optional - Can machine dry - Coolant improves finish, extends tool life

**Tool Life:** Similar to brass - 500-2000 parts typical (depending on operation) - Much better than ceramics, worse than aluminum

### Achievable Tolerances

- Diameter: +/-0.001" easily
- Flatness: 0.001" per inch
- Surface finish: Ra 32-63 muin standard, Ra 8-16 muin with fine finish pass

### Applications

- Vacuum feedthroughs (zero porosity)
- Electrical insulators (high voltage)
- Precision fixtures and jigs
- Prototypes (machine quickly, test, then replicate in harder ceramic)

## Cost Comparison

### Green + Fire + Grind (Standard Process)

**Example:** 2" diameter x 0.500" thick alumina disc, +/-0.001" tolerance

1. **Green machining:** \$50 (fast)
2. **Sintering:** \$30 (batch process, many parts)
3. **Finish grinding:** \$150 (slow, diamond wheel)
4. **Total:** \$230 per part

### Fire + Extensive Grinding (No Green Machining)

**Same part,** starting from fired blank:

1. **Rough grinding:** \$200
2. **Finish grinding:** \$150
3. **Total:** \$350 per part

**Savings:** \$120 per part (35%) with green machining

## Machining Cost Drivers

1. **Time** (dominant): Ceramic grinding = 10-100x slower than metals
2. **Tooling**: Diamond wheels \$50-500 each, wear out
3. **Equipment**: Precision grinders \$50,000-500,000
4. **Setup**: Fixturing brittle parts requires care
5. **Scrap**: Cracked parts = total loss

## Economic Strategies

**Maximize Green Machining**: - Do as much as possible before firing - Compensate for shrinkage - Accept fired grinding only for critical features

**Net Shape Sintering**: - Mold to near-final shape - Minimize all machining (green and fired) - Requires expensive tooling (justified for high volume)

**Batch Processing**: - Grind multiple parts together - Magnetic chuck with array of parts - Amortize setup time

## Safety Considerations

### Ceramic Dust

**Hazards**: - **Silica** (in many ceramics): Silicosis (lung disease) - **Aluminum oxide**: Respiratory irritant - **Fine particles**: <10  $\mu\text{m}$  respirable (deep lung penetration)

**Exposure Limits** (OSHA PEL): - Crystalline silica:  $0.05 \text{ mg}/\text{m}^3$  (8-hour TWA) - Aluminum oxide:  $15 \text{ mg}/\text{m}^3$  (total dust)

**Controls**: - Wet grinding (dust suppression) - Local exhaust ventilation (LEV) - HEPA filtration - Respiratory protection (N95 minimum for dry processes)

### Grinding Wheel Safety

**Burst Hazard**: Diamond wheels can break - Overspeed (exceed rated RPM) -> centrifugal failure - Impact damage -> weakened wheel - Improper mounting -> stress concentration

**Prevention**: - **Ring test before mounting**: Tap wheel, listen for clear ring (not dull thud = crack) - Never exceed rated RPM - Use wheel guards - Wear face shield

**Maximum RPM**:

$$N_{max} = \frac{12 \times SFPM_{max}}{\pi \times D}$$

**Example**: 6" wheel, rated 6500 SFPM max

$$N_{max} = \frac{12 \times 6500}{\pi \times 6} = 4138 \text{ RPM}$$

Running at 5000 RPM -> 21% overspeed -> UNSAFE

### Handling Ceramic Parts

**Brittle Fracture**: Drop from 6" can shatter part

**Handling**: - Soft gloves (reduce grip stress concentration) - Padded work surfaces - Store in compartmented trays (not loose in box)

**Cleaning**: Avoid ultrasonic on thin/complex parts (vibration can crack)

## Summary

Machining ceramics requires specialized processes:

**Key Methods:** 1. **Green machining:** Fast, conventional tools, before sintering 2. **Diamond grinding:** Slow, expensive, after sintering (primary method) 3. **Ultrasonic machining:** Complex shapes, very slow 4. **Lapping/polishing:** Flat, smooth surfaces

**Critical Factors:** - Very slow material removal (0.0001-0.001" per pass) - Diamond tooling mandatory (except green state) - Light cuts essential (prevent chipping/cracking) - Coolant critical (thermal shock prevention)

**Cost Drivers:** - Time (10-100x slower than metals) - Tooling (diamond wheels expensive) - Scrap risk (parts crack easily)

**Strategy:** Maximize green machining, minimize fired grinding

**Next:** Dust collection and safety systems for advanced materials

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**Next:** 17.6 Dust Collection and Safety Systems

## 17.6 Dust and Fume Control Systems

### Why Advanced Materials Require Special Dust Control

#### Health Hazards of Composite and Ceramic Dust

**Carbon Fiber Dust:** - Particle size: 5-10  $\mu\text{m}$  (respirable, reaches deep lungs) - IARC classification: Group 2B (possibly carcinogenic) - Fiber geometry: Length  $\gg$  diameter (similar to asbestos concern) - Irritant: Skin, eyes, respiratory tract - Conductivity: Can cause electrical shorts in equipment

**Glass Fiber (GFRP):** - Acute: Skin irritation (“itching”), eye irritation - Chronic: Possible lung scarring (fibrosis) - Diameter: 5-15  $\mu\text{m}$  fibers respirable - Less hazardous than carbon fiber but still significant

**Ceramic Dust:** - **Crystalline silica:** Silicosis (progressive, incurable lung disease) - Latency: 10-30 years from exposure to symptoms - OSHA PEL: 0.05  $\text{mg}/\text{m}^3$  (very low permissible exposure) - **Aluminum oxide:** Respiratory irritant, possible lung disease - **Zirconia:** Metal fume when laser cutting (inhalation hazard)

**Resin Systems** (composites): - **Epoxy dust:** Sensitizer (allergic reactions develop over time) - **Phenolic:** Formaldehyde release during cutting - **Polyester:** Styrene fumes (VOCs)

#### Equipment Contamination

**Carbon Fiber Conductivity:** - Fine carbon dust is electrically conductive - Can short circuit electronics (CNC controllers, drives) - Accumulation on spindle = bearing failure - Static discharge risk (explosion in high concentration)

**Abrasive Wear:** - Ceramic dust extremely hard (alumina 9 on Mohs scale) - Accelerates wear on machine ways, ballscrews, bearings - Contaminates lubricants  $\rightarrow$  grinding paste effect - Spindle seals critical to protect bearings

**Cost of Poor Dust Control:** - Machine rebuild: \$5,000-50,000 - Spindle replacement: \$2,000-20,000 - Lost productivity during repairs - Health claims (workers’ compensation)

### Dust Collection System Design

#### Capture Methods

**Downdraft Table:** Air drawn down through perforated surface - Face velocity: 100-150 FPM minimum - Captures dust before becoming airborne - Large CFM requirements (500-5000+ CFM)

**Spindle Shroud/Hood:** Surrounds cutting tool - Lower CFM (50-200 CFM typical) - Very effective (close to source) - Can combine with downdraft

**Full Enclosure:** Machine fully enclosed - Negative pressure containment - Best contamination control - More expensive, less access

#### Filter Selection

**HEPA Filters Required:** - 99.97% efficiency at 0.3  $\mu\text{m}$  - Mandatory for carbon fiber, silica dust - Cartridge filters with pulse cleaning

**Wet Collection:** - Water scrubs dust from air - Good for fine particles - Sludge disposal required - Higher maintenance

### Personal Protective Equipment

**Respiratory Protection:** - P100 half-mask minimum for production work - PAPR for comfort during extended use - N95 inadequate for carbon fiber/ceramic grinding

**Skin Protection:** - Gloves (nitrile) - Long sleeves (fiber protection) - Disposable coveralls (Tyvek) for heavy exposure

**Eye Protection:** - Safety glasses with side shields (minimum) - Face shield for visible dust clouds

## Machine Protection

**Spindle Sealing:** - Positive pressure purge air system - Prevents dust entering bearings - Critical for equipment longevity

**Way Covers:** - Bellows protect linear guides - Wipers remove dust from ways - Essential for abrasive materials

**Electronics Protection:** - Sealed control cabinets - Positive pressure with filtered air - Prevents carbon fiber shorts

## Maintenance and Housekeeping

**Daily:** - HEPA vacuum machine surfaces (never blow with air) - Empty dust collector hopper - Damp wipe control surfaces

**Weekly:** - Check filter pressure drop - Inspect capture hoods for blockage

**Monthly:** - Deep clean machine internals - Inspect spindle seals

**Annually:** - Replace filter cartridges - Air sampling (exposure monitoring) - Blower maintenance

## Regulatory Compliance

**OSHA Standards:** - Silica standard (29 CFR 1910.1053) for ceramics - Respiratory protection program required - Exposure monitoring and recordkeeping

**Exposure Limits (8-hour TWA):** - Crystalline silica:  $0.05 \text{ mg/m}^3$  - Carbon fiber:  $5 \text{ mg/m}^3$  (NIOSH recommendation) - Aluminum oxide:  $5 \text{ mg/m}^3$  (respirable)

## Cost Analysis

**Small Shop System:** \$3,000-9,000 - Downdraft table, blower, filters - Installation and ductwork

**Operating Cost:** \$2,800-4,000/year - Electricity, filters, maintenance

**Cost of Not Having System:** - Spindle failure: \$2,000-20,000 - Health claims: \$50,000-500,000+ - OSHA fines: \$15,000+ per violation

## Summary

Dust control is mandatory for advanced materials machining: - Health hazards: Carbon fiber (carcinogen), silica (silicosis), glass fiber (irritant) - Equipment protection: Fine dust destroys bearings, shorts electronics - System requirements: HEPA filtration, proper capture velocity, maintenance - PPE: P100 respirator minimum for production work - Compliance: OSHA silica standard, respiratory protection program

**Next:** Surface finishing techniques for composites and ceramics

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**Next:** 17.7 Surface Finishing Techniques

## 17.7 Surface Finishing Techniques for Advanced Materials

### Why Surface Finish Matters for Advanced Materials

#### Mechanical Performance

##### Strength Dependency on Surface Quality:

Ceramics and composites are extremely sensitive to surface defects:

$$\sigma_{fracture} = \frac{K_{IC}}{\sqrt{\pi \times a}}$$

where: -  $\sigma_{fracture}$  = fracture strength -  $K_{IC}$  = fracture toughness (material property) -  $a$  = crack/defect size

**Example** (alumina ceramic): - Material: 99% alumina,  $K_{IC} = 4 \text{ MPa}\sqrt{\text{m}}$  - As-ground surface:  $a = 50 \text{ }\mu\text{m}$  surface cracks - Strength:  $4 / \sqrt{\pi \times 0.00005} = \mathbf{320 \text{ MPa}}$  - Polished surface:  $a = 5 \text{ }\mu\text{m}$  - Strength:  $4 / \sqrt{\pi \times 0.000005} = \mathbf{1,010 \text{ MPa}}$

##### 3x strength improvement from polishing!

**Composites:** - Rough surfaces = exposed fibers = moisture ingress - Delamination starts at rough edges - Surface finish affects fatigue life significantly

#### Functional Requirements

**Sealing Surfaces:** - O-ring grooves in ceramics:  $R_a < 32 \text{ }\mu\text{m}$  required - Rough surface = leak paths

**Optical Applications:** - Ceramic windows:  $R_a < 5 \text{ }\mu\text{m}$  for optical clarity - Polishing removes subsurface damage (scattering sites)

**Aesthetic Applications:** - Carbon fiber parts: High-gloss finish expected - Surface finish = quality perception

**Biomedical:** - Implants:  $R_a < 10 \text{ }\mu\text{m}$  reduces bacterial adhesion - Smooth surface = better tissue integration

### Composite Surface Finishing

#### Post-Machining Defects

**Fuzzing:** Loose fibers standing up from surface - Cause: Dull tools, improper feed direction - Appearance: Hairy, uneven

**Delamination:** Plies separating at edges - Cause: Exit tear-out, improper support - Visible as gaps between layers

**Fiber Pullout:** Holes where fibers torn out - Cause: Cutting against fiber direction, dull tools - Affects surface appearance and strength

**Resin Smearing:** Melted resin covering fibers - Cause: Excessive heat (dull tool, high speed) - Appearance: Glossy, gummy surface

#### Abrasive Finishing Methods

##### Sanding:

**Grit Progression:** 1. **80-120 grit:** Remove machining marks, level surface 2. **180-220 grit:** Smooth surface, remove 80-120 scratches 3. **320-400 grit:** Pre-finishing 4. **600-800 grit:** Final dry sanding

**Sanding Technique:** - Always sand in direction of fibers (if unidirectional weave) - Light pressure (heavy = heat = resin smearing) - Dust extraction mandatory (carbon fiber dust hazardous)

**Hand Sanding vs Machine:** - Hand: Better control, less heat, slower - Random-orbital sander: Faster, consistent, heat risk with prolonged contact - Avoid belt sanders (too aggressive, heat buildup)

**Wet Sanding** (600+ grit): - Water lubricates, cools, captures dust - Achieves smoother finish (grit cuts cooler) - Final step before polishing

**Sanding Block:** Use firm backing - Prevents sander following contours of defects - Maintains flatness

**Example Sanding Schedule:** - As-machined CFRP: Ra 200 muin, visible machining marks - 120 grit, 5 min -> Ra 100 muin - 220 grit, 5 min -> Ra 50 muin - 400 grit, 5 min -> Ra 25 muin - 800 grit wet, 10 min -> Ra 15 muin

**Abrasive Pads** (Scotch-Brite type): - Coarse (maroon): Equivalent to 120-150 grit - Medium (gray): 220-320 grit - Fine (white): 600-800 grit - Conformable (good for curved surfaces) - Less aggressive than sandpaper (good for light cleanup)

### Trimming and Edge Finishing

**Router Trimming:** - Clean up rough-cut edges - Diamond-coated or carbide compression bits - Light passes (0.020-0.050" per pass) - Prevents delamination at exit

**File Finishing:** - Fine-cut mill bastard file - Remove burrs, smooth edges - Cut on push stroke only (don't drag back)

**Edge Sanding:** - Sandpaper on hard block - Slightly bevel sharp edges (0.005-0.010" radius) - Prevents ply lifting at edges

### Polishing Composites

#### Polishing Compounds:

**Rubbing Compound** (coarse polish): - Abrasive: Aluminum oxide, ~3000 grit equivalent - Removes fine scratches from 800-grit sanding - Apply by hand (soft cloth) or machine (foam pad)

**Polishing Compound** (fine polish): - Abrasive: Finer particles, ~6000 grit equivalent - Brings out gloss - Multiple applications improve finish

**Example:** - 3M Perfect-It Rubbing Compound -> Polishing Compound - Each step: Apply, rub/buff, wipe clean, inspect

**Machine Polishing:** - Rotary buffer: 1500-2500 RPM with foam pad - Light pressure (let abrasive do work, not pressure) - Keep moving (avoid heat buildup in one spot) - Heat = resin softening = smearing

**Result:** High-gloss finish, Ra 5-10 muin achievable

### Clear Coating

**Purpose:** - Protect surface from UV (resin degrades in sunlight) - Fill minor imperfections - Enhance gloss - Seal against moisture

#### Types:

**Epoxy Clear Coat:** - Thick build (10-20 mils) - Excellent UV protection - Hard, durable - Requires mixing (2-part) - Application: Brush, roller, or pour-on

**Polyurethane:** - Good UV resistance - Flexible (less brittle than epoxy) - 1-part or 2-part - Application: Spray or brush

**Automotive Clear Coat** (2K urethane): - Professional-grade finish - Excellent gloss and durability - Requires spray gun - Expensive equipment

**Application Process:** 1. **Surface prep:** 400-600 grit wet sand 2. **Clean:** Isopropyl alcohol, lint-free cloth 3. **First coat:** Thin, seal surface 4. **Additional coats:** 2-5 coats (per product instructions) 5. **Cure:** Full cure 24-72 hours (temperature-dependent) 6. **Wet sand:** 1000-2000 grit (if imperfections) 7. **Final polish:** Rubbing compound -> polishing compound

**Result:** Mirror-like finish, protected surface

### Filled Edge Finishing

**Problem:** Carbon fiber weave creates voids at cut edges - Hollow appearance - Not sealed against moisture

**Solution:** Fill voids before finishing

**Filler Options:** - **Epoxy resin:** Thin enough to penetrate voids, strong - **CA glue (cyanoacrylate):** Fast cure, hard, easy to sand - **Body filler:** Fast, easy to shape, less strong

**Process:** 1. Apply filler to edge (brush or scrape) 2. Allow to cure 3. Sand flush with surface (80-120 grit) 4. Refill low spots (may require 2-3 applications) 5. Final sand smooth (220-400 grit) 6. Proceed with normal finishing (sanding, polishing)

**Result:** Solid-looking edge, sealed against moisture

## Ceramic Surface Finishing

### Post-Grinding Finish Improvement

**As-Ground Surface:** - Diamond grinding: Ra 20-100 muin typical - Visible grinding marks - Subsurface damage: 5-50 mum deep (microcracks)

#### Improving Finish:

**Fine Grinding:** - Switch to finer grit wheel (600-1200 grit) - Very light depth of cut (0.0001-0.0002" per pass) - Lower table speed (longer dwell time) - Achieves Ra 10-20 muin

**Honing:** - Bonded abrasive stones (similar to grinding, finer grit) - Reciprocating motion - Good for cylindrical surfaces (bearing bores) - Achieves Ra 5-15 muin

### Lapping Ceramics

**Process:** Loose abrasive slurry on lap plate

**Equipment:** - Lap plate: Cast iron or glass (flat reference surface) - Abrasive: Diamond paste or suspension - Pressure: Part pressed onto lap (by hand or weighted)

**Grit Progression:** 1. **30 mum diamond:** Remove grinding damage, flatten - Removal rate: 0.001-0.002"/10 min - Finish: Ra 40 muin 2. **15 mum diamond:** Remove 30 mum scratches - Removal rate: 0.0005"/10 min - Finish: Ra 20 muin 3. **9 mum diamond:** Pre-polish - Removal rate: 0.0002"/10 min - Finish: Ra 10 muin 4. **3 mum diamond:** Final lap - Removal rate: 0.0001"/10 min - Finish: Ra 5 muin

**Technique:** - Figure-8 motion (distributes wear evenly on lap) - Rotate part periodically (prevents part cupping) - Wet surface (slurry must stay liquid) - Clean between grits (contamination ruins finer steps)

**Lap Maintenance:** - Lap wears (becomes concave in center) - Re-flatten with coarser grit or use three-plate method - Three-plate: Lap plates lap each other flat

**Applications:** - Gage blocks, precision spacers - Seal faces (mechanical seals) - Optical flats (when followed by polishing)



## Polishing Ceramics

**Objective:** Mirror finish, minimal subsurface damage

**Polishing vs Lapping:** - Lapping: Bonded or loose abrasive, removes measurable stock - Polishing: Very fine abrasive, removes scratches only

**Polishing Cloth:** - Soft pad (neoprene, felt, polyurethane foam) - Conforms to surface - Less aggressive than hard lap

**Abrasives:** - **3 µm diamond paste:** Pre-polish - **1 µm diamond paste:** Polish - **0.25 µm diamond paste:** Fine polish - **0.05 µm colloidal silica:** Final polish (chemical-mechanical)

**Colloidal Silica:** - Suspension of ultra-fine silica particles - Mild chemical action (dissolves strained surface layer) - Mechanical action (abrasion) - Removes subsurface damage without creating new damage - Achieves Ra <5 µin (mirror finish)

**Example Polishing Schedule (alumina):** 1. **9 µm diamond on hard pad**, 5 min -> Ra 15 µin 2. **3 µm diamond on medium pad**, 5 min -> Ra 8 µin 3. **1 µm diamond on soft pad**, 10 min -> Ra 4 µin 4. **0.05 µm colloidal silica**, 15 min -> Ra 2 µin, mirror finish

**Machine Polishing:** - Vibratory polisher: Part and abrasive vibrate together (batch process) - Rotary polisher: Part rotates against polishing pad - Automatic (less labor, consistent results)

**Hand Polishing:** - More control - Small parts or low volume - Tedious for large areas

**Cleaning Between Steps:** - Ultrasonic cleaner (water + detergent) - Removes previous grit (prevents contamination) - Dry thoroughly before next step

## Thermal Treatments for Surface Improvement

**Annealing** (stress relief):

**Purpose:** Remove residual tensile stress from grinding - Tensile stress reduces strength (cracks open easier) - Annealing relaxes stress

**Process:** - Heat to 50-70% of sintering temperature - Hold 1-4 hours - Slow cool (prevent thermal shock)

**Example** (99% alumina): - Sintering temp: 1600-1700 degC - Annealing temp: 1000-1200 degC - Hold: 2 hours - Cool rate: 100 degC/hour (slow)

**Result:** 10-30% strength increase

**Glazing** (surface melting):

**Purpose:** Melt thin surface layer -> smooth, sealed surface

**Process:** - Rapid heating of surface only (laser, flame) - Surface melts, flows, re-solidifies - Removes surface roughness

**Challenges:** - Thermal shock (can crack part) - Microstructure change (melted layer different properties) - Difficult to control

**Limited Use:** Research, specialized applications

## Edge Quality and Deburring

### Composite Edge Deburring

**Fuzzing Removal:** - Light sanding (220-320 grit) - Abrasive pad (fine) - Rub edge gently (don't round over excessively)

**Delaminated Edge Repair:** 1. Clean gap (compressed air) 2. Inject thin epoxy or CA glue 3. Clamp plies together (C-clamp with soft jaws) 4. Cure, sand flush

**Sharp Edge Treatment:** - Slight radius (0.010-0.020") prevents ply lifting - Sandpaper on hard block - 45 deg chamfer or radius

### Ceramic Edge Deburring

**Edge Chipping** (common defect): - Grinding creates tiny chips at edges - Weakens edge, stress concentration

**Diamond Honing:** - Fine diamond stone (600-1200 grit) - Lightly chamfer edge (0.002-0.005") - Removes fragile edge chips

**Vibratory Deburring:** - Parts + ceramic media + compound in vibrating tub - Tumbles parts, rounds edges - Hours to days (slow) - Good for batch processing

**Hand Deburring:** - Diamond file or stone - Careful chamfering by hand - Small parts, low volume

## Surface Finish Measurement

### Contact Methods

**Profilometer** (stylus type): - Stylus drags across surface - Measures vertical displacement (surface profile) - Output: Ra, Rz, profile graph

**Advantages:** - Quantitative (Ra value) - Industry standard

**Disadvantages:** - Slow (point-by-point measurement) - Stylus can damage soft materials - Expensive (\$5,000-50,000)

**Ra Definition:** Average roughness

$$Ra = \frac{1}{L} \int_0^L |y(x)| dx$$

where  $y(x)$  = height deviation from mean line

**Interpretation:** - Ra 125 muin: Rough (as-ground with coarse wheel) - Ra 63 muin: Medium (standard machined surface) - Ra 32 muin: Smooth (ground with fine wheel) - Ra 16 muin: Fine (polished) - Ra 4 muin: Very fine (mirror finish)

### Non-Contact Methods

**Optical Profilometer:** - Light interference measures surface height - Very fine resolution (<1 nm vertical) - No contact (doesn't damage surface) - 3D surface map - Expensive (\$50,000-200,000)

**Visual Inspection:** - Comparator blocks (surface roughness standards) - Match part to standard block by feel/appearance - Inexpensive, subjective

### Gloss Measurement

**Glossmeter:** - Measures specular reflection (shininess) - Units: Gloss units (GU) - 60 deg angle standard

**Interpretation:** - <10 GU: Matte - 10-70 GU: Semi-gloss - >70 GU: High gloss - >90 GU: Mirror-like

**Use for Composites:** - Verifies polishing quality - Consistent finish batch-to-batch - Cost: \$500-3,000

## Process Recommendations by Application

### Aerospace Structural (CFRP)

**Requirements:** Fatigue resistance, moisture protection

**Process:** 1. Machine with sharp diamond tooling 2. Sand: 120 -> 220 -> 400 grit 3. Seal edges with epoxy filler 4. Apply structural primer (corrosion protection) 5. Paint (UV protection, identification)

**Result:** Sealed, protected surface (not high-gloss)

### **Automotive Aesthetic (CFRP)**

**Requirements:** Show-quality finish

**Process:** 1. Machine with compression cutter (minimal delamination) 2. Fill edges (epoxy or CA glue) 3. Sand: 120 -> 220 -> 400 -> 800 wet 4. Rubbing compound (machine buff) 5. Polishing compound 6. Clear coat (3-5 coats) 7. Wet sand clear coat (1500-2000 grit) 8. Final polish (compound)

**Result:** Mirror finish, deep gloss

### **Ceramic Bearing Race**

**Requirements:** Low friction, high wear resistance

**Process:** 1. Diamond grind (220 grit wheel) 2. Fine grind (600 grit wheel) 3. Lap (9 mum -> 3 mum diamond) 4. Polish (1 mum diamond -> 0.05 mum colloidal silica) 5. Ultrasonic clean 6. Inspect (profilometer, Ra <4 muin)

**Result:** Ultra-smooth, mirror finish, minimal subsurface damage

### **Ceramic Seal Face**

**Requirements:** Flat, smooth (sealing)

**Process:** 1. Grind flat (220 grit wheel) 2. Lap (30 mum -> 15 mum -> 9 mum diamond) 3. Inspect flatness (optical flat, <0.0001") 4. Polish (3 mum -> 0.05 mum colloidal silica) 5. Inspect finish (Ra <5 muin)

**Result:** Flat, leak-free sealing surface

## **Cost and Time Analysis**

### **Composite Finishing Time**

**Example:** 12" x 12" carbon fiber panel, show-quality finish

- Machining (router trim): 5 min
- Edge filling: 10 min (plus cure time)
- Sanding (120-800 grit): 45 min
- Machine polishing: 30 min
- Clear coat (3 coats): 60 min (plus cure time between coats)
- Final wet sand & polish: 60 min
- **Total hands-on time:** 3.5 hours
- **Total elapsed time:** 1-2 days (curing)

**Labor cost** @ \$30/hour: \$105 **Materials** (sandpaper, compounds, clear coat): \$20-40 **Total:** \$125-145 per panel

### **Ceramic Finishing Time**

**Example:** 2" diameter alumina disc, mirror finish

- Diamond grinding (rough): 15 min
- Diamond grinding (fine): 30 min
- Lapping (30 -> 15 -> 9 -> 3 mum): 60 min
- Polishing (1 -> 0.05 mum): 45 min
- Cleaning between steps: 15 min
- **Total:** 2.75 hours

**Labor cost** @ \$40/hour: \$110 **Materials** (diamond paste, colloidal silica): \$10-20 **Equipment** (grinder, lap, polish): Amortized over many parts **Total**: \$120-130 per part

**Compare to:** As-ground part (~\$50) -> Finishing adds \$70-80

## Summary

Surface finishing transforms machined advanced materials into functional, beautiful parts:

**Composites:** - Sanding: 120 -> 220 -> 400 -> 800 grit removes machining marks - Polishing: Rubbing compound -> polishing compound achieves high gloss - Clear coating: Protects from UV, enhances appearance - Edge filling: Seals voids, improves appearance

**Ceramics:** - Fine grinding: 600-1200 grit improves as-ground surface - Lapping: 30 -> 15 -> 9 -> 3  $\mu\text{m}$  diamond flattens, smooths - Polishing: 1 -> 0.05  $\mu\text{m}$  achieves mirror finish, removes subsurface damage - Annealing: Relieves stress, increases strength 10-30%

**Finish Affects Performance:** - Ceramics: 3x strength increase (polished vs as-ground) - Composites: Fatigue life, moisture resistance depend on finish

**Measurement:** - Profilometer: Quantitative Ra measurement - Glossmeter: Verifies polish quality - Visual: Comparator blocks (low-cost alternative)

**Cost:** \$100-150 per part typical (labor + materials)

**Next:** Quality control and inspection methods for advanced materials

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**Next:** 17.8 Quality Control and Inspection

## 17.8 Quality Control and Inspection for Advanced Materials

### Unique Inspection Challenges

#### Hidden Defects in Composites

**Internal Delamination:** - Plies separated inside part (not visible externally) - Causes: Excessive cutting forces, improper cure, impact damage - Catastrophic failure mode (sudden strength loss) - **Cannot detect by visual inspection**

**Voids:** - Air pockets within laminate - Causes: Improper layup, incomplete resin infiltration - Reduce strength 5-30% depending on void content - Not visible once cured

**Fiber Misalignment:** - Fibers not oriented as designed - Causes: Shifting during layup, improper cutting - Reduces strength in load direction

#### Hidden Defects in Ceramics

**Subsurface Cracks:** - Microcracks below surface from grinding - Not visible to eye (1-50  $\mu\text{m}$  deep) - Propagate under load  $\rightarrow$  fracture - Reduce strength 20-50%

**Inclusions:** - Foreign particles in ceramic body - Causes: Contamination during powder processing - Stress concentrators (failure initiation sites)

**Density Variations:** - Non-uniform sintering - Weak regions more porous - Can fracture prematurely

### Dimensional Inspection

#### Composites Dimensional Challenges

**Moisture Absorption:** - Epoxy composites absorb 1-5% water by weight - Swelling: 0.1-0.5% dimensional change - Reversible (part shrinks when dried) - **Measure in controlled environment** (50% RH, 68  $^{\circ}\text{F}$  standard)

**Thermal Expansion:** - Carbon fiber: Near-zero or negative CTE (along fiber) - Epoxy matrix: Positive CTE ( $\sim 50$  ppm/  $^{\circ}\text{C}$ ) - Part dimensions temperature-dependent - **Measure at reference temperature** (68  $^{\circ}\text{F}$  typical)

**Springback:** - Residual stresses cause shape change after machining - Parts drift hours to days after cutting - **Allow stabilization time before final inspection**

#### Ceramics Dimensional Challenges

**Firing Shrinkage:** - Green  $\rightarrow$  fired: 15-20% linear shrinkage typical - Variation:  $\pm 0.5\%$  batch-to-batch - Cannot hold tight tolerances without fired grinding

**Thermal Expansion** (during measurement): - Alumina: 8 ppm/  $^{\circ}\text{C}$  - If measured at 80  $^{\circ}\text{F}$  instead of 68  $^{\circ}\text{F}$ : - 2.000" part grows 0.000192" (0.0002") - Significant for tight tolerances ( $\pm 0.001$ " )

**Grinding Burn:** - Overheating during grinding  $\rightarrow$  localized expansion during measurement - False reading (part appears oversize) - Cools, shrinks  $\rightarrow$  actually undersize

#### Measurement Tools

**Calipers/Micrometers:** - Standard tools ( $\pm 0.001$ " accuracy) - **Caution with ceramics:** Point contact stress  $\rightarrow$  can chip edge - Use flat anvils (not pointed)

**Coordinate Measuring Machine (CMM):** - Probe touches part, records 3D coordinates - Accuracy:  $\pm 0.0001$ " (high-end machines) - Software calculates dimensions, GD&T

**CMM Challenges with Advanced Materials:** - Composite surfaces soft (probe deforms surface slightly)  
- Ceramic surfaces chip easily (probe contact force) - Solution: Non-contact probe (optical, laser)

**Optical Comparator:** - Projects magnified shadow of part onto screen - Compare to overlay (drawing) -  
Good for edge inspection, small features - No contact (gentle on fragile parts)

**Laser Scanner:** - Non-contact 3D measurement - Fast (thousands of points per second) - Good for complex shapes - Expensive (\$20,000-200,000)

### GD&T Considerations

**Flatness** (ceramics): - Lapping achieves 0.0001" easily - Measure with optical flat (interference fringes) -  
One fringe = 0.000012" deviation

**Parallelism** (ceramics): - CMM or indicator on surface plate - Achievable: +/-0.0001" for lapped ceramics

**Profile** (composites): - Complex 3D shapes common - CMM or laser scanner required - Compare scan to CAD model

## Non-Destructive Testing (NDT)

### Ultrasonic Inspection (UT)

**Principle:** Sound waves reflect from defects

**Process:** 1. Transducer emits ultrasonic pulse (1-10 MHz) 2. Pulse travels through part 3. Reflects from back surface or internal defect 4. Transducer receives echo 5. Time delay indicates defect depth

**Pulse-Echo Mode:**

```
Transducer -----> |           | <----- Echo from back
                    |   Part   |
                    |   -     | <----- Echo from void (earlier)
                    |           |
```

**Through-Transmission Mode:**

```
Transmitter -->|           |--> Receiver
              |   Part   |
              |   -     | = Void blocks signal (no signal at receiver)
```

**Detects:** - Delamination (composites) - Voids (composites) - Cracks (ceramics) - Porosity, inclusions

**Advantages:** - Internal defects detected - Quantitative (defect size, depth) - Relatively fast

**Limitations:** - Requires coupling (water, gel between transducer and part) - Complex shapes difficult -  
Interpretation requires training - Equipment: \$5,000-50,000

**C-Scan Output:** - 2D map of part (color-coded) - Good regions: One color - Defects: Different color (amplitude drop) - Easy visualization

**Example:** - Carbon fiber panel, 0.250" thick - 5 MHz transducer - Scan 12" x 12" panel: 10-30 minutes -  
Defect >0.050" diameter detectable

### Radiography (X-Ray, CT)

**Principle:** X-rays pass through part, absorbed by defects differently

**2D Radiography** (X-ray image): - X-ray source on one side, film/detector on other - Dense regions (inclusions) appear lighter - Voids appear darker - Similar to medical X-ray

**Computed Tomography (CT):** - Multiple X-ray images from different angles - Computer reconstructs 3D model - Slice through part virtually (see internal features) - Very detailed

**Detects:** - Voids, porosity - Inclusions (dense particles) - Cracks (if oriented correctly) - Fiber orientation (composites)

**Advantages:** - Excellent visualization (CT) - No contact - Permanent record (images)

**Limitations:** - Expensive (CT: \$100,000-1,000,000) - Slow (CT: 30 min to several hours per part) - Radiation safety (shielding, licensing) - Composites: Low contrast (carbon and resin similar density)

**Applications:** - High-value parts (aerospace) - Failure analysis - Production CT for critical parts (automotive, medical)

### **Thermography (Infrared Inspection)**

**Principle:** Heat flow disrupted by defects

**Active Thermography:** 1. Heat part (flash lamp, hot air) 2. Infrared camera observes cooling 3. Defects (voids, delaminations) cool differently 4. Thermal image shows defects

**Passive Thermography:** - Part in service generates heat - Monitor with IR camera - Hot spots indicate problems (friction, overload)

**Detects:** - Delamination (composites) - Voids near surface (<0.125" deep) - Poor bond lines

**Advantages:** - Fast (seconds to minutes) - Large area inspection (full panel) - Non-contact - Good for composites (UT sometimes difficult with complex weaves)

**Limitations:** - Shallow defects only - Requires thermal contrast - Equipment: \$10,000-100,000 (IR camera)

### **Tap Test (Coin Test)**

**Principle:** Sound changes when striking defect

**Process:** - Tap part surface with coin, small hammer, or automated tapper - Listen to sound - Solid area: Clear, ringing tone - Delaminated area: Dull, dead sound (energy absorbed by gap)

**Advantages:** - Simple, cheap (coin) - Fast - Effective for large defects (>0.25" diameter)

**Limitations:** - Subjective (depends on operator hearing) - Small defects undetectable - No permanent record

**Automated Tap Testing:** - Mechanical tapper + microphone - Computer analyzes sound frequency - More objective than manual - Equipment: \$5,000-20,000

### **Visual and Optical Inspection**

**Borescope** (fiber optic scope): - Inspect internal passages, holes - Diameter: 2-10 mm typical - Composites: Look for delamination at edges, fiber pullout

**Dye Penetrant** (ceramics): 1. Apply bright dye to surface 2. Dye seeps into cracks (capillary action) 3. Wipe surface clean 4. Apply developer (pulls dye back out) 5. Cracks visible as bright lines

**Detects:** Surface-breaking cracks only (not internal)

**Advantages:** Simple, cheap, very sensitive (0.001" wide cracks)

**Limitations:** Surface only, messy (cleaning required)

**Microscopy:** - Optical microscope: 10-1000x magnification - Inspect edge quality, fiber orientation, crack size - Scanning Electron Microscope (SEM): >10,000x magnification - Fracture surface analysis (failure mode) - Fiber-matrix adhesion quality - Very expensive (>\$100,000)

## Mechanical Testing

### Destructive Testing (Lot Sampling)

**Purpose:** Verify material properties (not every part, but sample)

**Tensile Test** (composites): - ASTM D3039 (flat coupon test) - Measures: Ultimate tensile strength, modulus, strain-to-failure - Verify: Material meets specification - Frequency: Each lot of material, or quarterly

**Flexural Test** (ceramics): - ASTM C1161 (4-point bend) - Measures: Flexural strength (modulus of rupture) - Ceramics too brittle for tensile test - Frequency: Each batch of parts (5-10 samples typical)

**Example** (alumina): - Specification:  $\sigma_{\text{flexural}} > 400 \text{ MPa}$  - Test 10 samples per batch - All pass -> accept batch - Any fail -> investigate (possibly reject batch)

**Weibull Statistics:** - Ceramics have large strength variation (brittle fracture) - Strength described by Weibull distribution - Requires many samples (20-30) for accurate characterization - Used for design (account for probability of failure)

### Non-Destructive Strength Testing

**Proof Testing:** - Load part to fraction of design load (80-90%) - If survives -> acceptable - If fails -> defective (removed from population) - Increases reliability of fleet (weak parts eliminated)

**Application:** - Ceramic parts for critical service (engine components) - Composite pressure vessels

**Limitation:** Some parts damaged (subcritical crack growth) even if pass test

## Surface Quality Inspection

### Surface Roughness

**Profilometer** (stylus type): - Quantitative Ra measurement - Industry standard - Verifies grinding/polishing quality

**Comparator Blocks:** - Machined samples with known Ra - Fingernail test: Drag fingernail across sample, then part - Match feel/appearance - Inexpensive, subjective

**Typical Specifications:** - Composite machined edge:  $Ra < 63 \mu\text{in}$  - Ceramic ground surface:  $Ra < 32 \mu\text{in}$  - Ceramic polished seal face:  $Ra < 5 \mu\text{in}$

### Gloss (Composites)

**Glossmeter:** - Measures specular reflection (shininess) - Verifies polish quality - Batch consistency

**Example Specification:** - Carbon fiber show panel:  $> 85 \text{ GU @ } 60^\circ$  (high gloss)

### Edge Quality (Composites)

**Visual Inspection:** - Magnification: 5-10x (handheld magnifier) - Look for: - Delamination (gap between plies) - Fiber pullout (holes) - Fuzzing (loose fibers) - Resin voids (gaps)

**Accept/Reject Criteria** (example): - Delamination  $> 0.010''$  long: Reject - Fiber pullout  $> 0.020''$  diameter: Reject - Fuzzing: Acceptable if removable by light sanding

### Subsurface Damage (Ceramics)

**Indentation Test:** - Vickers indenter pressed into surface - Observe cracks radiating from indentation - Crack length indicates subsurface damage depth

**Example:** - As-ground: Cracks extend  $50 \mu\text{m}$  from indentation - Polished: Cracks extend  $5 \mu\text{m}$  - Conclusion: Polishing removed  $45 \mu\text{m}$  damaged layer



**Etching:** - Chemical etch reveals grain structure - Grinding damage visible as disturbed grain layer - Metallographic technique (destructive, sample only)

## In-Process Monitoring

### Tool Wear Monitoring

**Why Critical for Advanced Materials:** - Rapid tool wear (abrasive fibers, hard ceramics) - Dull tool -> poor quality (delamination, chipping) - Quality degrades before tool “feels” dull

**Dimensional Trending:** - Measure first part, every 10th part, last part in run - Plot dimension vs part number - Trend shows tool wear - Example: Hole diameter increases 0.001” over 50 parts -> tool wearing

**Visual Inspection:** - Microscope (10-20x) inspect cutting edge - Compare to new tool - Replace when wear land >0.010” (composites), >0.005” (ceramics)

**Tool Life Tracking:** - Log tool hours (spindle on-time) - Replace at predetermined interval (before failure) - Example: Diamond endmill in CFRP, 20 hours life

### Part Count to Failure

**Establish Tool Life:** - Run tools to failure (quality degrades) - Count parts machined - Set replacement interval at 70-80% of failure count - Example: Tool fails at part 150 -> replace at part 100

### Force/Vibration Monitoring

**Dynamometer:** Measures cutting forces - Dull tool -> higher forces - Threshold alarm (force exceeds limit -> stop) - Expensive (\$10,000-50,000), research/production use

**Accelerometer:** Detects vibration - Chatter -> poor finish, tool damage - Monitor vibration amplitude - Stop machine if excessive

**Spindle Load Monitoring:** - CNC monitors spindle current - Dull tool -> higher current (more resistance) - Alarm if current exceeds limit - Built into some CNCs

## Statistical Process Control (SPC)

### Control Charts

**X-bar Chart** (average): - Plot average dimension of sample (5 parts) - Control limits: +/-3 standard deviations from mean - Points outside limits -> process out of control

**R Chart** (range): - Plot range (max - min) of sample - Monitors variation (consistency) - Increasing range -> tool wear, process instability

**Example** (ceramic grinding): - Dimension: 2.000 +/-0.002” - Sample: Measure 5 parts every 30 minutes - Plot average and range - Trend shows process drift (tool wear) - Adjust before parts go out of tolerance

### Cpk (Process Capability)

**Definition:** How well process fits within tolerance

$$C_{pk} = \min \left( \frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

where: -  $USL$  = Upper Specification Limit -  $LSL$  = Lower Specification Limit -  $\mu$  = Process mean -  $\sigma$  = Process standard deviation

**Interpretation:** -  $C_{pk} < 1.0$ : Process produces defects (cannot meet tolerance) -  $C_{pk} = 1.33$ : Acceptable (4-sigma process) -  $C_{pk} = 1.67$ : Good (5-sigma process) -  $C_{pk} > 2.0$ : Excellent (6-sigma process)

**Example:** - Tolerance:  $2.000 \pm 0.002$ " (LSL = 1.998, USL = 2.002) - Process mean:  $\mu = 2.000$ " - Process std dev:  $\sigma = 0.0005$ " -  $C_{pk} = (2.002 - 2.000) / (3 \times 0.0005) = 1.33$  (acceptable)

**If Cpk Low:** - Reduce variation (better machine, sharper tools, stable environment) - Center process (adjust offsets) - Widen tolerance (negotiate with customer)

## Quality Documentation

### Inspection Reports

**First Article Inspection (FAI):** - Complete dimensional inspection of first production part - Verify all features meet drawing - AS9102 form (aerospace) or similar - Submitted to customer for approval

**In-Process Inspection:** - Frequency: Per control plan (e.g., every 10 parts) - Record dimensions, visual inspection results - Batch traceability (lot numbers, material certs)

**Final Inspection:** - Before shipping - Verify critical dimensions, visual quality - Certificate of Conformance (CoC)

### Material Certifications

**Composites:** - Prepreg lot number, cure date - Mechanical test results (tensile, flexural) - Traceability to raw material batch

**Ceramics:** - Powder lot number - Sintering parameters (temperature, time) - Density, grain size - Mechanical properties (flexural strength, hardness)

**Retention:** - Aerospace: Permanent (life of aircraft) - Automotive: 15 years typical - General industry: 5-10 years

### NDT Reports

**Ultrasonic C-Scan:** - Image of part (color-coded defect map) - Operator notes (defect locations, sizes) - Accept/reject decision with criteria

**Radiography:** - X-ray images (film or digital) - Defect call-outs (circles, arrows) - Interpretation by certified technician (Level II or III per ASNT)

## Acceptance Criteria Development

### Defining Defect Limits

**Cosmetic vs Structural:** - Cosmetic: Visible defects, don't affect strength (scratches, color variations) - Structural: Affect performance (cracks, delamination, porosity)

**Example** (composite panel): - Scratches  $< 0.010$ " deep: Acceptable (cosmetic) - Surface porosity  $< 0.020$ " diameter,  $< 5$  per square inch: Acceptable - Delamination any size: Reject (structural)

**Engineering Analysis:** - FEA (finite element analysis) with defect modeled - Stress concentration factor - Determine safe defect size

**Example** (ceramic): - Part design:  $\sigma_{max} = 200$  MPa in service - Material strength: 400 MPa (safety factor 2.0) - Analysis: 0.1 mm surface crack  $\rightarrow$  stress concentration 2.5x  $\rightarrow$   $\sigma_{local} = 500$  MPa - Conclusion: Reject parts with cracks  $> 0.1$  mm

### Industry Standards

**Composites:** - ASTM D standards (test methods) - SAE, AMS specs (aerospace materials) - Customer specifications (often proprietary)

**Ceramics:** - ASTM C standards (ceramic materials, test methods) - ISO standards (international) - Military specs (MIL-STD)

**NDT:** - ASTM E standards (NDT procedures) - ASNT (American Society for Nondestructive Testing) certification

## Cost of Quality

### Inspection Cost

**In-Process Inspection:** - Operator measures dimensions: 2-5 min per part - @ \$25/hour -> \$0.83-2.08 per part

**Final Inspection:** - CMM inspection (complex part): 30-60 min - @ \$50/hour (technician + equipment) -> \$25-50 per part

**NDT (Ultrasonic C-Scan):** - Setup + scan: 30 min per part - @ \$60/hour -> \$30 per part - Equipment amortization: \$10-20 per part - **Total:** \$40-50 per part

**Cost Drivers:** - Inspection frequency (every part vs sampling) - Complexity (number of features) - Equipment cost (CMM, NDT)

### Cost of Poor Quality

**Scrap** (defect found before shipping): - Material cost - Machining cost - Example: \$500 part, scrap rate 5% -> \$25 per good part (scrap cost allocated)

**Rework** (defect repairable): - Labor to fix (sanding, filling, re-grinding) - Example: 30 min @ \$30/hour = \$15 - If 10% require rework -> \$1.50 per part average

**Return/Warranty** (defect found by customer): - Replacement part: \$500 - Shipping: \$50 - Customer downtime: \$1,000-10,000 (lost revenue) - Reputation damage: Unquantifiable - **Far exceeds scrap cost**

**Liability** (part fails in service): - Injury, property damage - Legal costs, settlements: \$100,000-10,000,000+ - Aerospace, medical: Extremely high stakes

**Conclusion:** Inspection is cheap compared to failures

## Summary

Quality control for advanced materials requires specialized techniques:

**Inspection Challenges:** - Hidden defects (delamination, subsurface cracks) - Dimensional instability (moisture, thermal effects) - Fragile (contact measurement difficult)

**Dimensional Inspection:** - CMM, optical measurement (non-contact preferred) - Control environment (temperature, humidity) - GD&T: Flatness, parallelism achievable to  $\pm 0.0001$ " (ceramics)

**Non-Destructive Testing:** - Ultrasonic: Detects internal delamination, voids, cracks - Thermography: Fast, large area, shallow defects - Radiography/CT: Excellent visualization, slow, expensive - Tap test: Simple, effective for large delaminations

**Mechanical Testing:** - Destructive sampling: Verify material properties - Proof testing: Remove weak parts from population

**Surface Quality:** - Profilometer: Ra measurement (objective) - Glossmeter: Polish quality (composites) - Visual: Edge quality, defects

**In-Process Monitoring:** - Tool wear trending (dimensional, visual) - SPC charts: Detect process drift

**Documentation:** - FAI, in-process reports, material certs - NDT reports with images - Traceability (aerospace: permanent records)

**Cost:** - Inspection: \$5-50 per part (method-dependent) - Poor quality: \$500-1,000,000+ per failure - **In-  
spection is cheap insurance**

**Next:** Safety and health hazards in advanced materials machining

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**Next:** 17.9 Safety and Health Hazards

## 17.9 Safety and Health Hazards in Advanced Materials Machining

### Respiratory Hazards

#### Carbon Fiber Dust

**Particle Characteristics:** - Size: 5-10  $\mu\text{m}$  diameter, length  $\gg$  diameter (fiber-like) - **Respirable:** Particles  $<10 \mu\text{m}$  reach deep lung (alveoli) - Geometry: Similar to asbestos (length:diameter ratio  $>3:1$ ) - Biopersistent: Body cannot break down or clear effectively

#### Health Effects:

**Acute** (short-term): - Cough, throat irritation - Difficulty breathing (if heavy exposure) - Eye irritation (fibers scratch cornea)

**Chronic** (long-term): - **Possible carcinogen:** IARC Group 2B (possibly carcinogenic to humans) - Lung fibrosis (scarring): Similar mechanism to asbestos - No definitive epidemiological studies (material relatively new) - Precautionary principle: Treat as serious hazard

**Occupational Exposure Limit:** - **No OSHA PEL** (not yet regulated) - NIOSH recommends:  **$5 \text{ mg}/\text{m}^3$**  (8-hour TWA) for respirable particles - UK HSE:  **$2 \text{ fibers}/\text{cm}^3$**  (for fibers  $>5 \mu\text{m}$  long) - Some companies: Self-impose  $1\text{--}2 \text{ mg}/\text{m}^3$  (more conservative)

**Exposure Scenarios:** - Routing carbon fiber: Dust clouds visible  $\rightarrow$  very high exposure - Grinding: Fine dust generation (worst case) - Hand sanding: Localized dust (still hazardous)

**Example Measurement:** - Routing CFRP without dust collection - Breathing zone sample:  $15 \text{ mg}/\text{m}^3$  - **3x over NIOSH recommendation**  $\rightarrow$  unacceptable

#### Glass Fiber Dust

**Particle Characteristics:** - Diameter: 5-15  $\mu\text{m}$  (respirable) - Length: Broken fibers during cutting - Less biopersistent than carbon fiber (body can dissolve slowly)

#### Health Effects:

**Acute:** - Skin irritation (“itching”): Fibers embed in skin - Eye irritation: Mechanical scratch - Respiratory irritation: Cough, sore throat

**Chronic:** - Lung fibrosis (possible): Less evidence than asbestos - IARC Group 3: Not classifiable as carcinogenic (insufficient evidence) - Still hazardous (respiratory irritant)

**Occupational Exposure Limit:** - OSHA PEL:  **$15 \text{ mg}/\text{m}^3$**  (total dust),  **$5 \text{ mg}/\text{m}^3$**  (respirable) - ACGIH TLV:  **$5 \text{ mg}/\text{m}^3$**  (respirable),  **$1 \text{ fiber}/\text{cm}^3$**  (fibers  $>5 \mu\text{m}$ )

**Less Hazardous than Carbon Fiber:** But still requires dust control

#### Ceramic Dust

##### Crystalline Silica (quartz):

Many ceramics contain silica: - Porcelain: 20-40% silica - Alumina (lower grades): 1-10% silica impurity - Silicon carbide, silicon nitride: Oxidize to silica at high temp

**Health Effects:** - **Silicosis:** Progressive, irreversible lung disease - Inhaled silica particles engulfed by macrophages - Macrophages die, release fibrotic agents - Lung tissue scars (fibrosis) - Breathing capacity declines - Can progress even after exposure stops - **No cure** - Latency: 10-30 years from exposure to symptoms - Increased risk: Lung cancer, tuberculosis, autoimmune diseases

**Occupational Exposure Limit:** - OSHA PEL:  **$0.05 \text{ mg}/\text{m}^3$**  (respirable crystalline silica, 8-hour TWA) - **Extremely low** (one of the strictest limits) - Difficult to achieve without engineering controls

**Silica Rule** (29 CFR 1910.1053): - Requires: Exposure assessment, engineering controls, medical surveillance - Recordkeeping: 30 years - Applies to: Any operation generating silica dust (grinding, sawing)

#### Other Ceramic Dusts:

**Aluminum oxide** (alumina): - OSHA PEL: **15 mg/m<sup>3</sup>** (total dust), **5 mg/m<sup>3</sup>** (respirable) - Respiratory irritant (less hazardous than silica) - Chronic exposure: Possible lung disease (aluminosis, rare)

**Zirconia** (zirconium oxide): - OSHA PEL: **5 mg/m<sup>3</sup>** (respirable) - Generally low toxicity - Thermal processes (laser cutting): Releases zirconium fumes (more hazardous)

**Beryllium oxide** (BeO): - **Extremely toxic**: Beryllium disease (chronic berylliosis) - OSHA PEL: **0.2 mg/m<sup>3</sup>** (0.0002 mg/m<sup>3</sup>!) - Specialized handling, rarely used in general machining

#### Resin Dust and Fumes

**Epoxy Dust** (cured): - Respiratory irritant - **Sensitizer**: Repeated exposure -> allergic reactions develop - Initial: No reaction - After weeks/months: Sudden allergic response (rash, asthma) - Once sensitized: Permanent (even tiny exposure causes reaction) - OSHA PEL: Variable by specific epoxy (0.1-5 mg/m<sup>3</sup> typical)

**Phenolic Resin** (grinding/cutting): - Releases formaldehyde (VOC) - OSHA PEL (formaldehyde): **0.75 ppm** (8-hour TWA) - Carcinogen (IARC Group 1) - Requires ventilation

**Polyester Resin**: - Releases styrene (VOC) during cure and thermal decomposition - OSHA PEL (styrene): **100 ppm** (8-hour TWA) - Short-term: Dizziness, headache - Long-term: Neurological effects

#### Skin Hazards

##### Fiber Irritation

**Carbon Fiber**: - Stiff, sharp fibers (7 µm diameter) - Embed in skin -> itching, rash - Difficult to remove (too small to see/grasp) - Splinters work deeper over time

**Glass Fiber**: - Similar to carbon fiber (worse due to larger diameter) - “Fiberglass itch” well-known

**Prevention**: - Long sleeves (tightly woven fabric) - Gloves (nitrile, leather) - Wash exposed skin (cold water, don’t rub -> drives fibers deeper) - Adhesive tape: Dab on skin, lift fibers

##### Chemical Exposure

**Uncured Resins**: - Epoxy: Sensitizer (skin contact -> allergic dermatitis) - Once sensitized: Blistering rash from minimal exposure - Prevention: Gloves (nitrile), avoid skin contact

**Solvents** (cleaning, laminating): - Acetone: Dries skin (dermatitis) - Methyl ethyl ketone (MEK): Skin irritant, absorbed through skin - Prevention: Gloves, minimize contact

**Coolants** (machining): - Water-based coolants: Can harbor bacteria (dermatitis) - Oil-based: Clog pores (acne-like rash) - Prevention: Barrier cream, wash after exposure

#### Eye Hazards

**Particulate Hazards**: - Carbon fiber: Scratches cornea (painful, slow healing) - Ceramic dust: Abrasive (grinding particles) - Flying chips (rare with composites, common with ceramics)

**Chemical Splash**: - Resins (epoxy, polyester): Irritant, can cause damage - Coolants: Bacterial contamination (infection risk)

**Prevention**: - Safety glasses with side shields (minimum) - Face shield (when dust clouds visible) - Goggles (chemical splash protection)

**First Aid:** - Eye flush station (15 minutes continuous rinse) - Seek medical attention (foreign body, chemical exposure)

## Fire and Explosion Hazards

### Carbon Fiber Conductivity

**Electrical Hazard:** - Carbon fiber is electrically conductive (similar to graphite) - Fine dust settles on electronics - Can cause short circuits: - CNC controller failure - Motor drive damage - Instrumentation malfunction

**Static Electricity:** - Carbon fiber dust generates static during handling - Accumulation -> spark discharge - Dust cloud ignition possible

**Prevention:** - Grounded equipment (dissipate static) - Sealed control cabinets (positive pressure) - Dust collection (prevent accumulation) - No open flames in machining area

### Dust Explosions

**Combustible Dust:** - Organic materials (resin dust, wood) form explosive mixtures - Carbon fiber dust: Combustible - Fine particles + oxygen + ignition source = explosion

**Explosion Pentagon:** 1. Combustible dust 2. Oxygen (air) 3. Ignition source (spark, hot surface) 4. Confinement (dust cloud in enclosed space) 5. Dispersion (dust suspended, not settled)

**Remove any one factor** -> no explosion

**Historical Incidents:** - Dust explosions in woodworking, grain handling (devastating) - Composites: Lower risk (less fuel) but still possible

**Prevention:** - Dust collection (remove fuel) - Eliminate ignition sources (no sparking tools, hot work permits) - Vent enclosures (prevent confinement) - Housekeeping (prevent dust accumulation on surfaces)

**OSHA Combustible Dust NEP** (National Emphasis Program): - Inspections targeting dust hazards - Fines for non-compliance (\$15,000-150,000 per violation)

### Resin Fires

**Uncured Resins:** - Flammable liquids (epoxy, polyester) - Flash point: 200-400 degF (moderate) - Storage: Flammable cabinet

**Cured Resin Dust:** - Combustible (burns if ignited) - Lower risk than uncured (not liquid)

**Prevention:** - No smoking, no open flames - Grounded containers (static dissipation) - Minimize inventory (less fuel)

## Chemical Hazards

### Volatile Organic Compounds (VOCs)

**Sources:** - Resin systems during cure (styrene, formaldehyde) - Solvents (acetone, MEK) for cleaning - Thermal decomposition during machining (burnt resin)

**Health Effects:** - Short-term: Dizziness, headache, nausea - Long-term: Liver/kidney damage, neurological effects - Some VOCs: Carcinogens (formaldehyde, benzene)

**Control:** - Ventilation (local exhaust capture fumes) - Activated carbon filters (VOC adsorption) - Minimize heating (lower speeds, sharp tools)

**Monitoring:** - PID (photoionization detector): Real-time VOC measurement - Air sampling: Lab analysis (identify specific compounds)

## Coolant Contamination

**Bacterial Growth:** - Water-based coolants: Ideal growth medium (warm, nutrients) - Bacteria, fungi, endotoxins - Inhalation: Hypersensitivity pneumonitis (“machine operator’s lung”) - Skin contact: Dermatitis, infections

**Control:** - Biocides (maintain proper concentration) - Change coolant regularly (don’t just top off) - Clean sump (remove residue, biofilm) - Monitor pH, concentration

**Alternative:** Dry machining or mist systems (less bacteria risk)

## Noise Hazards

**High-Speed Machining:** - Composites: 85-95 dBA typical (routing, sawing) - Ceramics: 90-100 dBA (grinding) - OSHA PEL: **90 dBA** (8-hour TWA) - Exceeding PEL -> hearing protection required

**Hearing Protection:** - Earplugs: 15-30 dB reduction (NRR rating) - Earmuffs: 20-35 dB reduction - Combination (plugs + muffs): 30-40 dB reduction

**Engineering Controls:** - Enclosures (reduce noise at source) - Vibration damping (machine mounts)

**Hearing Conservation Program** (if >85 dBA): - Audiometric testing (annual) - Training (noise hazards) - Hearing protection provided

## Ergonomic Hazards

**Repetitive Motion:** - Hand sanding composites (hours) - Carpal tunnel syndrome, tendonitis

**Awkward Postures:** - Reaching into enclosures - Working on large panels (overhead, bent)

**Vibration** (hand tools): - Grinders, sanders - Hand-arm vibration syndrome (HAVS): Numbness, loss of dexterity

**Prevention:** - Rotate tasks (vary motions) - Ergonomic tools (padded handles, low vibration) - Mechanical assists (lifts for heavy parts)

## Machine Hazards

**Rotating Tooling:** - Spindle speeds: 10,000-60,000 RPM typical - Contact -> severe injury (cuts, amputations)

**Safeguarding:** - Interlocks (door open -> spindle stops) - Guards (fixed or adjustable barriers) - E-stop accessible

**Coolant Spray:** - High-pressure mist (chip clearing) - Eye hazard (bacterial contamination, particulates) - Prevention: Enclosed machining, eye protection

## Emergency Procedures

### Dust Fire

**Small Fire** (trash can size): - ABC fire extinguisher - Evacuate if grows - **Do not use water** (spreads burning dust)

**Large Fire:** - Evacuate immediately - Call 911 - Do not re-enter

**Prevention:** - Housekeeping (no dust accumulation) - Empty dust collectors daily - No ignition sources



## Chemical Spill

**Small Spill** (<1 gallon): - Absorbent (spill pads, kitty litter) - Dispose per regulations (hazardous waste if flammable)

**Large Spill:** - Evacuate area - Call emergency response (hazmat team) - Contain if safe (absorbent booms)

**Resin Spill:** - Epoxy (uncured): Absorbent, dispose as hazardous waste - Ventilate area (fumes)

## Exposure Incident

**Skin Contact** (resin, solvent): - Wash immediately (soap, water, 15 minutes) - Medical attention if irritation persists

**Eye Contact:** - Eye wash station (15 minutes continuous rinse) - Hold eyelids open (flush under lids) - Seek medical attention (even if feels better)

**Inhalation** (dust, fumes): - Move to fresh air - Seek medical attention if symptoms (difficulty breathing, chest pain) - Provide MSDS to medical personnel

**Ingestion** (rare): - Do not induce vomiting - Call poison control (1-800-222-1222)

## Personal Protective Equipment (PPE)

### Respiratory Protection Program

**Required Elements** (OSHA 29 CFR 1910.134): 1. **Written program:** Procedures, responsibilities 2. **Hazard assessment:** Measure exposure, determine required protection 3. **Respirator selection:** Match to hazard 4. **Fit testing:** Annual (quantitative or qualitative) 5. **Medical evaluation:** Physician clears employee to wear respirator 6. **Training:** Proper use, limitations, maintenance 7. **Maintenance:** Cleaning, storage, inspection, repair

### Respirator Selection:

| Hazard                             | Respirator Type                | Notes                           |
|------------------------------------|--------------------------------|---------------------------------|
| Carbon fiber dust (light exposure) | N95 filtering facepiece        | Disposable, limited protection  |
| Carbon fiber dust (production)     | P100 half-mask                 | Reusable, 99.97% filtration     |
| Ceramic dust (silica)              | P100 half-mask or PAPR         | PAPR for extended use (comfort) |
| VOCs (fumes from resins)           | Organic vapor cartridge + P100 | Combination filter              |
| High dust concentration            | PAPR                           | Powered air (positive pressure) |

**Fit Testing:** - Qualitative: Taste test (saccharin, bittering agent) - Quantitative: PortaCount (measures leakage) - Must pass annually - Facial hair breaks seal (not allowed)

**Medical Evaluation:** - Questionnaire (heart/lung conditions) - Physician determines if safe to wear respirator - Some conditions preclude use (respirator increases work of breathing)

## Skin Protection

**Gloves:** - Nitrile: Chemical resistance (resins, solvents) - Leather: Abrasion resistance (handling parts) - Cotton: Absorbs sweat, comfort (over nitrile)

**Sleeves:** - Long-sleeve shirt (tightly woven) - Prevents fiber embedment - Disposable sleeves (Tyvek) for heavy exposure

**Coveralls** (Tyvek): - Disposable - Prevents contamination of street clothes - Remove before leaving work area (don't take fibers home)

### Eye/Face Protection

**Safety Glasses:** - Side shields required - Polycarbonate lenses (impact resistant) - Prescription available

**Face Shield:** - Full face coverage - Use over safety glasses (not replacement) - Visible dust clouds, chemical splash risk

**Goggles:** - Sealed (chemical splash protection) - Anti-fog coating - Uncomfortable (use when necessary)

## Training and Communication

### Hazard Communication (HazCom)

**OSHA 29 CFR 1910.1200:** Right to know about chemical hazards

**Requirements:** 1. **Written program:** How hazards communicated 2. **Labels:** Containers labeled (chemical name, hazards) 3. **Safety Data Sheets (SDS):** Available for all chemicals 4. **Training:** Employees trained on hazards, protective measures

**SDS Sections:** - Section 2: Hazards (GHS pictograms) - Section 8: Exposure limits (PEL, TLV) - Section 8: Personal protection (respirator, gloves) - Section 11: Toxicology (health effects)

**Example** (carbon fiber MSDS): - Section 2: Health hazard (respiratory) - Section 8: OEL 5 mg/m<sup>3</sup>, P100 respirator recommended - Section 11: Possible carcinogen (IARC 2B)

### Employee Training

**Topics:** 1. **Hazards:** What materials are hazardous, health effects 2. **Engineering controls:** Dust collection, ventilation 3. **PPE:** What to use, how to wear, limitations 4. **Procedures:** Safe work practices, housekeeping 5. **Emergency:** Fire, spill, exposure response

**Frequency:** - Initial (before working with materials) - Annual refresher - After incident or procedure change

**Documentation:** - Sign-in sheets (proof of training) - Retain per OSHA (3-30 years depending on standard)

### Medical Surveillance

**Required by OSHA** (silica rule): - Baseline exam (before exposure) - Annual chest X-ray (detect silicosis early) - Spirometry (lung function test) - Physician review, recommendations

**Company Medical Surveillance** (proactive): - Even if not OSHA-required - Detect disease early (treatable stages) - Document exposure (workers' comp protection)

## Regulatory Compliance Summary

| Regulation                          | Scope                       | Requirements                                                                      |
|-------------------------------------|-----------------------------|-----------------------------------------------------------------------------------|
| 29 CFR 1910.1053 (Silica)           | Crystalline silica exposure | Exposure assessment, controls, medical surveillance, training, records (30 years) |
| 29 CFR 1910.134 (Respiratory)       | Respirator use              | Written program, fit testing, medical evaluation, training                        |
| 29 CFR 1910.1200 (HazCom)           | Chemical hazards            | SDS, labels, training                                                             |
| 29 CFR 1910.1000 (Air Contaminants) | General air quality         | Exposure limits (PELs)                                                            |

| Regulation                       | Scope               | Requirements                                                 |
|----------------------------------|---------------------|--------------------------------------------------------------|
| 29 CFR 1910.95 (Noise)           | Noise >85 dBA       | Hearing conservation program (testing, protection, training) |
| 29 CFR 1910.147 (Lockout/Tagout) | Machine maintenance | Energy isolation procedures                                  |

**Inspections and Fines:** - OSHA inspections: Complaint-based, referral, or programmed (silica NEP)  
- Violations: Serious (\$16,131 per violation), willful (\$161,323 max), repeat (double) - Silica rule highly enforced (priority)

## Summary

Machining advanced materials creates serious health and safety hazards:

**Respiratory Hazards** (most serious): - Carbon fiber: Possibly carcinogenic (IARC 2B), biopersistent fibers  
- Silica (ceramics): Silicosis (deadly lung disease), OSHA PEL 0.05 mg/m<sup>3</sup> - Exposure limits very low -> engineering controls mandatory - Respirator: P100 half-mask or PAPR required for production work

**Skin/Eye Hazards:** - Fibers: Embed in skin (itching), scratch eyes - Resins: Sensitizers (allergic reactions), eye irritants - PPE: Gloves, long sleeves, safety glasses/face shield

**Fire/Explosion:** - Carbon fiber: Conductive (shorts circuits), combustible dust - Dust explosions: Possible with organic dusts - Prevention: Housekeeping, dust collection, no ignition sources

**Compliance:** - OSHA Silica Rule (29 CFR 1910.1053): Medical surveillance, 30-year records - Respiratory Protection Program: Fit testing, medical eval, training - HazCom: SDS, labels, training

**Cost of Non-Compliance:** - Fines: \$16,000+ per violation - Workers' comp claims: \$50,000-500,000 (occupational disease) - Lawsuits: Millions (if employee disabled/killed)

**Engineering controls** (dust collection, ventilation) are most effective and economical long-term solution.

**Next:** Maintenance and tool management for advanced materials machining

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**Next:** 17.10 Maintenance and Tool Management